

**EXPLICIT SUB-WEYL BOUND FOR THE RIEMANN ZETA
FUNCTION**

DISSERTATION

Presented in partial fulfillment of the requirements for the degree Doctor of Philosophy
in the Graduate School of The Ohio State University

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The Ohio State University
2021

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ABSTRACT

In this thesis we obtain the following explicit sub-Weyl bound for the Riemann zeta function: $|\zeta(1/2 + it)| \leq 307.098t^{27/164}$ for $t \geq 3$. Applications of this bound include improvement to Turing's method as done by Trudgian, and improvement to explicit zero free regions as done by Kadiri–Lumley–Ng. We discuss these applications in Sections 1.3 and 1.4 respectively.

Using the Riemann–Siegel formula, one obtains the following explicit bound on zeta (a so-called convexity bound): If $t \geq 3$, then $|\zeta(1/2 + it)| \leq 4t^{1/4}/(2\pi)^{1/4}$. Using the van der Corput method, Cheng and Graham gave an explicit Weyl-bound on zeta, which was later improved by Hiary who obtained: If $t \geq 3$, then $|\zeta(1/2 + it)| \leq 0.77t^{1/6} \log t$. We discuss this further in Section 2.2.1. There was no explicit sub-Weyl bound on zeta, which is to say no bound where the t exponent is less than $1/6$. The exponent $1/6$ (called Weyl-exponent) is significant. Even using implicit big- $\mathcal{O}$ constants, breaking the $1/6$ -barrier presented significant difficulties, leading to the development of the powerful method of exponent pairs.

Our main result advances explicit bounds for $|\zeta(1/2 + it)|$ beyond the Weyl-exponent $1/6$. So we find a quadruple of numbers (a, κ, C, t_0) , with $a < 1/6$, such that $|\zeta(1/2 + it)| \leq Ct^a \log^\kappa t$ for $t \geq t_0$. Part of our motivation is to remove the $\log t$ term from the previous explicit bounds, that is to enable $\kappa = 0$. This is significant in practice since for a long range of t the benefit of removing the $\log t$ term is more substantial than lowering the exponent a by a small amount.

To obtain our main result, we develop explicit processes from the method of exponent pairs. The method of exponent pairs furnishes bounds of the form $\sum_{n \in I} e^{2\pi i f(n)} \ll P^k N^l$ where (l, k) is the “exponent pair,” I is a subinterval of $[N, 2N]$, $P \ll |f'(x)| \ll P$ over $x \in I$, and under certain additional conditions on the real-valued function f . One can apply two processes: The A -process which transforms the exponential sum using the Weyl differencing, and the B -process which transforms the exponential sum using the Poisson summation. Each time one of these processes is applied to an exponent pair, the process returns another exponent pair. We discuss the theory of exponent pairs briefly in Section 1.5.

Cheng and Graham derived an explicit version of the process A , and an explicit version of the process B . Hiary derived an explicit version of the process AB . In this thesis we derive explicit versions of the following processes:

- A^2B process stated in Lemma 2.14.
- A^3B process stated in Lemma 2.15.
- B process stated in Theorem 2.31.

Using the above listed explicit processes, we derive an explicit version of the process ABA^3B . When applied to the trivial exponent pair $(0, 1)$, this process returns $|\zeta(1/2+it)| \ll t^{27/164}$. This exponent $27/164$ is approximately 0.16463 which is smaller than the Weyl exponent $1/6$ which is approximately 0.16666.

DEDICATION

Dedicated to my grandpa, *Dadu*

ACKNOWLEDGEMENTS

First and foremost, I would like to thank my advisor, Ghaith Hiary for introducing me to this problem and for without him this work would not have been possible. His patience, encouragement and honest feedback has helped me grow as a researcher.

To Roman Holowinsky for helping me and other graduate students successfully navigate the job market and guiding us for post graduate opportunities in industry.

To Niko Schonsheck and Duncan Clark for shaping the way I teach and for numerous talks we had regarding being a better teacher.

To Dan Boros and teaching staff at the Math Department for always being encouraging in my teaching endeavors.

Special thanks to the following people who supported me outside of academia during my graduate school journey: Niko Schonsheck and Duncan Clark for our climbing and lifting sessions, Andreu Ferré Moragues for discussing math along with listening to my frustrations over Friday lunch. A huge thank you to Brittney Wilson for her unreserved support, and care.

This list would be incomplete without massive thank you to my parents and wonderful sister for their numerous sacrifices and unconditional love for me. Their work ethic has been a constant source of inspiration. Without them, I do not think I would have been the person I am today.

Throughout my academic journey, whether it be in India or the U.S, I have been extremely fortunate to receive guidance and encouragement from numerous friends, teachers, and family members and to whom I wish to extend my thanks. I have surely missed many individuals who helped me get to this stage but to them I am always grateful.

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CHAPTER 1

INTRODUCTION

1.1 The Riemann Zeta Function

The Riemann zeta function [Dav00] is defined for $s = \sigma + it$ by the Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \sigma > 1. \quad (1.1)$$

It can be continued analytically to the whole complex plane except for a simple pole at $s = 1$. Riemann achieved this using an integral representation for the zeta function [Edw01]

$$\zeta(s) = \frac{\Gamma(1-s)}{2\pi i} \int_{+\infty}^{+\infty} \frac{(-x)^s}{e^x - 1} \frac{dx}{x}. \quad (1.2)$$

where the definition of $(-x)^s$ is $e^{s \log(-x)}$ with $-\pi < \Im(\log(-x)) < \pi$. The limits of integration are intended to indicate a path of integration which begins at $+\infty$, moves to the left down the positive real axis, circles the origin in positive (counterclockwise) direction, and returns up the the positive real axis to $+\infty$. This path of integration must be taken to be slightly above the real axis as it descends from $+\infty$ to 0 and slightly below the real axis as it goes from 0 back to $+\infty$ since $\log(-x)$ is not defined for positive real x .

From (1.2) one obtains a relationship between $\zeta(s)$ and $\zeta(1-s)$ which is known as the *functional equation of the zeta function* given by

$$\zeta(s) = \chi(s) \zeta(1-s), \quad (1.3)$$

where $\chi(s) = \pi^{s-\frac{1}{2}} \frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})}$.

Although the series in (1.1) does not converge for $\sigma \leq 1$, $\zeta(s)$ can still be approximated by a partial Dirichlet sum beyond the region $\sigma > 1$. For instance, using Euler–Maclaurin summation formula found in [Mur08, Theorem 2.1.9], we obtain the following result:

Theorem 1.1. $\zeta(s)$ extends to an analytic function in the domain $\sigma > 0$, $s \neq 1$ by

$$\zeta(s) = \sum_{n=1}^{N-1} n^{-s} + \frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} - s \int_N^{\infty} x^{-s-1} \left(x - [x] - \frac{1}{2} \right) dx \quad (1.4)$$

where $N \geq 2$.

Moreover, the idea of estimating the size of the zeta function by a partial Dirichlet sum plays an important role in obtaining explicit bounds for zeta on the $\sigma = 1/2$ -line, also known as the *critical line*. One of the well-known unsettled problem in number theory that concerns the growth of the zeta function on the critical line is the *Lindelöf Hypothesis* [Ivi03] which states that for every $\epsilon > 0$

$$\zeta(1/2 + it) = \mathcal{O}_\epsilon(t^\epsilon) \quad (1.5)$$

where the notation¹ $\mathcal{O}_\epsilon(\cdot)$ means that the implicit constants depend on ϵ . This result follows from another well-known conjecture known as the *Riemann Hypothesis* [Ivi03] that states that all non-trivial zeros of $\zeta(s)$ lie on the critical line. It is not known if the Lindelöf hypothesis implies the Riemann hypothesis.

A lot of results can be found in the literature that describe progress towards (1.5). One of the early results in this direction is given by the Euler–Maclurian formula. This formula implies that $\zeta(1/2 + it) = \mathcal{O}(t^{1/2})$ for $t \geq 1$. By using the Riemann–Siegel formula one can obtain an improved bound of $\zeta(1/2 + it) = \mathcal{O}(t^{1/4})$. Even more, the Weyl–Hardy–Littlewood method described in [Tit86] exploits cancellation in the main sum of the Riemann–Siegel formula, giving us a refined bound of $\zeta(1/2 + it) = \mathcal{O}(t^{1/6} \log^{3/2} t)$. The van der Corput method in [GK91] sharpens this bound further to $\zeta(1/2 + it) = \mathcal{O}(t^{1/6} \log t)$.

The exponent $1/6$ (called *Weyl-exponent*) which appears in the bounds above is substantially hard to improve. The method of exponent pairs described in [GK91] has been successful in improving this bound and in turn gives us an exponent of $27/164$ (called *sub-Weyl exponent*), that is $\zeta(1/2 + it) = \mathcal{O}(t^{27/164})$. In this thesis, we derive an explicit version of this sub-Weyl estimate; that is we find explicit constants C and t_0 such that

$$|\zeta(1/2 + it)| \leq C t^{27/164}, \quad t \geq t_0. \quad (1.6)$$

This explicit sub-Weyl estimate has direct applications to the Turing method and explicit zero-density estimates for $\zeta(s)$ which are described in Sections 1.3 and 1.4 respectively.

1.2 Known Explicit Bounds on $|\zeta(1/2 + it)|$

In this section, we state some progress that has been made towards obtaining explicit bounds for the Riemann zeta function on the critical line. To achieve such bounds, one typically uses the Riemann–Siegel formula, an explicit version of which is given in formula (3.1). This formula reduces the problem of bounding zeta to estimating an exponential sum of length approximately $\sqrt{t}$.

¹Here we use the standard Landau notation in number theory for $f(x) = \mathcal{O}(g(x))$, equivalently $f \ll g$ to mean that there exists some absolute constant $C > 0$ such that $|f(x)| \leq C|g(x)|$, for all $x > x_0$.

In 1968, Lehman derived an explicit convexity bound in [Leh70] using the Riemann–Siegel formula

$$|\zeta(1/2 + it)| \leq \frac{4}{(2\pi)^{1/4}} t^{1/4}, \quad t \geq 0.2. \quad (1.7)$$

Although it seems from [Tit36] that in 1935 Titchmarsh was aware of a version of the Riemann–Siegel formula, which was later simplified in [Tur53] by Turing in 1952, it was Lehman who first obtained such an explicit bound.

Moreover, similar bounds improving the exponent on t have also been obtained in [CG04; PT15a] which are of the form

$$|\zeta(1/2 + it)| \leq At^{\theta_1}(\log t)^{\theta_2}, \quad t \geq t_0 \quad (1.8)$$

where $A, \theta_1, \theta_2, t_0$ are absolute constants. However, explicit estimates of the above form are difficult to produce since attempts at improving θ_1 typically increase A 's value significantly which is undesirable for computation purposes [PT15a].

At this stage we record explicit bounds of the form (1.8) as they appear in literature. However, the forthcoming four explicit bounds are all based on an erroneous result in the work of Cheng-Graham [CG04, Lemma 2 and 3]. We discuss and fix this error in Section 2.2.1 and also revise the most recent bound of the form (1.8) in literature. But, for the sake of historical records, we state them here as they appear in their published version. With this in mind, we note that an explicit result of the form (1.8) is given by Cheng and Graham in [CG04] namely,

$$|\zeta(1/2 + it)| \leq 3t^{1/6} \log t, \quad t \geq e. \quad (1.9)$$

However, as noted in [PT15a] this upper bound is smaller than the one in (1.7) only when $t \geq 1.4 \times 10^{21}$. Thus, the constant in estimate (1.9) went through a series of improvements: first by Trudgian in [Tru14] who showed

$$|\zeta(1/2 + it)| \leq 2.38t^{1/6} \log t, \quad t \geq e.$$

Later, Platt and Trudgian in [PT15a] improved the result to

$$|\zeta(1/2 + it)| \leq 0.732t^{1/6} \log t, \quad t \geq 2. \quad (1.10)$$

Finally, the most recent improvement was made by Hiary in [Hia16] where he simplified the method by Platt and Trudgian to obtain

$$|\zeta(1/2 + it)| \leq 0.63t^{1/6} \log t, \quad t \geq 3.$$

In this thesis we improve the exponent of t in the above estimate and, more importantly, remove the $\log t$ factor.

1.3 Turing's Method

Improvements in the explicit bounds of the Riemann zeta function are used in verification of the Riemann Hypothesis upto a certain height through Turing's method. Here, we briefly describe the main ingredients of this method.

We begin by noting that such a verification includes the following two steps:

1. Find a list of zeros of $\zeta(s)$ in a given range (t_1, t_2) on the critical line.
2. Verify that these are all the zeros of $\zeta(s)$ in that range.

The first step is usually carried out by investigating the sign changes for the Hardy Z -function which is defined as

$$Z(t) := \zeta(1/2 + it)e^{i\theta(t)},$$

$$\theta(t) := \Im \left\{ \log \Gamma \left(\frac{1}{4} + \frac{it}{2} \right) \right\} - \frac{t}{2\pi} \log(\pi).$$

For the second step, we use the Turing's method, which gives an upper bound on the number of zeros in a given range. This upper bound is achieved by obtaining an explicit upper bound on the quantity

$$\left| \int_{t_1}^{t_2} S(t) dt \right|.$$

where $S(t)$ is the argument of $\zeta(1/2 + it)$ defined by a continuous variation.

To further understand the advantage of using Turing's method, let us first define the zero counting function for $\zeta(s)$

$$N(T) := \#\{\rho = \beta + i\gamma : 0 < \beta < 1, 0 < \gamma < T, \zeta(\rho) = 0\}.$$

It is known that $N(T)$ satisfies the relation

$$N(T) = \frac{1}{\pi} \theta(T) + 1 + S(T). \tag{1.11}$$

From this we can interpret $S(t)$ as the error in the approximation $N(t) \sim \frac{1}{\pi} \theta(t) + 1$.

Moreover, to see (1.11) we define

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Using the functional equation and the argument principle we obtain

$$N(T) = \frac{1}{\pi} \Delta_L \arg \xi(s) \tag{1.12}$$

where L is the line segment from 2 to $2 + iT$ and then to $1/2 + iT$

Now, if one wants to find the number of zeros with ordinates in a given range (t_1, t_2) in the critical strip $0 < \sigma < 1$, then one can follow a similar idea as above for $T = t_1, t_2$ and then compute $N(t_2) - N(t_1)$. However, this proves to be cumbersome as one has to evaluate the zeta function on a line other than the critical line as seen in (1.12).

Hence, Turing introduced a new method to find the number of zeros in a given range where one needs to evaluate the zeta function only on the critical line. The idea of Turing's method is based on the following result by Littlewood [Lit24]

$$\int_0^T S(t) dt = \mathcal{O}(\log T). \quad (1.13)$$

This implies that the average of $S(t)$ over $0 \leq t \leq T$ approaches zero as $T \rightarrow \infty$;

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T S(t) dt = 0.$$

Turing proved an explicit version of (1.13) which can be found in [Tur53] and was later improved by Lehman [Leh70]. Turing's result states that for $t_2 > t_1 > 168\pi$ we have

$$\left| \int_{t_1}^{t_2} S(t) dt \right| \leq 2.30 + 0.128 \log \left(\frac{t_2}{2\pi} \right). \quad (1.14)$$

Improvement in explicit bounds for $|\zeta(1/2+it)|$ gives improvements in the explicit bound (1.14). To clarify the role of this bound in Turing's method, let us say that one missed a zero, $1/2 + it_0$ while locating the zeros of zeta in the range (t_1, t_2) ; that is, the computed value of $N(t)$ in the subinterval $(t_0, t_2) \subset (t_1, t_2)$, that is, $N(t_2) - N(t_0)$ is incorrect. Then the computed value of $S(t)$ will be off by at least 1 since $S(t)$ jumps by at least 1 every time it passes a zero. If we use this erroneous $S(t)$ to compute $|\int_{t_1}^x S(t) dt|$, then it will differ from the actual value by at least $(x - t_0)$ for $x > t_0$. Hence, the difference between the actual and the erroneous integrals will grow as a function of x . Thus, in the case of a missing zero, condition (1.14) will fail when $x - t_0 \geq c \log t_2$ for some small positive constant c . So to verify that one has indeed found all the zeros in the range (t_1, t_2) , it suffices to verify the condition (1.14) for $x - t_2 \leq c \log t_2$.

To summarize [Boo06]: Let t_0 be a large number that is not the ordinate of a zero. Suppose we have located several zeros of zeta between ordinates $t_0 - h$ and $t_0 + h$ for $h > 0$. In other words we have found small intervals (a_n, b_n) such that $Z(a_n)$ and $Z(b_n)$ has opposite signs. Define $N_{\text{left}}(t_0, t)$ for $t \leq t_0$ to be the step function that starts at zero at $t = t_0$ and decreases by 1 at each a_n as one moves to the left of t_0 . Similarly, define $N_{\text{right}}(t, t_0)$ for $t \geq t_0$ to be the step function that starts at zero at $t = t_0$ and increases by 1 at each b_n as one moves to the right of t_0 . Thus, we have

$$N(t) \leq N(t_0) + N_{\text{left}}(t_0, t), \quad t \leq t_0, \quad (1.15)$$

$$N(t) \geq N(t_0) + N_{\text{right}}(t_0, t), \quad t \geq t_0. \quad (1.16)$$

From (1.15) and (1.16) we can bound $N(t_0)$ by integrating:

$$N(t_0)h + \int_{t_0}^{t_0+h} N_{\text{right}}(t_0, t) dt \leq \int_{t_0}^{t_0+h} \frac{1}{\pi} \theta(t) + 1 dt + \int_{t_0}^{t_0+h} S(t) dt, \quad (1.17)$$

$$N(t_0)h + \int_{t_0-h}^{t_0} N_{\text{left}}(t_0, t) dt \geq \int_{t_0-h}^t \frac{1}{\pi} \theta(t) + 1 dt + \int_{t_0-h}^{t_0} S(t) dt. \quad (1.18)$$

Thus, for h large enough if we have effective bounds for $\int_{t_1}^{t_2} S(t) dt$, then we can determine $N(t_2) - N(t_1)$ by carrying out the above process for $t_0 = t_1, t_2$.

Finally, Trudgian showed in [Tru14] that improvements in explicit bounds for the zeta function on the critical line improves the constants in (1.14). For completeness, we state Trudgian's result here which can be found in [Tru11, Theorem 2.12]

Theorem 1.2. *Let $t_2 > t_1 > t_0 > 168\pi$ and let the pair of numbers K, θ satisfy the relation that $|\zeta(1/2 + it)| \leq Kt^\theta$ for $t > t_0$. Also let $\mu = 3 \cdot 10^{-6}$. If the parameters c and d are chosen such that $1 < c \leq \frac{5}{4}$ and $\frac{1}{2} < d \leq 1$ then*

$$\left| \int_{t_1}^{t_2} S(t) dt \right| \leq a + b \log t_2$$

where a, b, c, d are defined in his paper and depend on K, θ .

1.4 Explicit Zero Density Estimates for Riemann Zeta Function

Although many interesting results in analytic number theory are proved assuming the Riemann Hypothesis, which still remains unsettled, it is important to investigate the behavior of possible nontrivial zeros of $\zeta(s)$ that lie off the critical line. Note that the Riemann Hypothesis has been verified until the height of 10^{10} on the critical line [Pla17], and until height 10^{13} in [Gou04], and no counter-examples were found in extensive calculations at large heights carried out in [BH18].

First, let us define a function that counts such zeros as:

$$N(\sigma, T) = \#\{\rho = \beta + i\gamma : \zeta(\rho) = 0, 0 < \gamma < T \text{ and } \sigma < \beta < 1\}.$$

Explicit non-trivial upper bounds on $N(\sigma, T)$ are called “zero-density estimates.” These bounds provide further evidence towards the Riemann Hypothesis. One of the earliest zero-density result was by Bohr and Landau in 1932 who showed that for $\sigma > \frac{1}{2}$

$$N(\sigma, T) = \mathcal{O}\left(\frac{T}{\sigma - \frac{1}{2}}\right). \quad (1.19)$$

for large T .

Ingham later improved this bound in 1937 [Ing37] and showed that for $\sigma \geq \frac{1}{2}$

$$N(\sigma, T) = \mathcal{O}(T^{(2+4c)(1-\sigma)} (\log T)^5) \quad (1.20)$$

assuming that $\zeta(1/2 + it) = \mathcal{O}(t^{c+\epsilon})$. The Lindelöf Hypothesis implies that $N(\sigma, T) = \mathcal{O}(T^{2(1-\sigma)+\epsilon})$, also known as the “Density Hypothesis.”

Although, one can find many improvements of the bound (1.19), there are very few explicit bounds for $N(\sigma, T)$ in the literature. One of the recent bounds that can be found in the literature is by Kadiri–Lumley–Ng [KLN18] which states:

Theorem 1.3. *Let $H_0 = 3.0610046 \cdot 10^{10}$, $\frac{10^9}{H_0} \leq k \leq 1$, $d > 0$, $H \in [1002, H_0)$, $\alpha > 0$, $\delta \geq 1$, $\eta_0 = 0.23622 \dots$, $1 + \eta_0 \leq \mu \leq 1 + \eta$, and $\eta \in (\eta_0, \frac{1}{2})$ be fixed. Let $\sigma > \frac{1}{2} + \frac{d}{\log H_0}$. Then*

$$N(\sigma, T) \leq \frac{\mathcal{C}_1}{2\pi d} (\log(kT))^{2\sigma} (\log T)^{5-4\sigma} T^{\frac{8}{3}(1-\sigma)} + \frac{\mathcal{C}_2}{2\pi d} (\log T)^2. \quad (1.21)$$

where the constants $\mathcal{C}_1$ and $\mathcal{C}_2$ are defined in [KLN18].

To prove this theorem Kadiri applies Ingham’s argument with a general weight function and uses the explicit Weyl bound for the zeta function proved by Hiary in [Hia16] to estimate a certain integral of the form

$$\int_H^T \log |h(\sigma' + it)| dt, \quad (1.22)$$

where $h(s) = \zeta(s)M(s)$, $M(s)$ is entire, $H \in (0, H_0)$ and $T > H_0$.

One can use the sub-Weyl bound to estimate (1.22) and investigate the possible improvements that can be made to Theorem 1.3.

1.5 The Method of Exponent Pairs

In this section we summarize the method of *exponent pairs* which is used to obtain upper bounds for exponential sums which are of the form $\sum_{n=N}^{2N} e(f(n)) \ll_{R,k,l} P^k N^l$. Here N is the length of the summation and $f(x)$ is a real-valued function with R derivatives and that

- $P \ll |f'(x)| \ll P$ and $P > 1/2$.
- for all $1 \leq r \leq R : N^{1-r} P \ll_r |f^{(r)}(x)| \ll_r N^{1-r} P$.
- $0 \leq k \leq 1/2 \leq l \leq 1$

In this case the pair (k, l) is called an *exponent pair*. For a detailed description of the method of exponent pairs we refer the reader to [GK91].

Note that $(0, 1)$ is an exponent pair and it is easy to show that the exponent pairs form a convex set. To find other exponent pairs we use van der Corput’s processes which reduces the problem of estimating a given exponential sum to estimating some other sum or family of sums. The two processes can be summarized as follows:

- Process A: Through the application of van der Corput's version of Weyl differencing lemma we transform an exponential sum into a family of exponential sums which are easier to estimate.

The following theorem which can be found in [Mon94] is known as Process A in literature:

Theorem 1.4 (Process A). *Let $f(x)$ be a real-valued function on $[a, b]$. Let $H \leq b - a$ be an integer, such that*

$$\sum_{h=1}^H \left| \sum_{n=a}^{b-h} e^{2\pi i(f(n+h)-f(n))} \right| \leq b - a \quad (1.23)$$

Then

$$\sum_{n=a}^b e^{2\pi i f(n)} \ll \frac{b-a}{H^{1/2}} \quad (1.24)$$

Process A gives us the following result [Ivi03, Lemma 2.8] to find additional exponent pairs given a pair (k, l) :

Theorem 1.5. *If (k, l) is an exponent pair then*

$$A(k, l) = \left(\frac{k}{2k+2}, \frac{1}{2} + \frac{l}{2k+2} \right)$$

is also an exponent pair.

The above result is obtained by applying the A -process, and then applying the exponent pair (k, l) to the new exponential sum.

- Process B: Through the application of Poisson Summation, we transform an exponential sum into another exponential sum plus error terms, and this new exponential sum is typically of shorter length.

The following theorem which can be found in [Tit86] is known as Process B in literature:

Theorem 1.6 (Process B). *Let $f(x)$ be a real function with derivatives up to the third order. Let $f'(x)$ be steadily decreasing in $[a, b]$ and $f'(b) = \alpha, f'(a) = \beta$. Let x_ν be defined by*

$$f'(x_\nu) - \nu = 0 \quad (\alpha < \nu \leq \beta) \quad (1.25)$$

Let

$$\lambda_2 \leq |f''(x)| \leq A_2 \lambda_2, \quad |f'''(x)| \leq A_3 \lambda_3 \quad (1.26)$$

Then

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = e^{-\pi i/4} \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i(f(x_\nu) - \nu x_\nu)}}{\sqrt{f''(x_\nu)}} + \mathcal{O}(\lambda_2^{-1/2}) + \mathcal{O}(\log(2 + (b-a)\lambda_2)) + \mathcal{O}((b-a)\lambda_2^{1/2}\lambda_3^{1/5}) \quad (1.27)$$

Similar to Process A , Process B also gives us a method to construct new exponent pairs given (k, l) an exponent pair. This is seen in the following result [Ivi03, Lemma 2.9]:

Theorem 1.7. *If (k, l) is an exponent pair then*

$$B(k, l) = \left(l - \frac{1}{2}, k + \frac{1}{2} \right)$$

is also an exponent pair.

The above result is obtained by applying the B -process, and then applying the exponent pair (k, l) to the new exponential sum.

Some examples of exponent pairs are: $B(0, 1) = (1/2, 1/2)$, $AB(0, 1) = (1/6, 2/3)$, $A^2B(0, 1) = (1/14, 11/14)$, $BABA^2B(0, 1) = (4/18, 11/18)$.

We end this section by stating an application of the theory of exponent pairs in obtaining asymptotic bounds for $\zeta(1/2 + it)$ which can be seen from the following theorem found in [GK91]:

Theorem 1.8. *Suppose that $t \geq 3$, and that (k, l) is an exponent pair such that $k + 2l \geq 3/2$. Let $\theta(k, l) = (2k + 2l - 1)/4$. Then*

$$\zeta(1/2 + it) \ll t^{\theta(k, l)} \log t \tag{1.28}$$

In particular, the exponent $27/164$ follows by taking $(k, l) = ABA^3B(0, 1) = (11/82, 57/82)$.

A possible future goal beyond this thesis is to obtain a useful explicit version of the process $A^{k-2}B$ for arbitrary integer $k \geq 2$. A sought long-term outcome is to enable helpful automation of proofs of explicit bounds on zeta. And our work in this thesis serves as the first step towards automating proofs for bounding the zeta function. This could serve as a model for automating proofs of explicit bounds on the critical line for other L -functions.

CHAPTER 2

ESTIMATING EXPONENTIAL SUMS AND INTEGRALS

2.1 Elementary Results

In this section we record explicit estimates for some well-known functions and sums in literature as we will make use of them in our calculations throughout the rest of the thesis.

Lemma 2.1. *For $n \geq 2$*

$$\psi(n) = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k} \quad (2.1)$$

where $\gamma = 0.57721 \dots$ is the Euler–Mascheroni constant.

Here $\psi(x)$ is the digamma function which is defined as the logarithmic derivative of the gamma function, that is, $\psi(x) = \frac{d}{dx} \ln(\Gamma(x)) = \frac{\Gamma'(x)}{\Gamma(x)}$.

The above lemma can be found in [AS92, Page 258]

Lemma 2.2. *For $x > 0$*

$$\psi(x) < \log(x) - \frac{1}{2x} - \frac{1}{12(x + 1/14)^2} \quad (2.2)$$

This lemma can be found in [Gor94, Theorem 5]

Lemma 2.3. *For $n \geq 2$*

$$\sum_{k=1}^n \frac{1}{k} < \log n + \gamma + \frac{1}{2n + 1/3} \quad (2.3)$$

The above lemma can be found in [Tot+92, E 3432]

Lemma 2.4 (Partial Summation). *Let $b_1 \geq b_2 \geq \dots \geq b_n \geq 0$, and $s_m = a_1 + a_2 + \dots + a_m$ where the a 's are any real or complex numbers. Then if $|s_m| \leq M$ ($m = 1, 2, \dots$),*

$$|a_1 b_1 + a_2 b_2 + \dots + a_n b_n| \leq b_1 M. \quad (2.4)$$

Lemma 2.5 (Square-root inequality). *Let $x_1, x_2, \dots \geq 0$. Then*

$$\sqrt{x_1 + x_2 + \dots} \leq \sqrt{x_1} + \sqrt{x_2} + \dots \quad (2.5)$$

Let us first give an explicit version of “Weyl differencing” lemma [Tit86, Lemma 5.10] which can be found in [PT15b, Lemma 2]

Lemma 2.6. *Let $f(n)$ be a real-valued function and M be a positive integer. Then*

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 \leq \left(1 + \frac{M-1}{L}\right) \frac{L^2}{M} + 2 \left(1 + \frac{M-1}{L}\right) \frac{L}{M} \sum_{m=1}^{M-1} \left(1 - \frac{m}{M}\right) \max_{K \leq L} |S'_m(K)| \quad (2.6)$$

where

$$S'_m(K) = \sum_{n=N+1}^{N+K} e^{2\pi i [f(n+m) - f(n)]} \quad (2.7)$$

Using the square-root inequality in (2.6) together with $1 - m/M \leq 1$ we get

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right| \leq w_1 \frac{L}{\sqrt{M}} + w_2 \left(\frac{L}{M} \sum_{m=1}^{M-1} \max_{K \leq L} \left| \sum_{n=N+1}^{N+K} e^{2\pi i (f(n+m) - f(n))} \right| \right)^{1/2} \quad (2.8)$$

where

$$w_1 = \sqrt{1 + \frac{M-1}{L}}; \quad w_2 = \sqrt{2 + \frac{2(M-1)}{L}} \quad (2.9)$$

In our calculations, we will eventually bound w_1, w_2 based on the choice of M .

Remark 2.7. Note that the Weyl differencing lemma in itself does not quite detect any cancellations in the original sum, but it lowers the “complexity” of the original sum. This is easily seen if one assumes that the phase function $f(x)$ is a polynomial of degree d , then applying the Weyl differencing lemma gives us an exponential sum with a phase function that is a polynomial of degree $d - 1$. Repeating this process, one eventually reduces to a geometric sum which is easily estimated. Moreover bounding an average of exponential sum with the differenced phase function $f(x + m) - f(x)$ is typically easier than bounding one exponential sum with phase function $f(x)$. This is because the differenced phase grows more slowly than the original phase and thus is easier to control.

2.2 Explicit $k = 2, 3, 4, 5$ Derivative Tests

In this section we derive explicit estimates for van der Corput’s $k = 2, 3, 4, 5$ derivative tests. One can refer to [Tit86, Theorem 5.9, 5.11, 5.13] for an asymptotic version of these tests. However, before we do this, we correct an important error in literature.

2.2.1 Erroneous Lemma in Literature and Explicit $k = 2, 3$ Derivative Tests

An important tool used to obtain explicit bounds on exponential sums, for instance the one corresponding to (2.49) with $a = \frac{3}{4}$ is an explicit version of van der Corput's second derivative test, that is explicit $k = 2$ derivative test. This result can be found in the work of Cheng-Graham in [CG04, Lemma 3]. However a computational flaw was discovered by Kevin Ford [For20] and Juan Arias de Reyna in [Rey20] with this Cheng-Graham lemma. This affects many explicit estimates in the literature such as [For02], [Tru14], [Hia16] to name a few. However, we note that even if (2.49) with $a = \frac{3}{4}$ were true, our result given in Theorem 2.16 gives an improvement on it for $t \geq 2.17 \times 10^3$.

For the remainder of the section we record the errors in the work of Cheng-Graham [CG04, Lemma 2 and 3]. We also revise another result in literature [Hia16, Lemma 1.2] affected due to these errors. We begin by providing corrected version of flawed Cheng-Graham lemma in [CG04, Lemma 2]. To do so, we first define $\|x\| := \min_{n \in \mathbb{Z}} |x - n|$ and we observe that $0 \leq \|x\| \leq 1/2$.

Lemma 2.8. *Suppose f is a continuously differentiable real-valued function with a monotonic derivative and $\|f'\| \geq U^{-1}$ for some positive real number U on the interval $(a, b]$. Then*

$$|S| = \left| \sum_{n \in (a, b]} e^{2\pi i f(n)} \right| \leq \frac{2}{\pi} U. \quad (2.10)$$

Remark 2.9. This result is often attributed to the works of Kuzmin-Landau [Kuz27; Lan28] in literature.

Proof. First we notice that $0 < U^{-1} \leq 1/2$. Next, we make note of the error in the proof of Lemma 2 in [CG04] and give a possible fix. This fix is based on the ideas found in [Mor58]. Most of the proof of Lemma 2 in [CG04] is valid except we note a typo on Page 1266 where the equality for $G(n) - G(n-1)$ should read

$$G(n) - G(n-1) = \frac{1}{2i} (\cot(\pi g(n-1)) - \cot(\pi g(n))) \quad (2.11)$$

However, there is a fatal flaw that originates in the first inequality on Page 1267 of [CG04] because of missing absolute values after the first two cotangent terms. That inequality should instead read:

$$\left| \sum_{n=L}^M e^{2\pi i f(n)} \right| \leq \frac{1}{2} (\cot(\pi g(L)) - \cot(\pi g(M-1))) + \left| \frac{1}{2} + \frac{i}{2} \cot(\pi g(L)) \right| + \left| \frac{1}{2} - \frac{i}{2} \cot(\pi g(M-1)) \right| \quad (2.12)$$

With this fix in mind, we now provide a possible way to finish Cheng-Graham proof correctly giving us (2.10). To do so, we note:

$$\left| \sum_{n=L}^M e^{2\pi i f(n)} \right| \leq \frac{1}{2} \left(\cot(\pi g(L)) + \frac{1}{\sin(\pi g(L))} \right) + \frac{1}{2} \left(\frac{1}{\sin(\pi g(M-1))} - \cot(\pi g(M-1)) \right) \quad (2.13)$$

$$= \frac{\cos(\pi g(L)) + 1}{2 \sin(\pi g(L))} + \frac{1 - \cos(\pi g(M-1))}{2 \sin(\pi g(M-1))} \quad (2.14)$$

$$= \frac{1}{2} \cot\left(\frac{\pi g(L)}{2}\right) + \frac{1}{2} \tan\left(\frac{\pi g(M-1)}{2}\right) \quad (2.15)$$

$$\leq \frac{1}{2} \cot\left(\frac{\pi U^{-1}}{2}\right) + \frac{1}{2} \tan\left(\frac{\pi(1-U^{-1})}{2}\right) \quad (2.16)$$

$$\leq \cot\left(\frac{\pi U^{-1}}{2}\right) \quad (2.17)$$

$$\leq \frac{2}{\pi} U \quad (2.18)$$

For proof readability we make several remarks regarding the above inequalities here. First note that going from (2.12) to (3.73) we use

$$0 < U^{-1} \leq g(L), \quad g(M-1) \leq 1 - U^{-1} < 1, \quad (2.19)$$

$$1 + i \cot(x) = \frac{ie^{-ix}}{\sin x}, \quad \sin(\pi x) > 0 \text{ for } x \in (0, 1). \quad (2.20)$$

Next to go from (3.73) to (2.14) we write $\cot(x)$ in terms of $\sin(x)$ and $\cos(x)$ and then gather like terms. To pass from (2.14) to (3.76) we use the following relation valid for $x \neq k\pi$ where $k \in \mathbb{Z}$:

$$\cot\left(\frac{x}{2}\right) = \frac{1 + \cos x}{\sin x}, \quad \tan\left(\frac{x}{2}\right) = \frac{1 - \cos x}{\sin x}.$$

To go from (3.76) to (2.16) we use bound on $g(L), g(M-1)$ in (2.19) along with the fact that on $(0, \pi/2)$, $\cot x$ is non-negative and decreasing function and $\tan x$ is non-negative and increasing function. For inequality (2.16) to (2.17) we use the relation below valid for $\theta \neq k\pi$ and $k \in \mathbb{Z}$:

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta).$$

And lastly inequality (2.18) follows from (2.17) from the fact that $\cot x \leq 1/x$ for $0 < x < \pi/2$. $\square$

Furthermore, for an alternate proof of Lemma 2.8 one can refer to [Ten15, Lemma 6.6] albeit we note a couple of typos in that proof¹, the estimate on $|c_n|$ in the beginning of the

¹These errors were also pointed out by Kevin Ford in [For20].

proof should include an equality because of equality in (2.20) and read:

$$|c_n| = |1 - c_n| = \frac{1}{2|\sin \pi y_n|} \leq \frac{1}{2 \sin \pi \vartheta}$$

Next, the inequality at the end of that proof should involve a negative sign between cotangent terms and read:

$$\left| \sum_{1 \leq n \leq N} e(x_n) \right| \leq \frac{1}{2} \cot(\pi y_1) - \frac{1}{2} \cot(\pi y_{N-1}) + |c_1| + |1 - c_{N-1}| \quad (2.21)$$

Moreover, Landau showed in [Lan28] that the constant $2/\pi$ in (2.10) is the best possible. For historical context behind this result we refer to Arias de Reyna's work in [Rey20] where additionally he gives an alternate proof of Lemma 2.8 above. Although we make note of a typo in [Rey20, Lemma 2(a)], where the inequality on b_k should read: $\pi\theta \leq b_k \leq \pi(1 - \theta)$.

Next, having corrected Lemma 2 in [CG04] we now correct Lemma 3 in [CG04] which is a crucial tool used in literature to obtain explicit estimates.

Lemma 2.10. *Assume that f is a real-valued function with two consecutive derivatives on $[N + 1, N + L]$. If there exists two real numbers $V < W$ with $W > 1$ such that*

$$\frac{1}{W} \leq |f''(x)| \leq \frac{1}{V} \quad (2.22)$$

for x on $[N + 1, N + L]$, then

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right| \leq 2 \left(\frac{L}{V} + 2 \right) \left(2\sqrt{\frac{W}{\pi}} + 1 \right) \quad (2.23)$$

Remark 2.11. Because of this correction, the leading term in the incorrect Cheng-Graham Lemma 3 is off by a factor of $\sqrt{2}$.

Proof. The proof in [CG04, Lemma 3] is modified as follows: The $k - 1$ sub-sums corresponding to the interval $[y_j, x_{j+1}]$ are bounded by $\frac{2}{\pi\Delta}$ instead of $1/(\pi\Delta) + 1$. And we take $\Delta = 1/\sqrt{\pi W}$ instead of $1/\sqrt{2\pi W}$. We also make note of a couple of typos in the proof: First, the estimate on k should be $k \leq L/V + 3$. Next, when estimating the sum trivially, the mean value theorem in this case should be applied to $(f')^{-1}$ instead of f^{-1} . $\square$

Moreover using Platt-Trudgian's observation in [PT15a, Lemma 1] we obtain a slight improvement to (2.23) in the form of

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right| \leq 2 \left(\frac{L-1}{V} + 2 \right) \left(2\sqrt{\frac{W}{\pi}} + \frac{1}{2} \right) + 1 \quad (2.24)$$

The corrected version (2.23) in turn gives us new constants in the explicit third derivative test found in [Hia16, Lemma 1.2] and again for completeness we state the revised version here.

Lemma 2.12. *Let $f(x)$ be a real-valued function with four continuous derivatives on $[N+1, N+L]$. Suppose there are constants $W_3 > 1$ and $\lambda_3 \geq 1$ such that $\frac{1}{W_3} \leq |f^{(3)}(x)| \leq \frac{\lambda_3}{W_3}$ for $N+1 \leq x \leq N+L$. If $\eta_3 > 0$, then*

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 \leq (LW_3^{-1/3} + \eta_3)(\alpha_3 L + \beta_3 W_3^{2/3}),$$

where

$$\alpha_3 = \frac{1}{\eta_3} + \frac{32\lambda}{15\sqrt{\pi}} \sqrt{\eta_3 + W_3^{-1/3}} + \frac{2\lambda_3\eta_3}{W_3^{1/3}} + \frac{2\lambda_3}{W_3^{2/3}},$$

$$\beta_3 = \frac{64}{3\sqrt{\pi}\sqrt{\eta_3}} + \frac{4}{W_3^{1/3}}.$$

Proof. We use (2.23) above in the proof of Lemma 1.2 in [Hia16] and replace the estimate for $|S'_m(L)|$ in equation (43) of that proof with

$$|S'_m(L)| \leq \frac{4\lambda_3 L \sqrt{m/W_3}}{\sqrt{\pi}} + \frac{2\lambda_3 L m}{W_3} + \frac{8\sqrt{W_3/m}}{\sqrt{\pi}} + 4. \quad (2.25)$$

□

This revised version of the explicit third derivative test in Lemma 2.12 in turn changes the explicit van der Corput bound derived by Hiary in [Hia16, Theorem 1.1]. In that paper, Hiary obtained $|\zeta(1/2 + it)| \leq 0.63t^{1/6} \log t$ at the time. This bound was an improvement to Platt-Trudgian's result in [PT15a, Theorem 1] that stated $|\zeta(1/2 + it)| \leq 0.732t^{1/6} \log t$. With the revision provided in Lemma 2.12, the estimate obtained by Hiary in [Hia16] now becomes $|\zeta(1/2 + it)| \leq 0.77t^{1/6} \log t$ and the Platt-Trudgian, 0.732, bound is no longer valid since it also uses the erroneous lemma. Here we record the complete list of changes required in [Hia16] to achieve the $0.77t^{1/6} \log t$ bound:

- Eq (4): replace 0.63 with 0.77 in Theorem 1.1.
- Page 526: $|\zeta(1/2 + iT_0)| \leq 1.033 \times 10^7$, replace 521 with 636, replace 507 with 625.
- Eq (9): $c_0 = 0.77$, $t_0 = 2 \times 10^{10}$.
- Lemma 2.4: $a_1 = 0.754348$, $a_2 = 0.197545$, $a_3 = -2.912371$.
- Page 530: $\eta_0 = (15\sqrt{\pi}/32)^{2/3}$.
- Page 531: $r_0 = 19$.

Remark 2.13. We make an important observation regarding the constants α_3, β_3 in Lemma 2.12 which is since α_3, β_3 are decreasing function in W_3 and $W_3 > 1$ we get that the estimates:

$$\alpha_3 \leq \widetilde{\alpha}_3 = \frac{1}{\eta_3} + \frac{32\lambda}{15\sqrt{\pi}}\sqrt{\eta_3 + 1} + 2\lambda_3\eta_3 + 2\lambda_3, \quad \beta_3 \leq \widetilde{\beta}_3 = \frac{64}{3\sqrt{\pi}\sqrt{\eta_3}} + 4.$$

Note that we will obtain our explicit result for k derivative test by using the $k - 1$ test. Hence, for clarity we label the constants and other quantities in these results in such a way that the subscripts indicate the derivative test they arise from. With Lemma 2.12 at hand we are now ready to find explicit versions of fourth and fifth derivative tests.

2.2.2 Explicit $k = 4, 5$ Derivative Tests

Referring back to the theory of exponent pairs, we note that in Lemma 2.14 we make the process A^2B explicit.

Lemma 2.14. *Let $f(x)$ be a real-valued function with four continuous derivatives on $[N + 1, N + L]$. Suppose there are constants $W_4 > 1$ and $\lambda_4 \geq 1$ such that $\frac{1}{W_4} \leq |f^{(4)}(x)| \leq \frac{\lambda_4}{W_4}$ for $N + 1 \leq x \leq N + L$. If $\eta_4 > 0$, then*

$$|S(L)|^2 := \left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 \leq (LW_4^{-1/7} + \eta_4)(\alpha_4 L + \gamma_4 \sqrt{L}W_4^{2/7} + \beta_4 W_4^{3/7}),$$

where

$$\alpha_4 = \frac{1}{\eta_4} + \frac{72}{91}\sqrt{\widetilde{\alpha}_3}(\eta_4 + W_4^{-1/7})^{1/6}, \quad \gamma_4 = \frac{72}{55}\sqrt{\widetilde{\beta}_3}\eta_4^{-1/6} + \sqrt{\eta_3\widetilde{\alpha}_3}W_4^{-1/7}, \quad \beta_4 = \frac{18}{20}\sqrt{\widetilde{\beta}_3\eta_3}\eta_4^{-1/3}.$$

Proof. We will use the Weyl-van der Corput Lemma in Cheng and Graham [CG04, Lemma 5], but use the form given at the bottom of page 1273 as well as a further refinement by Platt and Trudgian [PT15a, Lemma 2]. In all, if M is a positive integer, then

$$|S(L)|^2 \leq (L + M - 1) \left(\frac{L}{M} + \frac{2}{M} \sum_{m=1}^M \left(1 - \frac{m}{M} \right) |S'_m(L)| \right), \quad (2.26)$$

where

$$S'_m(L) = \sum_{r=N+1}^{N+L-m} e^{2\pi i (f(r+m) - f(r))}. \quad (2.27)$$

Here, we can assume that $m < L$ and $L > 1$. Otherwise, the sum $S'_m(L)$ is empty and does not contribute to the upper bound.

Now, let $g(x) := f(x+m) - f(x)$ where $N+1 \leq x \leq N+L-m$. Then $g'''(x) = f'''(x+m) - f'''(x)$. Hence, using the mean value theorem we obtain $g'''(x) = mf^{(4)}(\xi)$ for some $\xi \in (x, x+m) \subset [N+1, N+L]$. Next, using the given bound on $f^{(4)}(x)$, we deduce that

$$\frac{m}{W_4} \leq |g'''(x)| \leq \frac{m\lambda_4}{W_4}, \quad (N+1 \leq x \leq N+L-m). \quad (2.28)$$

Applying Lemma 2.12 to bound $|S'_m(L)|^2$, multiplying the terms out and then using the inequalities:

$$\begin{aligned} \sqrt{x_1 + x_2 + \dots} &\leq \sqrt{x_1} + \sqrt{x_2} + \dots, \quad \text{where } x_1, x_2, \dots \geq 0 \\ \alpha_3 &\leq \tilde{\alpha}_3 \text{ and } \beta_3 \leq \tilde{\beta}_3, \quad L-m < L \text{ for } m \geq 1. \end{aligned} \quad (2.29)$$

$$|S'_m(L)|^2 < \sqrt{\tilde{\alpha}_3} L W_4^{-1/6} m^{1/6} + \sqrt{\tilde{\beta}_3} L W_4^{1/6} m^{-1/6} + \sqrt{\eta_3 \tilde{\alpha}_3 L} + \sqrt{\tilde{\beta}_3 \eta_3 W_4^{1/3}} m^{-1/3}. \quad (2.30)$$

Next, let us bound $\sum_{m=1}^M \left(1 - \frac{m}{M}\right) |S'_m(L)|$ using (2.30) and the below estimate valid for $-1 < q < 1$:

$$\sum_{m=1}^M \left(1 - \frac{m}{M}\right) m^q \leq \frac{M^{q+1}}{(q+1)(q+2)}. \quad (2.31)$$

To prove the above estimate for $q \geq 0$ we use [CG04, Lemma 7] and for $q < 0$ we replace the sum with an integral. After obtaining such bounds we substitute them back in (2.26) and get

$$\begin{aligned} |S(L)|^2 &\leq (L+M-1) \left(\frac{L}{M} + \frac{72}{91} \sqrt{\tilde{\alpha}_3} L W_4^{-1/6} M^{1/6} + \frac{72}{55} \sqrt{\tilde{\beta}_3} L W_4^{1/6} M^{-1/6} \right. \\ &\quad \left. + \sqrt{\eta_3 \tilde{\alpha}_3 L} + \frac{18}{10} \sqrt{\tilde{\beta}_3 \eta_3 W_4^{1/3}} M^{-1/3} \right). \end{aligned} \quad (2.32)$$

Now we would like to make the first two terms in (2.32) of the same magnitude to minimize the rhs. This can be achieved if we choose $M = \lceil \eta_4 W_4^{1/7} \rceil$ for some free parameter $\eta_4 > 0$ that can be optimized. With this choice of M we obtain the inequality $\eta_4 W_4^{1/7} \leq M \leq \eta_4 W_4^{1/7} + 1$. Using this inequality and then factoring $W_4^{1/7}$ term from the first parenthesis and multiplying it in the second we deduce that

$$|S(L)|^2 \leq (L W_4^{-1/7} + \eta_4) W_4^{1/7} \left(\left(\eta_4^{-1} W_4^{-1/7} + \frac{72}{91} \sqrt{\tilde{\alpha}_3} \left(\frac{\eta_4 W_4^{1/7} + 1}{W_4} \right)^{1/6} \right) L \right. \quad (2.33)$$

$$\left. + \left(\frac{72}{55} \sqrt{\tilde{\beta}_3} \eta_4^{-1/6} W_4^{1/7} + \sqrt{\eta_3 \tilde{\alpha}_3} \right) \sqrt{L} + \frac{18}{10} \sqrt{\tilde{\beta}_3 \eta_3} \eta_4^{-1/3} W_4^{6/21} \right) \quad (2.34)$$

$$\leq (L W_4^{-1/7} + \eta_4) (\alpha_4 L + \gamma_4 \sqrt{L} W_4^{2/7} + \beta_4 W_4^{3/7}) \quad (2.35)$$

where α_4, γ_4 , and β_4 are defined as in the statement of the lemma. $\square$

Next, we obtain the explicit $k = 5$ derivative test and just as before we note that in Lemma 2.15 we make the process $A^3 B$ explicit.

Lemma 2.15. *Let $f(x)$ be a real-valued function with five continuous derivatives on $[N+1, N+L]$. Suppose there are constants $W_5 > 1$ and $\lambda_5 \geq 1$ such that $\frac{1}{W_5} \leq |f^{(5)}(x)| \leq \frac{\lambda_5}{W_5}$ for $N+1 \leq x \leq N+L$. If $\eta_5 > 0$, then*

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 \leq (LW_5^{-1/15} + \eta_5)(\alpha_5 L + \tau_5 L^{3/4} W_5^{2/15} + \gamma_5 \sqrt{L} W_5^{3/15} + \omega_5 L^{1/4} W_5^{1/5} + \beta_5 W_5^{4/15});$$

where

$$\begin{aligned} \alpha_5 &= \frac{1}{\eta_5} + \frac{392}{435} \sqrt{\tilde{\alpha}_4} (\eta_5 + W_5^{-1/15})^{1/14}, \quad \tau_5 = \frac{392}{351} \sqrt{\tilde{\gamma}_4} \eta_5^{-1/14}, \quad \gamma_5 = \frac{98}{78} \sqrt{\beta_4} \eta_5^{-1/7} + \sqrt{\eta_4 \tilde{\alpha}_4} W_5^{-2/15}, \\ \omega_5 &= \frac{98}{78} \sqrt{\eta_4 \tilde{\gamma}_4} \eta_5^{-1/7}, \quad \beta_5 = \frac{392}{275} \sqrt{\eta_4 \beta_4} \eta_5^{-3/14} \end{aligned} \quad (2.36)$$

Proof. The proof of this lemma is very similar to Lemma 2.14 where we first bound $|S'_m(L)|$ by letting $g(x) := f(x+m) - f(x)$ where $N+1 \leq x \leq N+L-m$. Then $g^{(4)}(x) = f^{(4)}(x+m) - f^{(4)}(x)$ from which we can deduce using the mean value theorem and given bound on $f^{(5)}(x)$ that

$$\frac{m}{W_5} \leq |g^{(4)}(x)| \leq \frac{m\lambda_5}{W_5}, \quad (N+1 \leq x \leq N+L-m). \quad (2.37)$$

Applying the result for fourth derivative stated in Lemma 2.14 along with the estimates

$$\alpha_4 \leq \tilde{\alpha}_4 = \frac{1}{\eta_4} + \frac{72}{91} \sqrt{\tilde{\alpha}_3} (\eta_4 + 1)^{1/6}, \quad \gamma_4 \leq \tilde{\gamma}_4 = \frac{72}{55} \sqrt{\tilde{\beta}_3} \eta_4^{-1/6} + \sqrt{\eta_3 \tilde{\alpha}_3} \quad (2.38)$$

and inequalities similar to (2.29) to bound $|S'_m|^2$, we get

$$\begin{aligned} |S'_m(L)| &\leq \sqrt{\tilde{\alpha}_4} L W_5^{-1/14} m^{1/14} + \sqrt{\tilde{\gamma}_4} L^{3/4} W_5^{1/14} m^{-1/14} + \sqrt{\beta_4} L W_5^{1/7} m^{-1/7} \\ &\quad + \sqrt{\eta_4 \tilde{\alpha}_4} L + \sqrt{\eta_4 \tilde{\gamma}_4} L^{1/4} W_5^{1/7} m^{-1/7} + \sqrt{\eta_4 \beta_4} W_5^{3/14} m^{-3/14}. \end{aligned} \quad (2.39)$$

Now we bound $\sum_{m=1}^M \left(1 - \frac{m}{M}\right) \left| \sum_{r=N+1}^{N+L-m} e^{2\pi i g(r)} \right|$ using (2.39) and (2.31) and then substitute them in an expression like (2.26) to get

$$\begin{aligned} \left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 &\leq (L+M-1) \left(\frac{L}{M} + \frac{392}{435} \sqrt{\tilde{\alpha}_4} L W_5^{-1/14} M^{1/14} + \frac{392}{351} \sqrt{\tilde{\gamma}_4} L^{3/4} W_5^{1/14} M^{-1/14} \right. \\ &\quad + \frac{98}{78} \sqrt{\beta_4} L W_5^{1/7} M^{-1/7} + \sqrt{\eta_4 \tilde{\alpha}_4} L + \frac{98}{78} \sqrt{\eta_4 \tilde{\gamma}_4} L^{1/4} W_5^{1/7} M^{-1/7} \\ &\quad \left. + \frac{392}{275} \sqrt{\eta_4 \beta_4} W_5^{3/14} M^{-3/14} \right) \end{aligned} \quad (2.40)$$

Now we would like to make the first two terms in (2.40) of the same magnitude to minimize the rhs in (2.40). This can be achieved if we choose $M = \lceil \eta_5 W_5^{1/15} \rceil$ for some free parameter

$\eta_5 > 0$ that can be optimized. Next, using the inequality $\eta_5 W^{1/15} \leq M \leq \eta_5 W_5^{1/15} + 1$ obtained because of the choice of M and using similar algebraic manipulations as in proof of Lemma 2.14 we deduce that

$$\begin{aligned} \left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 &\leq (LW_5^{-1/15} + \eta_5) \left(\left(\eta_5^{-1} + \frac{392}{435} \sqrt{\tilde{\alpha}_4} (\eta_5 + W_5^{-1/15})^{1/14} \right) L \right. \\ &\quad + \frac{392}{351} \sqrt{\tilde{\gamma}_4} \eta_5^{-1/14} L^{3/4} W_5^{2/15} + \left(\frac{98}{78} \sqrt{\beta_4} \eta_5^{-1/7} + \sqrt{\eta_4 \tilde{\alpha}_4} W_5^{-2/15} \right) \sqrt{L} W_5^{1/5} \\ &\quad \left. + \frac{98}{78} \sqrt{\eta_4 \tilde{\gamma}_4} \eta_5^{-1/7} L^{1/4} W_5^{1/5} + \frac{392}{275} \sqrt{\eta_4 \beta_4} \eta_5^{-3/14} W_5^{4/15} \right). \end{aligned} \quad (2.41)$$

This finally gives us:

$$\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 \leq (LW_5^{-1/15} + \eta_5) (\alpha_5 L + \tau_5 L^{3/4} W_5^{2/15} + \gamma_5 \sqrt{L} W_5^{3/15} + \omega_5 L^{1/4} W_5^{1/5} + \beta_5 W_5^{4/15}) \quad (2.42)$$

where $\alpha_5, \tau_5, \gamma_5, \omega_5$ and β_5 are defined as in the statement of the lemma. $\square$

To summarize, Lemmas 2.12, 2.14 and 2.15 are explicit versions of processes AB, A^2B, A^3B respectively in the theory of exponent pairs respectively. Moreover these lemmas give a saving of $\approx W^n$ when compared to the trivial bound, which is $\left| \sum_{n=N+1}^{N+L} e^{2\pi i f(n)} \right|^2 \leq L^2$. Note that here $n = -\frac{1}{3}, -\frac{1}{7}$ and $-\frac{1}{15}$ for Lemmas 2.12, 2.14 and 2.15. In application, it is often unclear on how to choose the correct derivative test to obtain an estimate. For instance as it will be seen later, the choice of W , dictates the length of the interval over which the k -th derivative test is applied while bounding the zeta function. This along with the method used to estimate the initial sum determines the derivative test to be applied.

2.3 An Application of $k = 5$ Derivative Test

2.3.1 Motivation and Background

The study of the growth rate of $\zeta(1+it)$ has been of great interest because of its application in estimating $S(T)$ as shown in [Tru14] and computing zero free regions for the Riemann zeta function.

In 1900 Mellin [Mel02] was the first to obtain a result in this direction and showed that for real t bounded away from 0 that

$$\zeta(1+it) = \mathcal{O}(\log |t|) \quad (2.43)$$

In 1921, Weyl improved (2.43) using Weyl's sums in [Wey21] to

$$\zeta(1+it) = \mathcal{O}\left(\frac{\log t}{\log \log t}\right) \quad (2.44)$$

and was in turn improved upon by Vinogradov which can be found in [Tit86, Theorem 6.14] to

$$\zeta(1+it) = \mathcal{O}(\log^{3/4} t \log^{3/4} \log t) \quad (2.45)$$

Several authors namely Flett [Fle50], Walfisz [Wal58], and Korobv [Kor58b] between 1950-58 obtained bounds of the form $\mathcal{O}(\log^{3/4} t \log \log^{1/2+\epsilon} t)$, $\mathcal{O}(\log^{3/4} t \log \log^{1/2} t)$, $\mathcal{O}(\log^{5/7+\epsilon} t)$ respectively. Moreover, authors such as Vinogradov [Vin58], Korobov [Kor58c] [Kor58a] in 1958 and Richert [Ric67] in 1967 gave the best known unconditional estimate

$$\zeta(1+it) = \mathcal{O}(\log^{2/3} t) \quad (2.46)$$

There are also several conditional estimates known for $\zeta(1+it)$. One such estimate was given by Littlewood in 1912 [Lit12], assuming the Lindelöf hypothesis and showed that

$$\zeta(1+it) = \mathcal{O}(\log \log t \log \log \log t) \quad (2.47)$$

He further improved upon this result in 1928 [Lit28] and provided the best known conditional bound assuming the Riemann Hypothesis (RH) that states

$$\zeta(1+it) = \mathcal{O}(\log \log t) \quad (2.48)$$

In addition to the asymptotic behavior of $\zeta(1+it)$, many explicit bounds of the form

$$|\zeta(1+it)| \leq a \log t, \text{ for } t \geq t_0 \quad (2.49)$$

have also been derived for it. One of the earliest known results is given by Landau [Lan03] in 1903 where he shows $a = 2$, $t_0 = 10$. Backlund [Bac16] in 1918 improved this result to $a = 1$ and $t_0 > 50$ and this t_0 was lowered by Trudgian in [Tru14] to $2.001\dots$ and in the same paper he showed² that $a = \frac{3}{4}$ and $t_0 = 3$.

The best known explicit bounds for large t are of the form

$$|\zeta(1+it)| \leq A \log^{2/3} t, \text{ for } t \geq t_0 \quad (2.50)$$

In 1967, Richert first obtained³ (2.50) in [Ric67] for unknown constant A which was computed in 1985 by Ellison [EE85] to be 2100 with $t_0 = 3$ and in 1995 Cheng [Che99] improved it to 175 with $t_0 = 2$. More recent improvements have been given by Ford in [For02] where he showed $A = 72.6$ which Trudgian improved to $A = 62.6$ with $t_0 = 3$ in [Tru14]. We

²This explicit estimate with $a = 3/4$ is obtained as an application of an explicit van der Corput test using the second derivative derived by Cheng-Graham in [CG04, Lemma 3]. However, that Cheng-Graham result is now known to be incorrect. This seems to be an irrecoverable error and the $|\zeta(1+it)|$ estimate in [Tru14] no longer holds. This is elaborated further in section 2.2.1.

³In fact, Richert obtained a much more general result in [Ric67] that states $|\zeta(\sigma+it)| < ct^{100(1-\sigma)^{3/2}} \log^{2/3} t$ for $1/2 \leq \sigma \leq 1$, $t \geq 2$, and an unknown constant c .

also note that a conditional explicit estimate of the form (2.48) can be found in [LLS15] (assuming RH).

Moreover, bounds such as (2.50) seem to improve over bounds like (2.49) only when t is astronomically large because of extremely large A value. Hence, it is worth obtaining good explicit bounds of the form (2.49) for computational purposes when t is relatively small. Keeping this in mind we have the following theorem

Theorem 2.16. *If $t \geq 3$, then*

$$|\zeta(1+it)| \leq \min\left(\log t, \frac{1}{2} \log t + 1.93, \frac{1}{5} \log t + 44.02\right) \quad (2.51)$$

In particular for $t \geq 8.261 \dots \times 10^{60}$

$$|\zeta(1+it)| \leq \frac{1}{5} \log t + 44.02 \quad (2.52)$$

2.3.2 Preparatory Lemmas

We begin this section by finding good explicit upper and lower bound for the gamma function.

Lemma 2.17. *If $s = \sigma + it$ where $\sigma > 0$ and $t > 0$ then we have*

$$\sqrt{2\pi}e^{-\pi t/2}\xi_1(\sigma, t)t^{\sigma-1/2} \leq |\Gamma(s)| \leq \sqrt{2\pi}e^{-\pi t/2}\xi_2(\sigma, t)t^{\sigma-1/2} \quad (2.53)$$

where

$$\begin{aligned} \xi_1(\sigma, t) &= \exp\left(-\frac{\sigma}{12t^2} - \frac{\pi}{24t} - \frac{\sigma^3}{3t^2}\right), \\ \xi_2(\sigma, t) &= \exp\left(\left(\sigma - \frac{1}{2}\right)\frac{\sigma^2}{2t^2} + \frac{\sigma}{12t^2} + \frac{\pi}{24t}\right). \end{aligned}$$

Proof. The proof follows a similar strategy as in [Hab05]. By Stirling's formula for complex values s such that $-\pi + \delta \leq \arg s \leq \pi - \delta$ given in [Tit58, Page 151] we have

$$\log \Gamma(s) = \left(s - \frac{1}{2}\right) \log s - s + \frac{\log(2\pi)}{2} - \int_0^\infty \frac{\{x\} - 1/2}{x+s} dx. \quad (2.54)$$

This gives

$$\log |\Gamma(s)| = \left(\sigma - \frac{1}{2}\right) \log |s| + t \left(-\arctan\left(\frac{t}{\sigma}\right) - \frac{\sigma}{t}\right) + \frac{\log(2\pi)}{2} - \Re\left(\int_0^\infty \frac{\{x\} - 1/2}{x+s} dx\right). \quad (2.55)$$

Using integration by parts we have

$$\int_0^\infty \frac{\{x\} - 1/2}{x+s} dx = -\frac{1}{12s} + \frac{1}{2} \int_0^\infty \frac{\{x\}^2 - \{x\} + 1/6}{(x+s)^2} dx$$

Hence we have using triangle inequality and $\sigma \geq 0$

$$\left| \Re \left(\int_0^\infty \frac{\{x\} - 1/2}{x+s} dx \right) \right| \leq \frac{\sigma}{12|s|^2} + \frac{1}{12} \int_0^\infty \frac{dx}{(x+\sigma)^2 + t^2} \leq \frac{\sigma}{12t^2} + \frac{\pi}{24t} \quad (2.56)$$

Also, since $0 \leq \log(1+x) \leq x$ for $x \geq 0$ we also have

$$0 \leq \log|s| - \log t = \frac{1}{2} \log \left(1 + \frac{\sigma^2}{t^2} \right) \leq \frac{\sigma^2}{2t^2}, \quad \text{for } \sigma \geq 0 \quad (2.57)$$

Moreover note that since $\sigma > 0$

$$-\frac{\pi}{2} - \frac{\sigma^3}{3t^3} \leq -\arctan\left(\frac{t}{\sigma}\right) - \frac{\sigma}{t} = \arctan\left(\frac{\sigma}{t}\right) - \frac{\pi}{2} - \frac{\sigma}{t} \leq -\frac{\pi}{2} \quad (2.58)$$

where the inequalities follows from for $x \geq 0$

$$x - \frac{1}{3}x^3 \leq \arctan(x) \leq x$$

and the equality in the middle is because

$$\arctan(x) + \arctan(1/x) = \frac{\pi}{2}, \quad \text{for } x > 0$$

Thus substituting (2.57), (2.58), (2.56) in (2.55) we obtain

$$\log |\Gamma(s)| \leq \log \sqrt{2\pi} + \left(\sigma - \frac{1}{2} \right) \log t - \frac{\pi}{2}t + \left(\sigma - \frac{1}{2} \right) \frac{\sigma^2}{2t^2} + \frac{\sigma}{12t^2} + \frac{\pi}{24t} \quad (2.59)$$

$$\log |\Gamma(s)| \geq \frac{\log(2\pi)}{2} + \left(\sigma - \frac{1}{2} \right) \log t - \frac{\pi}{2}t - \frac{\sigma^3}{3t^2} - \frac{\sigma}{12t^2} - \frac{\pi}{24t} \quad (2.60)$$

Therefore exponentiating on both sides we get

$$\sqrt{2\pi}e^{-\pi t/2}\xi_1(\sigma, t)t^{\sigma-1/2} \leq \Gamma(s) \leq \sqrt{2\pi}e^{-\pi t/2}\xi_2(\sigma, t)t^{\sigma-1/2}$$

where

$$\xi_1(\sigma, t) = \exp \left(-\frac{\sigma}{12t^2} - \frac{\pi}{24t} - \frac{\sigma^3}{3t^2} \right) \quad (2.61)$$

$$\xi_2(\sigma, t) = \exp \left(\left(\sigma - \frac{1}{2} \right) \frac{\sigma^2}{2t^2} + \frac{\sigma}{12t^2} + \frac{\pi}{24t} \right) \quad (2.62)$$

Here we remark that for $\sigma > 0, t \geq t_0 > 0$

$$\xi_1(\sigma, t) \geq \xi_1(\sigma, t_0), \quad \xi_2(\sigma, t) \leq \begin{cases} \frac{\sigma}{12t_0^2} + \frac{\pi}{24t_0} & \text{for } 0 < \sigma < \frac{1}{2} \\ \xi_2(\sigma, t_0) & \text{for } \sigma \geq \frac{1}{2} \end{cases} \quad (2.63)$$

□

We also remark that (2.53) is also valid for $t < 0$ provided that t is replace with $|t|$ since $\Gamma(\bar{s}) = \overline{\Gamma(s)}$ and $|\Gamma(\bar{s})| = |\Gamma(s)|$

Corollary 2.18. *We have for $\sigma \leq 0$ and $t > 0$*

$$\xi_3(\sigma, t, n) |\Gamma((\sigma + n) + it)| \leq |\Gamma(\sigma + it)| \leq \frac{1}{t^n} |\Gamma((\sigma + n) + it)|$$

where n is the smallest positive integer such that $\Re(\sigma + n) > 0$ and

$$\xi_3(\sigma, t, n) = \frac{1}{(-\sigma + t)(-\sigma - 1 + t) \dots (-\sigma - n + 1 + t)}.$$

Proof. The functional equation of gamma function states

$$\Gamma(z) = \frac{1}{z} \Gamma(z + 1), \quad z \in \mathbb{C} \setminus \{0, -1, -2, \dots\} \quad (2.64)$$

Now let $n \in \mathbb{Z}_{>0}$ be the smallest integer such that $\Re(z + n) > 0$ Thus using (2.64) we obtain the following relation:

$$|\Gamma(z)| = \frac{1}{|z(z+1)(z+2) \dots (z+n-1)|} |\Gamma(z+n)|.$$

Hence substituting $z = \sigma + it$ and using Lemma 2.17 with $\sqrt{(\sigma + l)^2 + t^2} \geq t$ for $l \in \{0, 1, \dots, n-1\}$ we get that

$$|\Gamma(\sigma + it)| \leq \frac{1}{t^n} |\Gamma(\sigma + n + it)|$$

And since $\sigma + l < 0$, we can find a lower bound using $\sqrt{(\sigma + l)^2 + t^2} \leq -(\sigma + l) + t$, for $l \in \{0, 1, \dots, n-1\}$ and thus

$$|\Gamma(\sigma + it)| \geq \frac{1}{(-\sigma + t)(-\sigma - 1 + t) \dots (-\sigma - n + 1 + t)} |\Gamma((\sigma + n) + it)|.$$

Hence the result follows. □

Corollary 2.19. *For $t > 0$ we have*

$$|\chi(1 + it)| \leq \frac{g(t)}{t^{1/2}} \quad (2.65)$$

where

$$g(t) = \sqrt{2\pi} \exp\left(\frac{5}{3t^2} + \frac{\pi}{6t}\right) \quad (2.66)$$

Proof. We have the following definition for $\chi(1 + it)$ as stated in [Tit86, Page 16] when $s = 1 + it$ is substituted

$$\chi(1 + it) = \pi^{1/2+it} \frac{\Gamma\left(\frac{-it}{2}\right)}{\Gamma\left(\frac{1+it}{2}\right)}. \quad (2.67)$$

In order to bound $|\chi(1+it)|$ using (2.67) we will first bound $\left|\Gamma\left(\frac{-it}{2}\right)\right|$ from above and $\left|\Gamma\left(\frac{1+it}{2}\right)\right|$ from below. First using Corollary 2.18 and then Lemma 2.17 along with the remark following it we have that

$$\left|\Gamma\left(-\frac{it}{2}\right)\right| \leq \frac{2\sqrt{\pi}}{t^{1/2}} e^{-\pi t/4} \xi_2(1, t/2), \quad \left|\Gamma\left(\frac{1+it}{2}\right)\right| \geq \sqrt{2\pi} e^{-\pi t/4} \xi_1(1/2, t/2) \quad (2.68)$$

which in turn gives us the result

$$|\chi(1+it)| \leq \sqrt{2\pi} \exp\left(\frac{5}{3t^2} + \frac{\pi}{6t}\right) \frac{1}{t^{1/2}}$$

where we used $n = 1$ and

$$\left|\frac{\xi_2(1, t/2)}{\xi_1(1/2, t/2)}\right| \leq \frac{\exp\left(\frac{4}{3t^2} + \frac{\pi}{12t}\right)}{\exp\left(-\frac{1}{3t^2} - \frac{\pi}{12t}\right)} = \exp\left(\frac{5}{3t^2} + \frac{\pi}{6t}\right), \quad \text{for } t > 0 \quad (2.69)$$

□

Theorem 2.20. For $t > 0$ and $n_1 = \lfloor \sqrt{t/2\pi} \rfloor$ we have

$$|\zeta(1+it)| \leq \left| \sum_{n=1}^{n_1} \frac{1}{n^{1+it}} \right| + \frac{g(t)}{t^{1/2}} \left| \sum_{n=1}^{n_1} \frac{1}{n^{-it}} \right| + \mathcal{R} \quad (2.70)$$

where

$$\mathcal{R} = \mathcal{R}(t) := \frac{1}{t^{1/2}} \left(\sqrt{\frac{\pi}{2}} + \frac{g(t)}{2} \right) + \frac{1}{t} \left(9\sqrt{\frac{\pi}{2}} + \frac{g(t)}{\sqrt{\pi(3-2\log 2)}} \right) + \frac{1}{t^{3/2}} \left(\frac{968\pi^{3/2} + g(t)242\pi}{700} \right).$$

and $g(t)$ is given by (2.66).

Proof. Note that from [Rey11] we have the following expression for $\zeta(s)$:

$$\zeta(s) = \mathcal{R}(s) + \chi(s) \overline{\mathcal{R}}(1-s), \quad (2.71)$$

where $\mathcal{R}(s)$ is defined by an integral as given in [Rey11] and

$$\overline{\mathcal{R}}(s) = \overline{\mathcal{R}(\overline{s})}, \quad \chi(s) = \pi^{s-1/2} \frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})}$$

Arias de Reyna showed in [Rey11] that with σ and t real and $t > 0$, and an integer $K \geq 0$ we have

$$\mathcal{R}(s) = \sum_{n=1}^N \frac{1}{n^s} + (-1)^{N-1} U a^{-\sigma} \left\{ \sum_{k=0}^K \frac{C_k(p)}{a^k} + R S_K \right\} \quad (2.72)$$

where

$$a := \sqrt{\frac{t}{2\pi}}, \quad N := \lfloor a \rfloor, \quad p := 1 - 2(a - N)$$

$$U := \exp\left(-i\left(\frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8}\right)\right)$$

and RS_K and $C_k(p)$ are defined in [Rey11, Page 999].

In particular substituting $s = 1 + it$ in (2.71) we obtain

$$|\zeta(1 + it)| \leq |\mathcal{R}(1 + it)| + |\chi(1 + it)| |\overline{\mathcal{R}}(-it)| \quad (2.73)$$

Thus to bound $|\zeta(1 + it)|$ we will first bound $|\mathcal{R}(1 + it)|$, $|\overline{\mathcal{R}}(-it)|$ using (2.72), triangle inequality and [Rey11] and combine it with the bound $|\chi(1 + it)|$ found in Corollary 2.19. Hence, after using (2.72) and triangle inequality we get

$$|\mathcal{R}(1 + it)| \leq \left| \sum_{n=1}^N \frac{1}{n^{1+it}} \right| + \left(\frac{2\pi}{t} \right)^{1/2} \left(\left| \sum_{k=0}^K \frac{C_k(p)}{a^k} \right| + |RS_K| \right). \quad (2.74)$$

It remains to bound the last two terms to the right of the inequality above. First, we will bound the second sum on the right hand side of (2.74) using $K = 1$, triangle inequality after expanding the sum and [Rey11, Theorem 4.1] with $\sigma = 1 > 0$, and $\Gamma(1/2) = \sqrt{\pi}$. We thus have

$$\left| \sum_{k=0}^1 \frac{C_k(p)}{a^k} \right| \leq |C_0(p)| + \frac{9}{2t^{1/2}} \leq \frac{1}{2} + \frac{9}{2t^{1/2}}. \quad (2.75)$$

The last inequality above follows from bounding $|C_0(p)|$ using [Rey11, Equation (5.2)] and [Rey11, Theorem 6.1].

Next to bound $|RS_K|$ we use [Rey11, Theorem 4.2], again with $K = 1$ and $\sigma = 1, t > 0$ and $\Gamma(1) = 1$. So we obtain

$$|RS_1| \leq \frac{242\pi}{700t} 2^{3/2}. \quad (2.76)$$

Now plugging in (2.75) and (2.76) in (2.74) we get

$$|\mathcal{R}(1 + it)| \leq \left| \sum_{n=1}^N \frac{1}{n^{1+it}} \right| + \frac{1}{t^{1/2}} \left(\frac{(2\pi)^{1/2}}{2} + \frac{968\pi^{3/2}}{700t} + \frac{9(2\pi)^{1/2}}{2t^{1/2}} \right) \quad (2.77)$$

Similarly we can a bound for $|\overline{\mathcal{R}}(-it)|$ by first observing $|\overline{\mathcal{R}}(-it)| = |\overline{\mathcal{R}(it)}| = |\mathcal{R}(it)|$ and using triangle inequality and (2.72) we get

$$|\overline{\mathcal{R}}(-it)| \leq \left| \sum_{n=1}^N \frac{1}{n^{it}} \right| + \left(\left| \sum_{k=0}^K \frac{C_k(p)}{a^k} \right| + |RS_K| \right) \quad (2.78)$$

Now to bound the second sum and $|RS_K|$ above we follow similar steps used to derive (2.75) and (2.76) with $\sigma = 0$ and $t > 0$ and obtain

$$|\overline{\mathcal{R}}(-it)| \leq \left| \sum_{n=1}^N \frac{1}{n^{it}} \right| + \frac{1}{2} + \frac{1}{\sqrt{\pi(3-2\log 2)}} \frac{1}{t^{1/2}} + \frac{242\pi}{700t} \quad (2.79)$$

Lastly combining (2.71), (2.77), (2.79), and the bound for $|\chi(1+it)|$ using Corollary 2.19 we have our theorem. $\square$

2.3.3 Proof of Theorem 2.16

First using Theorem 2.20 we get a significant improvement on known explicit bounds of the form (2.49) by estimating the two Dirichlet sums in (2.70) trivially using triangle inequality, (2.82), (2.66), and $\lfloor t^{1/2}/\sqrt{2\pi} \rfloor \geq t^{1/2}/\sqrt{2\pi} - 1$ to get the following bound valid for $t > 2\pi$:

$$|\zeta(1+it)| \leq \frac{1}{2} \log t + \gamma + \frac{\sqrt{2\pi}}{t^{1/2} - \sqrt{2\pi}} - \frac{1}{2} \log(2\pi) + \exp\left(\frac{5}{3t^2} + \frac{\pi}{6t}\right) + \mathcal{R} \quad (2.80)$$

To improve upon (2.80) we make use of Lemma 2.15. With this in mind, we first split the sums on the right of the inequality in (2.70) as follows:

$$|\zeta(1+it)| \leq \left| \sum_{n=1}^{\lfloor jt^{1/5} \rfloor - \delta_0} \frac{1}{n^{1+it}} \right| + \left| \sum_{n=\lfloor jt^{1/5} \rfloor}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} \frac{1}{n^{1+it}} \right| + \frac{g(t)}{t^{1/2}} \left| \sum_{n=1}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} \frac{1}{n^{-it}} \right| + \mathcal{R}. \quad (2.81)$$

where $j \in \mathbb{Z}_{>1}$ to be chosen later and

$$\delta_0 := \begin{cases} 1 & jt^{1/5} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

The first sum in (2.81) in the range $1 \leq n \leq \lfloor jt^{1/5} \rfloor - \delta_0$ is computed using triangle inequality and explicit bound on harmonic sum⁵ obtained using partial summation and stated in [Tru14]

$$\sum_{n \leq N} \frac{1}{n} \leq \log N + \gamma + \frac{1}{N}. \quad (2.82)$$

The second sum in the range $\lfloor jt^{1/5} \rfloor \leq n \leq \lfloor \sqrt{t/(2\pi)} \rfloor$ is divided into dyadic pieces where each of these pieces is estimated using the explicit fifth derivative test given by Lemma 2.15. Lastly, the third sum is bounded trivially using triangle inequality.

⁵We note here that a marginally better estimate than (2.82) can be found in [Fra19, Lemma 2.1] by improving upon the $1/N$ term. However due to the large magnitude of t used for numerical computation later in the proof we do not get a noticeable improvement than (2.82) in the constants obtained for $|\zeta(1+it)|$ bound. Hence, for proof readability we use the elementary estimate given in (2.82).

The dyadic subdivision for this second sum in (2.81) is carried out in the following manner and then using triangle inequality subject to $\epsilon > \frac{2}{jt^{1/5}} > 0$:

$$\sum_{n=\lceil jt^{1/5} \rceil}^{\lfloor \sqrt{t/2\pi} \rfloor} \frac{1}{n^{1+it}} = \sum_{r=0}^{R(\epsilon)-1} \sum_{n=\lceil (1+\epsilon)^r jt^{1/5} \rceil}^{\lfloor (1+\epsilon)^{r+1} jt^{1/5} \rfloor - \delta_r} \frac{1}{n^{1+it}} \quad (2.83)$$

where,

$$\delta_r := \begin{cases} 1 & (1+\epsilon)^{r+1} jt^{1/5} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

and

$$R(\epsilon) \leq \left\lfloor \frac{\frac{3}{10} \log t - \log \sqrt{2\pi} j}{\log(1+\epsilon)} \right\rfloor + 1 \quad (2.84)$$

along with $R(\epsilon) \geq 1$. Here, $R(\epsilon)$ gives the number of dyadic pieces of our main sum obtained using the parameter ϵ . This bound is obtained using the following inequality and solving for $R(\epsilon)$:

$$(1+\epsilon)^{R(\epsilon)-1} jt^{1/5} \leq \lceil (1+\epsilon)^{R(\epsilon)-1} jt^{1/5} \rceil \leq \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor \leq \sqrt{\frac{t}{2\pi}}.$$

Also note that that since $\epsilon > \frac{2}{jt^{1/5}}$ we have $\lceil (1+\epsilon)^r jt^{1/5} \rceil < \lfloor (1+\epsilon)^{r+1} jt^{1/5} \rfloor - \delta_r$.

To bound the sum on the right hand side in (2.83) we first use partial summation Lemma 2.4 and thus we get

$$S := \left| \sum_{r=0}^{R(\epsilon)-1} \sum_{n=\lceil (1+\epsilon)^r jt^{1/5} \rceil}^{\lfloor (1+\epsilon)^{r+1} jt^{1/5} \rfloor - \delta_r} \frac{1}{n^{1+it}} \right| \leq \sum_{r=0}^{R(\epsilon)-1} \frac{1}{\lceil (1+\epsilon)^r jt^{1/5} \rceil} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon)^r jt^{1/5} \rceil}^{\lceil (1+\epsilon)^{r+1} jt^{1/5} \rceil + \Delta - 1} n^{-it} \right| \quad (2.85)$$

where $L := L_r = \lfloor (1+\epsilon)^{r+1} jt^{1/5} \rfloor - \delta_r - \lceil (1+\epsilon)^r jt^{1/5} \rceil + 1$ where $L > 1$ because $\epsilon > \frac{2}{jt^{1/5}}$.

We will apply Lemma 2.15 to the inner sum in (2.85). To do so first note,

$$f(x) := \frac{-t}{2\pi} \log x, \quad \lceil (1+\epsilon)^r jt^{1/5} \rceil \leq x \leq \lceil (1+\epsilon)^{r+1} jt^{1/5} \rceil + (\Delta - 1),$$

then

$$f^{(4)}(x) := \frac{-12t}{\pi x^5}, \quad \lceil (1+\epsilon)^r jt^{1/5} \rceil \leq x \leq \lceil (1+\epsilon)^{r+1} jt^{1/5} \rceil + (\Delta - 1).$$

Now using $1 \leq \Delta \leq L$ we have,

$$\frac{12t}{\pi(\lceil (1+\epsilon)^r jt^{1/5} \rceil + \Delta - 1)^5} \leq |f^{(5)}(x)| \leq \frac{12t}{\pi \lceil (1+\epsilon)^r jt^{1/5} \rceil^5} \quad (2.86)$$

$$\frac{12t}{\pi(\lfloor (1+\epsilon)^{r+1} jt^{1/5} \rfloor)^4} \leq |f^{(5)}(x)| \leq \frac{12t}{\pi \lceil (1+\epsilon)^{r+1} jt^{1/5} \rceil^5} \quad (2.87)$$

$$\frac{1}{W} \leq |f^{(5)}(x)| \leq \lambda \frac{1}{W} \quad (2.88)$$

Thus, we set

$$W := W_r = \frac{\pi(\lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor)^5}{12t}, \quad \lambda := \lambda_r = \frac{(\lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor)^5}{(\lceil(1+\epsilon)^rjt^{1/5}\rceil)^5}$$

Using $\lceil(1+\epsilon)^rjt^{1/5}\rceil \leq \lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor - \delta_r \leq \lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor$ and $r \geq 0, j > 1$ we conclude $W > 1$ and $\lambda \geq 1$. Hence we can apply explicit fifth derivative Lemma 2.15 to the right-hand side sum in (2.85) and get

$$\begin{aligned} S \leq & \sum_{r=0}^{R(\epsilon)-1} \frac{1}{\lceil(1+\epsilon)^rjt^{1/5}\rceil} (\alpha_5 L^2 W^{-1/15} + \tau_5 L^{7/4} W^{1/15} + \gamma_5 L^{3/2} W^{2/15} + \omega_5 L^{5/4} W^{2/15} \\ & + \beta_5 L W^{3/15} + \eta_5 \alpha_5 L + \eta_5 \tau_5 L^{3/4} W^{2/15} + \eta_5 \gamma_5 L^{1/2} W^{3/15} + \eta_5 \omega_5 L^{1/4} W^{1/5} + \eta_5 \beta_5 W^{4/15})^{1/2} \end{aligned}$$

Factoring $L^2 W^{-1/15}$ gives us that

$$\begin{aligned} S \leq & \sum_{r=0}^{R(\epsilon)-1} \frac{\sqrt{L^2 W^{-1/15}}}{\lceil(1+\epsilon)^rjt^{1/5}\rceil} (\alpha_5 + \tau_5 L^{-1/4} W^{2/15} + \gamma_5 L^{-1/2} W^{3/15} + \omega_5 L^{-3/4} W^{3/15} \\ & + \beta_5 L^{-1} W^{4/15} + \eta_5 \alpha_5 L^{-1} W^{1/15} + \eta_5 \tau_5 L^{-5/4} W^{3/15} + \eta_5 \gamma_5 L^{-3/2} W^{4/15} \\ & + \eta_5 \omega_5 L^{-7/4} W^{4/15} + \eta_5 \beta_5 L^{-2} W^{5/15})^{1/2} \end{aligned} \quad (2.89)$$

Next let us bound $L^2 W^{-1/15}$ using the definition of L and W . With this in mind and using $\lfloor x \rfloor \leq x$, $\lceil x \rceil \geq x$ and $\delta_r \geq 0$ we obtain

$$L = \lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor - \delta_r - \lceil(1+\epsilon)^rjt^{1/5}\rceil + 1 \quad (2.90)$$

$$\leq \epsilon(1+\epsilon)^rjt^{1/5}\psi_{\epsilon,r,t} \quad (2.91)$$

where

$$\psi_{\epsilon,r,t} := 1 + \frac{1}{\epsilon(1+\epsilon)^rjt^{1/5}} \quad (2.92)$$

With this at hand we now bound $LW^{-1/15}$ from above using $\lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor \geq \lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor - \delta_r \geq \lceil(1+\epsilon)^rjt^{1/5}\rceil \geq (1+\epsilon)^rjt^{1/5}$ and get

$$LW^{-1/15} \leq \epsilon(1+\epsilon)^rjt^{1/5}\psi_{\epsilon,r,t} \left(\frac{\pi(\lfloor(1+\epsilon)^{r+1}jt^{1/5}\rfloor)^5}{12t} \right)^{-1/15} \quad (2.93)$$

$$\leq \epsilon\psi_{\epsilon,r,t} \left(\frac{12}{\pi} \right)^{1/15} j^{2/3} (1+\epsilon)^{2r/3} t^{1/5}. \quad (2.94)$$

Substituting (2.94) and (2.91) in (2.89) we have

$$S \leq \left(\frac{12}{\pi} \right)^{1/30} \epsilon \sum_{r=0}^{R(\epsilon)-1} \frac{\psi_{\epsilon,r,t}}{j^{1/6} (1+\epsilon)^{r/6}} \mathcal{B}_r \quad (2.95)$$

where

$$\begin{aligned} \mathcal{B}_r = & \left(\alpha_5 + \tau_5 \frac{W^{2/15}}{L^{1/4}} + \gamma_5 \frac{W^{3/15}}{L^{1/2}} + \omega_5 \frac{W^{3/15}}{L^{3/4}} + \beta_5 \frac{W^{4/15}}{L} + \eta_5 \alpha_5 \frac{W^{1/15}}{L} + \eta_5 \tau_5 \frac{W^{3/15}}{L^{5/4}} \right. \\ & \left. + \eta_5 \gamma_5 \frac{W^{4/15}}{L^{3/2}} + \eta_5 \omega_5 \frac{W^{4/15}}{L^{7/4}} + \eta_5 \beta_5 \frac{W^{5/15}}{L^2} \right)^{1/2} \end{aligned} \quad (2.96)$$

Now let us focus on bounding $\mathcal{B}_r$ using upper and lower bounds for W and L respectively.

We have the following upper bound for W :

$$W = \frac{\pi \lfloor (1+\epsilon)^{r+1} j t^{1/5} \rfloor^5}{12 t} \leq \frac{\pi}{12} (1+\epsilon)^5 j^5 (1+\epsilon)^{5r} \quad (2.97)$$

and using $\lfloor x \rfloor \geq x - 1$, $\lceil x \rceil \leq x + 1$ and $\delta_r \leq 1$

$$\begin{aligned} L &= \lfloor (1+\epsilon)^{r+1} j t^{1/5} \rfloor - \delta_r - \lceil (1+\epsilon)^r j t^{1/5} \rceil + 1 \\ &\geq (1+\epsilon)^r j t^{1/5} \left(\epsilon - \frac{2}{(1+\epsilon)^r j t^{1/5}} \right) \\ &\geq \phi_{\epsilon,j,t} (1+\epsilon)^r j t^{1/5} \end{aligned} \quad (2.98)$$

where $\phi_{\epsilon,j,t} = \left(\epsilon - \frac{2}{j t^{1/5}} \right)$ and recall that $\epsilon > \frac{2}{j t^{1/5}}$.

Now using (2.97) and (2.98) we can bound for a quantity in the form

$$\frac{W^{a/15}}{L^b} \leq \left(\frac{\pi}{12} \right)^{a/15} \frac{(1+\epsilon)^{a/3} j^{a/3-b}}{\phi_{\epsilon,j,t}^b} (1+\epsilon)^{(a/3-b)r} \frac{1}{t^{b/5}} \quad (2.99)$$

Using (2.99) we can bound the following quantities:

$$\frac{W^{2/15}}{L^{1/4}}, \frac{W^{1/5}}{L^{1/2}}, \frac{W^{1/5}}{L^{3/4}}, \frac{W^{4/15}}{L}, \frac{W^{1/15}}{L}, \frac{W^{1/5}}{L^{5/4}}, \frac{W^{4/15}}{L^{3/2}}, \frac{W^{4/15}}{L^{7/4}}, \frac{W^{1/3}}{L^2}$$

by substituting $a = 1, 2, 3, 4, 5$ and $b = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2$.

With (3.20), (2.99), the above mentioned a, b values and the inequality $\sqrt{x_1 + x_2 + \dots} \leq \sqrt{x_1} + \sqrt{x_2} + \dots$ for $x_1, x_2, \dots > 0$, we finally obtain the following complicated upper bound

for $\mathcal{B}_r$

$$\begin{aligned}
\mathcal{B}_r \leq & \sqrt{\alpha_5} + \left(\tau_5 \left(\frac{\pi}{12} \right)^{2/15} \frac{(1+\epsilon)^{2/3} j^{5/12}}{\phi_{\epsilon,j,t}^{1/4}} \frac{(1+\epsilon)^{5r/12}}{t^{1/20}} + \gamma_5 \left(\frac{\pi}{12} \right)^{1/5} \frac{(1+\epsilon) j^{1/2}}{\phi_{\epsilon,j,t}^{1/2}} \frac{(1+\epsilon)^{r/2}}{t^{1/10}} \right. \\
& + \omega_5 \left(\frac{\pi}{12} \right)^{1/5} \frac{(1+\epsilon) j^{1/4}}{\phi_{\epsilon,j,t}^{3/4}} \frac{(1+\epsilon)^{r/4}}{t^{3/20}} + \beta_5 \left(\frac{\pi}{12} \right)^{4/15} \frac{(1+\epsilon)^{4/3} j^{1/3}}{\phi_{\epsilon,j,t}} \frac{(1+\epsilon)^{r/3}}{t^{1/5}} \left. \right)^{1/2} \\
& + \left(\eta_5 \alpha_5 \left(\frac{\pi}{12} \right)^{1/15} \frac{(1+\epsilon)^{1/3}}{j^{2/3} \phi_{\epsilon,j,t}} \frac{1}{(1+\epsilon)^{2r/3} t^{1/5}} + \eta_5 \tau_5 \left(\frac{\pi}{12} \right)^{1/5} \frac{(1+\epsilon)}{j^{1/4} \phi_{\epsilon,j,t}^{5/4}} \frac{1}{(1+\epsilon)^{r/4} t^{1/4}} \right. \\
& + \eta_5 \gamma_5 \left(\frac{\pi}{12} \right)^{4/15} \frac{(1+\epsilon)^{4/3}}{j^{1/6} \phi_{\epsilon,j,t}^{3/2}} \frac{1}{(1+\epsilon)^{r/6} t^{3/10}} + \eta_5 \omega_5 \left(\frac{\pi}{12} \right)^{4/15} \frac{(1+\epsilon)^{4/3}}{j^{5/12} \phi_{\epsilon,j,t}^{7/4}} \frac{1}{(1+\epsilon)^{5r/12} t^{7/20}} \\
& \left. \eta_5 \beta_5 \left(\frac{\pi}{12} \right)^{1/3} \frac{(1+\epsilon)^{5/3}}{\phi_{\epsilon,j,t}^2 j^{1/3}} \frac{1}{(1+\epsilon)^{r/3} t^{2/5}} \right)^{1/2} \quad (2.100)
\end{aligned}$$

At this stage we can factor $\frac{(1+\epsilon)^{r/2}}{t^{1/20}}$ and $\frac{1}{(1+\epsilon)^{r/6} t^{1/5}}$ from the second and third square-root terms above and plugging the bound obtained for $\mathcal{B}_r$ into inequality (2.95) we obtain

$$S \leq \left(\frac{12}{\pi} \right)^{1/30} \frac{\epsilon}{j^{1/6}} \sum_{r=0}^{R(\epsilon)-1} \psi_{\epsilon,r,t} \left(\frac{\sqrt{\alpha_5}}{(1+\epsilon)^{r/6}} + \mathcal{E}_1 \frac{(1+\epsilon)^{r/12}}{t^{1/40}} + \mathcal{E}_2 \frac{1}{(1+\epsilon)^{r/4} t^{1/10}} \right) \quad (2.101)$$

where

$$\begin{aligned}
\mathcal{E}_1 &:= \mathcal{E}_1(t) = \left(\tau_5 \left(\frac{\pi}{12} \right)^{2/15} \frac{(1+\epsilon)^{2/3} j^{5/12}}{\phi_{\epsilon,j,t}^{1/4}} + \gamma_5 \left(\frac{\pi}{12} \right)^{1/5} \frac{(1+\epsilon) j^{1/2}}{\phi_{\epsilon,j,t}^{1/2}} \frac{1}{t^{1/20}} \right. \\
&\quad \left. + \omega_5 \left(\frac{\pi}{12} \right)^{1/5} \frac{(1+\epsilon) j^{1/4}}{\phi_{\epsilon,j,t}^{3/4}} \frac{1}{t^{1/10}} + \beta_5 \left(\frac{\pi}{12} \right)^{4/15} \frac{(1+\epsilon)^{4/3} j^{1/3}}{\phi_{\epsilon,j,t}} \frac{1}{t^{3/20}} \right)^{1/2} \\
\mathcal{E}_2 &:= \mathcal{E}_2(t) = \left(\eta_5 \alpha_5 \left(\frac{\pi}{12} \right)^{1/15} \frac{(1+\epsilon)^{1/3}}{j^{2/3} \phi_{\epsilon,j,t}} + \eta_5 \tau_5 \left(\frac{\pi}{12} \right)^{1/5} \frac{(1+\epsilon)}{j^{1/4} \phi_{\epsilon,j,t}^{5/4}} \frac{1}{t^{1/20}} \right. \\
&\quad + \eta_5 \gamma_5 \left(\frac{\pi}{12} \right)^{4/15} \frac{(1+\epsilon)^{4/3}}{j^{1/6} \phi_{\epsilon,j,t}^{3/2}} \frac{1}{t^{1/10}} + \eta_5 \omega_5 \left(\frac{\pi}{12} \right)^{4/15} \frac{(1+\epsilon)^{4/3}}{j^{5/12} \phi_{\epsilon,j,t}^{7/4}} \frac{1}{t^{3/20}} \\
&\quad \left. + \eta_5 \beta_5 \left(\frac{\pi}{12} \right)^{1/3} \frac{(1+\epsilon)^{5/3}}{\phi_{\epsilon,j,t}^2 j^{1/3}} \frac{1}{t^{1/5}} \right)^{1/2}
\end{aligned}$$

Next using the definition of $\psi_{\epsilon,r,t}$ from (2.92) we can write (2.101) as

$$S \leq \left(\frac{12}{\pi}\right)^{1/30} \frac{\epsilon}{j^{1/6}} \sum_{r=0}^{R(\epsilon)-1} \left(\frac{\sqrt{\alpha_5}}{(1+\epsilon)^{r/6}} + \frac{\mathcal{E}_1(1+\epsilon)^{r/12}}{t^{1/40}} + \frac{\mathcal{E}_2}{(1+\epsilon)^{r/4}t^{1/10}} \right) \\ + \left(\frac{12}{\pi}\right)^{1/30} \frac{1}{j^{7/6}} \sum_{r=0}^{R(\epsilon)-1} \left(\frac{\sqrt{\alpha_5}}{(1+\epsilon)^{7r/6}t^{1/5}} + \frac{\mathcal{E}_1}{(1+\epsilon)^{11r/12}t^{9/40}} + \frac{\mathcal{E}_2}{(1+\epsilon)^{5r/4}t^{3/10}} \right) \quad (2.102)$$

Let us estimate each of the above sums using the following inequality that is valid for $c, d > 0$ and using (2.84) :

$$\sum_{r=0}^{R(\epsilon)-1} \frac{1}{(1+\epsilon)^{cr}} \leq \frac{(1+\epsilon)^c}{(1+\epsilon)^c - 1}, \quad \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{dr} \leq \frac{(1+\epsilon)^d}{(2\pi)^{d/2}j^d((1+\epsilon)^d - 1)} t^{3d/10} \quad (2.103)$$

Finally we can obtain a bound for S using (2.102) and (2.103) :

$$S \leq d_1 \left(\sqrt{\alpha_5} \frac{(1+\epsilon)^{1/6}}{(1+\epsilon)^{1/6} - 1} + \frac{\mathcal{E}_1(1+\epsilon)^{1/12}}{(2\pi)^{1/24}j^{1/12}((1+\epsilon)^{1/12} - 1)} + \frac{\mathcal{E}_2(1+\epsilon)^{1/4}}{(1+\epsilon)^{1/4} - 1} t^{-1/10} \right) \\ + d_2 \left(\frac{\sqrt{\alpha_5}(1+\epsilon)^{7/6}}{(1+\epsilon)^{7/6} - 1} t^{-1/5} + \frac{\mathcal{E}_1(1+\epsilon)^{11/12}}{(1+\epsilon)^{11/12} - 1} t^{-9/40} + \frac{\mathcal{E}_2(1+\epsilon)^{5/4}}{(1+\epsilon)^{5/4} - 1} t^{-3/10} \right).$$

where

$$d_1 := \left(\frac{12}{\pi}\right)^{1/30} \frac{\epsilon}{j^{1/6}}, \quad d_2 := \left(\frac{12}{\pi}\right)^{1/30} \frac{1}{j^{7/6}}.$$

In the end we have

$$\left| \sum_{n=\lceil jt^{1/5} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} \frac{1}{n^{1+it}} \right| \leq A_1 t^{-3/10} + A_2 t^{-9/40} + A_3 t^{-1/5} + A_4 t^{-1/10} + A_5 \quad (2.104)$$

where

$$A_1 := \frac{d_2 \mathcal{E}_2(1+\epsilon)^{5/4}}{(1+\epsilon)^{5/4} - 1}, \quad A_2 := \frac{d_2 \mathcal{E}_1(1+\epsilon)^{11/12}}{(1+\epsilon)^{11/12} - 1}, \quad A_3 := \frac{d_2 \sqrt{\alpha_5}(1+\epsilon)^{7/6}}{(1+\epsilon)^{7/6} - 1}, \\ A_4 := \frac{d_1 \mathcal{E}_2(1+\epsilon)^{1/4}}{(1+\epsilon)^{1/4} - 1}, \quad A_5 := \frac{d_1 \sqrt{\alpha_5}(1+\epsilon)^{1/6}}{(1+\epsilon)^{1/6} - 1} + \frac{d_1 \mathcal{E}_1(1+\epsilon)^{1/12}}{(2\pi)^{1/24}j^{1/12}((1+\epsilon)^{1/12} - 1)}.$$

At this stage we bound the third sum in (2.81) trivially and thus we get the following bound valid for $t > (2/j)^5$

$$|\zeta(1+it)| \leq \gamma + \log j + \frac{1}{5} \log t + \frac{1}{jt^{1/5} - 2} + A_1 t^{-3/10} + A_2 t^{-9/40} + A_3 t^{-1/5} \\ + A_4 t^{-1/10} + A_5 + \frac{g(t)}{\sqrt{2\pi}} + \mathcal{O} \quad (2.105)$$

On the other hand, we also have the option of bounding the last sum in (2.81) using Lemma 2.15 and obtaining an estimate for it using the fact that $\left| \sum_{n=1}^N \frac{1}{n^{-it}} \right| = \left| \sum_{n=1}^N \frac{1}{n^{it}} \right|$ and following a similar interval split and then steps taken to estimate S as in (2.89). However, the improvements obtained in this case is negligible unless t is astronomically large. Hence we omit the details of such a computation here.

Finally, we choose the following values for our parameters via numerical experimentation:

$$\epsilon = 0.32, \quad j = 60, \quad \eta_3 = \left(\frac{15\sqrt{\pi}}{32} \right)^{2/3}, \quad \eta_4 = \left(\frac{91}{72\sqrt{23}} \right)^{6/7}, \quad \eta_5 = 2.2, \quad t_0 = 8 \times 10^{60}$$

Additionally, we use that

$$W \geq \frac{\pi j^5}{12}, \quad \lambda \leq (1 + \epsilon)^5, \quad \mathcal{C}_1 \leq \mathcal{C}_1(t_0), \quad \mathcal{C}_2 \leq \mathcal{C}_2(t_0), \quad g(t) \leq g(t_0), \quad \mathcal{R} \leq \mathcal{R}(t_0) \quad (2.106)$$

and plugging them into (2.105) we get

$$|\zeta(1 + it)| \leq 43.9259 + \frac{1}{60t^{1/5} - 2} + \frac{0.035264}{t^{3/10}} + \frac{0.255693}{t^{9/40}} + \frac{0.0552644}{t^{1/5}} + \frac{2.96078}{t^{1/10}} + \frac{1}{5} \log t \quad (2.107)$$

Now from (2.107) we deduce that

$$|\zeta(1 + it)| - \frac{1}{5} \log t \leq F(t). \quad (2.108)$$

where

$$F(t) := 43.9259 + \frac{1}{60t^{1/5} - 2} + \frac{0.035264}{t^{3/10}} + \frac{0.255693}{t^{9/40}} + \frac{0.0552644}{t^{1/5}} + \frac{2.96078}{t^{1/10}}$$

Note that $F(t)$ is decreasing in t and thus for $t \geq t_0$ and we have that

$$F(t) \leq F(t_0) \leq 44.02$$

giving us

$$|\zeta(1 + it)| \leq \frac{1}{5} \log t + 44.02 \quad (2.109)$$

Now for $t \geq 47$, we use (2.80) and get that

$$|\zeta(1 + it)| \leq \frac{1}{2} \log t + 1.93. \quad (2.110)$$

Lastly for $t \geq 3$ we also have

$$|\zeta(1 + it)| \leq \log t. \quad (2.111)$$

Ultimately combining (2.109), (2.110) and (2.111) for $t \geq 3$ we obtain the the following bound for the zeta function on 1-line:

$$|\zeta(1 + it)| \leq \min \left(\log t, \frac{1}{2} \log t + 1.93, \frac{1}{5} \log t + 44.02 \right) \quad (2.112)$$

2.3.4 Concluding Remarks on $|\zeta(1+it)|$ Estimate

Note that choosing a different set of parameters, one can obtain a slight improvement on the constant 44.02 in (2.51). However it seems that this constant cannot be improved beyond the Euler constant $\gamma = 0.57721\dots$ as the harmonic sums (2.82) are bounded above and below by at least $\log N + \gamma$. Moreover the leading constant $\frac{1}{5}$ in (2.52) is the best that can be obtained if one insists on using Lemma 2.15 (which is a fifth derivative test) and inequality (2.82) for harmonic sum, as done here. However, this $1/5$ could be improved if one uses higher explicit derivative test or if one finds a suitable method to take advantage of possible cancellations while estimating the first sum on the r.h.s in (2.81) instead of using triangle inequality followed by (2.82). Estimates on $|\zeta(1+it)|$ using higher explicit derivative tests are in preparation by the author.

2.4 Higher k Derivative Test

Granville and Ramaré have derived an explicit version of $A^{k-2}B$ process in [GR96, Proposition 8.2], but the constant attached to the main term in their bound is significantly larger, typically by a factor of $4 \sim 7$, than what we already have for $k = 1, \dots, 5$ and so is not affordable given our goal of improving on current explicit Weyl-bound even for moderately large t . Hence, it might be worth investigating if the methods described in previous sections for explicit fourth and fifth derivative tests could be generalized.

2.5 Explicit Estimates on Exponential Integrals

We begin the section by stating an extremely useful result, referred to as “complex mean value theorem for integrals” which is often used as a first step to obtain explicit estimates for complex integrals. This result can be found in [Bar76].

Theorem 2.21. *Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is monotone, $g : [a, b] \rightarrow \mathbb{C}$ is continuous, and $I = \int_a^b f(x)g(x) dx = |I|e^{i\theta}$. Then there exists a point $c \in [a, b]$ such that*

$$I = \left(f(a)\Re\left(e^{-i\theta} \int_a^c g(x) dx\right) + f(b)\Re\left(e^{-i\theta} \int_c^b g(x) dx\right) \right) e^{i\theta}$$

so that

$$|I| \leq \left| f(a) \int_a^c g(x) dx \right| + \left| f(b) \int_c^b g(x) dx \right|$$

Furthermore, if f is of constant sign and $|f|$ is decreasing, then there exists a point $c \in [a, b]$ such that

$$|I| = f(a)\Re\left(e^{-i\theta} \int_a^c g(x) dx\right) \leq \left| f(a) \int_a^c g(x) dx \right|$$

Now we state a sharper version of Lemma 4.2 in [Tit86].

Lemma 2.22. *Let $f(x)$ be a real differentiable function such that $f'(x)$ is monotonic and $f'(x) \geq m_1 > 0$, or $f'(x) \leq -m_1 < 0$, throughout the interval $[a, b]$. Then*

$$\left| \int_a^b e^{2\pi i f(x)} dx \right| \leq \frac{1}{\pi m_1} \quad (2.113)$$

Proof. The proof can be found in [Rog05, Lemma 3] where in that proof we scale the phase function by 2π and thus replace $f(x)$ by $2\pi f(x)$. $\square$

We use Lemma 2.22 above to provide an improved explicit version of Lemma 4.4 in [Tit86]. An explicit version of the following lemma can be found in [Tit86, Lemma 4.4], [Rog05, Lemma 4] with constants $4\sqrt{2}/\sqrt{\pi}$, and $4 \cdot 2^{1/6}/\sqrt{\pi}$ respectively.

Lemma 2.23. *Let $f(x)$ be a real function, twice differentiable and $f''(x) \geq m_2 > 0$, or $f''(x) \leq -m_2 < 0$, throughout the interval $[a, b]$. Then*

$$\left| \int_a^b e^{2\pi i f(x)} dx \right| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{m_2}} \quad (2.114)$$

Proof. The proof is similar as in [Tit86, Lemma 4.4] but using Lemma 2.22 above we find that the bound for intervals I_1 and I_3 changes to

$$|I_1|, |I_2| \leq \frac{1}{\pi m_2 \delta} \quad (2.115)$$

Thus, taking $\delta = 1/\sqrt{\pi m_2}$ we have our claim. $\square$

We can generalize Lemma 2.23 above for a k -times differentiable function where $k \geq 3$ which can be found in [Rog05, Lemma 4].

Now, with Lemmas 2.22 and 2.23 at hand, we make Lemma 4.6 in [Tit86] explicit. Before doing so, we mention the following result

Lemma 2.24. *Suppose $Y > 0$ then the following holds*

$$\int_Y^\infty \frac{e^{-iu}}{\sqrt{u}} du \leq \frac{2}{\sqrt{Y}} \quad (2.116)$$

Proof. Use integration by parts. $\square$

Lemma 2.25. *Let $f(x)$ be real, with derivatives up to the third order. Let*

$$0 < \lambda_2 \leq f''(x) < \mathcal{A}_2 \lambda_2 \quad \text{or} \quad 0 < \lambda_2 \leq -f''(x) < \mathcal{A}_2 \lambda_2;$$

$$|f'''(x)| \leq \mathcal{A}_3 \lambda_3$$

throughout the interval (a, b) . Let $f'(c) = 0$, where $c \in [a, b]$. Then

$$\int_a^b e^{2\pi i f(x)} dx = \frac{e^{2\pi i(f(c)-1/8)}}{\sqrt{f''(c)}} + \mathcal{J}_1(\lambda_2, \lambda_3) + \mathcal{J}_2(a, b, \lambda_2) \quad (2.117)$$

where

$$\begin{aligned} |\mathcal{J}_1(\lambda_2, \lambda_3)| &\leq \Psi_1 \mathcal{A}_3^{1/5} \lambda_2^{-4/5} \lambda_3^{1/5} \\ |\mathcal{J}_2(a, b, \lambda_2)| &\leq \min\left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(a)|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}}\right) + \min\left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(b)|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}}\right) \leq \frac{\Psi_2}{\sqrt{\lambda_2}} \end{aligned} \quad (2.118)$$

and the absolute constants associated with $\mathcal{J}_1(\lambda_2, \lambda_3)$ and $\mathcal{J}_2(a, b, \lambda_2)$ are

$$\Psi_1 = \frac{8}{(6\pi^3)^{1/5}} \approx 2.813 \dots; \quad \Psi_2 = \frac{8}{\sqrt{\pi}} \approx 4.513 \dots \quad (2.119)$$

Proof.

$$\int_a^b e^{2\pi i f(x)} dx = \int_a^{c-\delta} e^{2\pi i f(x)} dx + \int_{c-\delta}^{c+\delta} e^{2\pi i f(x)} dx + \int_{c+\delta}^b e^{2\pi i f(x)} dx := I_1 + I_2 + I_3 \quad (2.120)$$

We first suppose that $0 < \delta \leq \min(c - a, b - c)$. Through, this case we will obtain the first and second term on the right hand side in (2.117). In particular, estimating I_2 in (2.120) where the critical point c lies, will yield the first term and the second term, $\mathcal{J}_1(\lambda_2, \lambda_3)$, will be obtained using estimates on I_2, I_2, I_3 altogether.

We start by bounding $|I_1|$. To do so, we observe that $f'(x)$ is monotonic and thus apply Lemma 2.22 to obtain

$$|I_1| \leq \frac{1}{\pi |f'(c - \delta)|} \quad (2.121)$$

To finish bounding (2.121) we note that

$$|f'(c - \delta)| = \left| \int_{c-\delta}^c f''(t) dt \right| \geq \lambda_2 \delta \quad (2.122)$$

Hence, substituting (2.122) in (2.121) we get

$$|I_1| \leq \frac{1}{\pi \lambda_2 \delta} \quad (2.123)$$

Similar analysis shows that the bound in (2.123) also holds for $|I_3|$. Therefore, so far we have

$$|I_1| + |I_3| \leq \frac{2}{\pi \lambda_2 \delta} \quad (2.124)$$

Now, let's focus our attention on obtaining a bound for I_2 which we do so by first using the remainder theorem for a third-degree Taylor series approximation for $f(x)$ and get for $\theta \in [0, 1]$

$$2\pi i f(x) = 2\pi i f(c) + 2\pi i f'(c)(x - c) + \pi i f''(c)(x - c)^2 + \frac{\pi i}{3} f'''(c + \theta(x - c))(x - c)^3 \quad (2.125)$$

Hence substituting (2.125) in I_2 given in (2.120) and using $f'(c) = 0$ we have

$$I_2 = \int_{c-\delta}^{c+\delta} e^{2\pi i f(x)} dx = e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} e^{\frac{\pi i}{3} f'''(c+\theta(x-c))(x-c)^3} dx \quad (2.126)$$

At this stage, we remark that

$$e^{ix} = \int_0^x i e^{it} dt + 1 \quad (2.127)$$

Hence, letting

$$\vartheta(x) = \int_0^{\frac{\pi}{3} f'''(c+\theta(x-c))(x-c)^3} i e^{it} dt \quad (2.128)$$

we can rewrite the integral on the right hand side of (2.126) as

$$I_2 = e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} (1 + \vartheta(x)) dx \quad (2.129)$$

$$= e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} dx + e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} \vartheta(x) dx \quad (2.130)$$

Letting

$$I_{2,1} = e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} dx, \quad I_{2,2} = e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} \vartheta(x) dx \quad (2.131)$$

we rewrite (2.130) as

$$I_2 = I_{2,1} + I_{2,2} \quad (2.132)$$

Now we estimate I_2 by evaluating $I_{2,1}$ and $I_{2,2}$ instead and we will do this by using Lemma 2.24 and triangle inequality respectively. We start with $I_{2,1}$ first and note that

$$I_{2,1} = e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} dx = 2e^{2\pi i f(c)} \int_0^\delta e^{\pi i f''(c)x^2} dx \quad (2.133)$$

Using $f''(c) < 0$ and letting $u = \pi |f''(c)| x^2$, we modify the integral over $(0, \delta)$ on the right hand side of (2.133) as follows:

$$\int_0^\delta e^{\pi i f''(c)x^2} dx = \frac{1}{2\sqrt{\pi}\sqrt{|f''(c)|}} \int_0^{\pi |f''(c)|\delta^2} \frac{e^{-iu}}{\sqrt{u}} du \quad (2.134)$$

Now, to bound the integral on the right hand side of (2.134) we write that integral as a difference of two integrals given by

$$\int_0^{\pi |f''(c)|\delta^2} \frac{e^{-iu}}{\sqrt{u}} du = \int_0^\infty \frac{e^{-iu}}{\sqrt{u}} du - \int_{\pi |f''(c)|\delta^2}^\infty \frac{e^{-iu}}{\sqrt{u}} du \quad (2.135)$$

We observe that first integral on the right hand side of (2.135) evaluates to $\sqrt{\pi} e^{-\pi i/4}$. To bound the second integral on the right in (2.135) we use Lemma 2.24 and get

$$\int_{\pi |f''(c)|\delta^2}^\infty \frac{e^{-iu}}{\sqrt{u}} du = \varphi_2 \frac{2}{\sqrt{\pi}\sqrt{|f''(c)|}\delta} \quad (2.136)$$

where $|\varphi_2|$ is a complex number with $|\varphi_2| = 1$. Thus, substituting (2.135), and the observation made after it along with (2.136) in (2.134) we get

$$\int_0^\delta e^{\pi i f''(c)x^2} dx = \frac{e^{-\pi i/4}}{2\sqrt{|f''(c)|}} - \frac{\varphi_2}{\pi\delta|f''(c)|} \quad (2.137)$$

Substituting (2.137) in (2.133) and combining the exponentials we finally get that

$$I_{2,1} = \frac{e^{2\pi i(f(c)-1/8)}}{\sqrt{|f''(c)|}} - \frac{2\varphi_2}{\pi\delta|f''(c)|} \quad (2.138)$$

Thus, so far we have by substituting (2.138) in (2.132)

$$I_2 = \frac{e^{2\pi i(f(c)-1/8)}}{\sqrt{|f''(c)|}} - \frac{2\varphi_2}{\pi\delta|f''(c)|} + I_{2,2} \quad (2.139)$$

We now focus our attention on estimating $I_{2,2}$ and we start by applying triangle inequality and have

$$I_{2,2} = e^{2\pi i f(c)} \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} \vartheta(x) dx = \varphi_3 \left| \int_{c-\delta}^{c+\delta} e^{\pi i f''(c)(x-c)^2} \vartheta(x) dx \right| \leq 2\delta\varphi_3 |\vartheta(x)| \quad (2.140)$$

where φ_3 is complex unity $|\varphi_3| \leq 1$. Next, we bound $|\vartheta(x)|$ using (2.128) and triangle inequality, $|f''(x)| \leq \mathcal{A}_3\lambda_3$, and $x \leq c + \delta$ we get

$$|\vartheta(x)| \leq \left| \frac{\pi}{3} f'''(c + \theta(x-c))(x-c)^3 \right| \leq \frac{\pi}{3} \mathcal{A}_3\lambda_3\delta^3 \quad (2.141)$$

Substituting (2.141) in (2.140) we have

$$|I_{2,2}| \leq \frac{2\pi}{3} \mathcal{A}_3\lambda_3\delta^4 \quad (2.142)$$

Therefore, using (2.138) and (2.142) and applying triangle inequality to (2.132) along with noting $|f''(x)| \geq \lambda_2$ to get

$$|I_2| \leq \frac{1}{\sqrt{|f''(c)|}} + \frac{2}{\pi\delta\lambda_2} + \frac{2\pi}{3} \mathcal{A}_3\lambda_3\delta^4 \quad (2.143)$$

To choose δ we gather the terms involving δ in $|I_1|$, $|I_2|$, and $|I_3|$ using (2.124) and (2.143). These terms are

$$\frac{4}{\pi\delta\lambda_2}, \quad \frac{2\pi}{3} \mathcal{A}_3\lambda_3\delta^4 \quad (2.144)$$

and the two terms in (2.144) are of equal magnitude when

$$\delta = \left(\frac{6}{\pi^2 \mathcal{A}_3} \right)^{1/5} (\lambda_2\lambda_3)^{-1/5} \quad (2.145)$$

Hence, recalling (2.120) along with (2.139) we have that

$$\int_a^b e^{2\pi i f(x)} dx = \frac{e^{2\pi i(f(c)-1/8)}}{\sqrt{|f''(c)|}} + \mathcal{J}_1(\lambda_2, \lambda_3) \quad (2.146)$$

and gathering all the terms involving δ in I_1, I_2 and I_3 using (2.124) and (2.143) and applying triangle inequality and then substituting the value of δ found in (2.145) we get

$$|\mathcal{J}_1(\lambda_2, \lambda_3)| \leq \frac{8}{(6\pi^3)^{1/5}} \mathcal{A}_3^{1/5} \lambda_2^{-4/5} \lambda_3^{1/5} = \Psi_1 \mathcal{A}_3^{1/5} \lambda_2^{-4/5} \lambda_3^{1/5} \quad (2.147)$$

where

$$\Psi_1 = \frac{8}{(6\pi^3)^{1/5}} \approx 2.813 \dots \quad (2.148)$$

This finishes our analysis under the assumption that $0 < \delta \leq \min(c-a, b-c)$.

Now suppose that δ is large such that $b-\delta < c \leq b$. Hence we can rewrite (2.120) as

$$\int_a^b e^{2\pi i f(x)} dx = I_1 + I_2 - I_4 \quad (2.149)$$

where

$$I_4 = \int_b^{c+\delta} e^{2\pi i f(x)} dx \quad (2.150)$$

Using similar analysis used to derive (2.130) we also obtain a similar expressions for I_4 where the range of integration changes to $(b, c+\delta)$

$$I_4 = I_{4,1} + I_{4,2} \quad (2.151)$$

where

$$I_{4,1} = e^{2\pi i f(c)} \int_b^{c+\delta} e^{\pi i f''(c)(x-c)^2} dx, \quad I_{4,2} = e^{2\pi i f(c)} \int_b^{c+\delta} e^{\pi i f''(c)(x-c)^2} \vartheta(x) dx \quad (2.152)$$

Now to estimate I_4 we start with estimating $I_{4,1}$. To do so, we set

$$F(x) = \frac{1}{2} f''(c)(x-c)^2 \quad (2.153)$$

and obtain $|I_{4,1}|$ to be

$$|I_{4,1}| = \left| \int_b^{c+\delta} e^{2\pi i F(x)} dx \right| \quad (2.154)$$

We now obtain a bound for $|I_{4,1}|$ in two different ways by using Lemma 2.22 and Lemma 2.23.

To apply Lemma 2.22, we use (2.153) to get $|F'(x)| = |f''(c)|(x-c) \geq \lambda_2(b-c)$ and that $F'(x)$ is monotonic. Thus Lemma 2.22 gives us

$$|I_{4,1}| \leq \frac{1}{\pi} \frac{1}{\lambda_2(b-c)} \quad (2.155)$$

We finish the bound for $|I_{4,1}|$ by applying mean value theorem over (b, c) to $f(x)$ and obtain

$$\frac{f'(b) - f'(c)}{b - c} = f''(\xi), \quad \text{for } \xi \in (b, c) \quad (2.156)$$

Note that $f'(c) = 0$ and taking absolute values in (2.156) along with $|f''(x)| \leq \mathcal{A}_2 \lambda_2$ we get

$$b - c \geq \frac{|f'(b)|}{\mathcal{A}_2 \lambda_2} \quad (2.157)$$

Thus substituting (2.157) in (2.155) we finally get

$$|I_{4,1}| \leq \frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(b)|} \quad (2.158)$$

To use Lemma 2.23, we use (2.153) and $|f''(x)| \geq \lambda_2$ and note that

$$|F''(x)| = |f''(c)| \geq \lambda_2 \quad (2.159)$$

Hence with (2.159) at hand we apply Lemma 2.23 to (2.154) and obtain

$$|I_{4,1}| = \left| \int_b^{c+\delta} e^{2\pi i F(x)} dx \right| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \quad (2.160)$$

Thus, gathering (2.158) and (2.160) we conclude

$$|I_{4,1}| \leq \min \left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(b)|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \right) \quad (2.161)$$

Next, we deal with $I_{4,2}$ by subtracting it with the integral $I_{2,2}$ which is over $(c - \delta, c + \delta)$ obtained in (2.131) and the resulting integral can be bounded by $|I_{2,2}|$. Hence, in this case (2.161) is the only additional term obtained besides the one obtained in (2.146).

Similar estimate as in (2.161) holds for the case $a < c \leq a + \delta$ and in that case when have the extra contributing term of:

$$\min \left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(a)|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \right) \quad (2.162)$$

In all, collecting (2.146) and the contribution from extreme cases given in (2.161) and (2.162) we prove our lemma. $\square$

2.6 Explicit Estimates on Exponential Sums

We make Lemma 4.7 in [Tit86] explicit which will allow us to transform an exponential sum into sum of exponential integrals.

Lemma 2.26. *Let $f(x)$ be a real function with a continuous and steadily decreasing derivative $f'(x)$ in (a, b) with $f'(b) = \alpha, f'(a) = \beta$. Then for $0 < \eta < 1$ we have*

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha - \eta \leq \nu < \beta + \eta} \int_a^b e^{2\pi i (f(x) - \nu x)} dx + \mathfrak{J}_1 \log(\beta - \alpha + 4) + \mathfrak{J}_2 \quad (2.163)$$

where

$$|\mathfrak{J}_1| \leq \frac{3}{\pi}; \quad |\mathfrak{J}_2| \leq \frac{4}{\pi} \frac{1}{\eta} + \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{13}{7} \right) \quad (2.164)$$

Proof. We start with equation (2.1.2) in [Tit86] and get

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + 2\pi i \int_a^b (x - [x] - 1/2) f'(x) e^{2\pi i f(x)} dx + C \quad (2.165)$$

where

$$C = (a - [a] - 1/2) e^{2\pi i f(a)} - (b - [b] - 1/2) e^{2\pi i f(b)} \quad (2.166)$$

Without loss of generality we can assume $\eta - 1 < \alpha \leq \eta$. If this is not the case then we can replace $f(x)$ with $f(x) - kx$ where $k = [\alpha]$ or $[\alpha] - 1$ and use the argument for the former case to achieve our result.

We now focus on bounding the second integral on the right hand side of (2.165). To do so, we first note that using (2.1.7) in [Tit86], we have the Fourier series

$$x - [x] - \frac{1}{2} = -\frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{\sin 2\nu\pi x}{\nu}, \quad x \notin \mathbb{Z} \quad (2.167)$$

Thus, substituting (2.167) in the second integral on the right hand side in (2.165) and noting that the Fourier series is boundedly convergent. so that we may multiply by an integrable function and interchange the integral and sum to get

$$2\pi i \int_a^b (x - [x] - 1/2) f'(x) e^{2\pi i f(x)} dx = -2i \sum_{\nu=1}^{\infty} \int_a^b \frac{\sin 2\pi\nu x}{\nu} f'(x) e^{2\pi i f(x)} dx \quad (2.168)$$

Now using the relation

$$e^{-2\pi i \nu x} - e^{2\pi i \nu x} = -2i \sin(2\pi \nu x) \quad (2.169)$$

we rewrite the integral on the right hand side in (2.168) and combine the exponential terms in the integrand to get

$$\begin{aligned} 2\pi i \int_a^b (x - [x] - 1/2) f'(x) e^{2\pi i f(x)} dx &= \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i (f(x) - \nu x)} dx \\ &\quad - \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i (f(x) + \nu x)} dx \end{aligned} \quad (2.170)$$

and the two sums on the right hand side in (2.170) converge as we will soon see.

Letting

$$G_1 := \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x)-\nu x)} dx, \quad G_2 := \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x)+\nu x)} dx \quad (2.171)$$

allows us to rewrite (2.165) as follows:

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + G_1 - G_2 + C \quad (2.172)$$

The series G_2 is typically easier to estimate as the function $f(x) + \nu x$ does not have a critical point in the interval (a, b) .

At this stage, we recall that $\eta - 1 < \alpha \leq \eta$ which implies $-1 < \alpha \leq 1$ and $\beta \geq \alpha$ which implies $\beta > -1$ also. Hence we have the following possibilities for α, β : $\alpha \geq 0, \beta \geq 0$; $\alpha \leq 0, \beta \leq 0$; $\alpha \leq 0, \beta \geq 0$.

We start with estimating G_2 first and we do so by rewriting that integral as

$$|G_2| = \left| \sum_{\nu=1}^{\infty} \frac{1}{2\pi\nu} \int_a^b \frac{f'(x)}{f'(x) + \nu} d(e^{2\pi i(f(x)+\nu x)}) \right| \quad (2.173)$$

Before beginning our computation for bounding G_2 we note the following useful estimate

$$\left| \int_a^b d(e^{2\pi i(f(x)\pm\nu x)}) \right| \leq 2 \quad (2.174)$$

We apply the complex mean value theorem from Theorem 2.21 part (a) and then use (2.174) to get

$$\begin{aligned} |G_2| &\leq \left| \sum_{\nu=1}^{\infty} \frac{1}{2\pi\nu} \frac{f'(a)}{f'(a) + \nu} \int_{\xi_1}^b d(e^{2\pi i(f(x)+\nu x)}) \right| + \left| \sum_{\nu=1}^{\infty} \frac{1}{2\pi\nu} \frac{f'(b)}{f'(b) + \nu} \int_a^{\xi_2} d(e^{2\pi i(f(x)+\nu x)}) \right| \\ &\leq \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\beta|}{\nu(\beta + \nu)} + \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\alpha|}{\nu(\alpha + \nu)} \end{aligned} \quad (2.175)$$

where $\xi_1, \xi_2 \in [a, b]$.

We pull out the first terms in the series on the right hand side in (2.175) and obtain

$$|G_2| \leq \frac{1}{\pi} \left(\frac{|\beta|}{\beta + 1} + \frac{|\alpha|}{\alpha + 1} \right) + \frac{1}{\pi} \sum_{\nu=2}^{\infty} \frac{|\beta|}{\nu(\beta + \nu)} + \frac{1}{\pi} \sum_{\nu=2}^{\infty} \frac{|\alpha|}{\nu(\alpha + \nu)} \quad (2.176)$$

Now, we bound the first term on the right hand side in (2.176) but noting that

$$\frac{|\beta|}{\beta + 1} \leq \begin{cases} \frac{1}{\eta} & \text{if } \beta < 0 \\ 1 & \text{if } \beta \geq 0 \end{cases}$$

where the inequality for $\beta < 0$ follows because this implies $|\beta| \leq 1$ and also $\beta + 1 > \eta$ since $-1 < \eta - 1 < \alpha \leq \beta < 0$. The inequality for $\beta \geq 0$ is valid because in this case $|\beta| = \beta < \beta + 1$. Hence we conclude that

$$\frac{|\beta|}{\beta + 1} \leq \max\left(\frac{1}{\eta}, 1\right) \leq \frac{1}{\eta} \quad (2.177)$$

where the last inequality above follows because $\eta < 1$. Similarly we argue that

$$\frac{|\alpha|}{\alpha + 1} \leq \frac{1}{\eta} \quad (2.178)$$

Next we observe that the first sum on the right hand side in (2.176) can be bounded by another sum where $1 \leq \nu < \infty$ and the denominator is replaced by $|\beta| + \nu$. We use a similar argument to bound the second sum on the right hand side in (2.176). Hence with this observation at hand along with substituting (2.177), (2.178) in (2.176) we get

$$|G_2| \leq \frac{2}{\pi} \frac{1}{\eta} + \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\beta|}{\nu(|\beta| + \nu)} + \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\alpha|}{\nu(|\alpha| + \nu)} \quad (2.179)$$

And finally, to bound the sums on the right hand side in (2.179) we use the digamma function relation stated in (2.1) and then bound it using (2.2) to get

$$\frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\beta|}{\nu(|\beta| + \nu)} = \frac{1}{\pi} (\psi(|\beta| + 1) + \gamma) \leq \frac{1}{\pi} \log(|\beta| + 1) + \frac{\gamma}{\pi} \quad (2.180)$$

Similarly we obtain

$$\frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\alpha|}{\nu(|\alpha| + \nu)} \leq \frac{1}{\pi} \log(|\alpha| + 1) + \frac{\gamma}{\pi} \leq \frac{1}{\pi} \log 2 + \frac{\gamma}{\pi} \quad (2.181)$$

where the last inequality above follows because $|\alpha| \leq 1$. Therefore, substituting (2.180) and (2.181) in (2.179) we obtain

$$|G_2| \leq \frac{2}{\pi} \frac{1}{\eta} + \frac{1}{\pi} (\log 2 + 2\gamma) + \frac{1}{\pi} \log(|\beta| + 1) \quad (2.182)$$

Lastly to replace $|\beta| + 1$ inside the logarithm in (2.182) we note that for $\beta > -1$ we have $|\beta| + 1 \leq \max\{2, \beta + 1\} \leq \beta + 3$. Hence, with this we have our estimate for $|G_2|$ as

$$|G_2| \leq \frac{2}{\pi} \frac{1}{\eta} + \frac{1}{\pi} (\log 2 + 2\gamma) + \frac{1}{\pi} \log(\beta + 3). \quad (2.183)$$

Next, we focus on bounding $|G_1|$ which we recall from (2.171) that.

$$G_1 := \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x) - \nu x)} dx \quad (2.184)$$

To do so, we split the range of ν as: $1 \leq \nu < \beta + \eta$ and $\beta + \eta \leq \nu < \infty$. We first bound the series in the latter range as $f(x) - \nu x$ does not have a critical point in that range. Thus, using triangle inequality we can rewrite G_1 in this range as

$$G_1 = \sum_{1 \leq \nu < \beta + \eta} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x) - \nu x)} dx + G_{1,2} \quad (2.185)$$

where

$$G_{1,2} := \sum_{\beta + \eta \leq \nu < \infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x) - \nu x)} dx = \sum_{\beta + \eta \leq \nu < \infty} \frac{1}{2\pi i \nu} \int_a^b \frac{f'(x)}{f'(x) - \nu} d(e^{2\pi i(f(x) - \nu x)}) \quad (2.186)$$

At this stage, we make the important observation that we may assume that $\beta + \eta \geq 1$ to avoid shifting the range of summation in (2.185) to $0 \leq \nu < \infty$ which is problematic.

In case of $\beta + \eta < 1$ we note that the first sum on the right hand side in (2.185) is empty and thus would not yield any contribution. And we change the range of summation in $|G_{1,2}|$ to be $1 \leq \nu \leq \infty$ which in turn gives us $|G_1| = |G_{1,2}|$. Hence the two possible ranges of summation for $G_{1,2}$ are: $\beta + \eta \leq \nu \leq \infty$ and $1 \leq \nu \leq \infty$ in case of $\beta + \eta \geq 1$ and $\beta + \eta < 1$ respectively. With this observation at hand, we are now ready to bound $|G_{1,2}|$. We start with applying the second mean value theorem given by Theorem 2.21 part (a) and then using the estimate given in (2.174) and obtain for $n_0 = \max\{1, \beta + \eta\}$ that

$$\begin{aligned} |G_{1,2}| &= \left| \sum_{n_0 \leq \nu < \infty} \frac{1}{2\pi \nu} \int_a^b \frac{f'(x)}{f'(x) - \nu} d(e^{2\pi i(f(x) - \nu x)}) \right| \\ &\leq \frac{1}{\pi} \sum_{n_0 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} + \frac{1}{\pi} \sum_{n_0 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \end{aligned} \quad (2.187)$$

Let us start by estimating the first sum on the right hand side in (2.187) and we do so by considering the sum of its first term and the rest of the sum and note

$$\sum_{n_0 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \frac{|\beta|}{n_0(n_0 - \beta)} + \sum_{n_1 + 1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \quad (2.188)$$

where n_1 is the smallest integer such that $n_1 \geq \max\{1, \beta + \eta\}$. To bound the first term on the right hand side in (2.188) we note that

$$n_0(n_0 - \beta) = \max\{1, \beta + \eta\} \max\{1 - \beta, \eta\} \geq \max\{1, \beta + \eta\} \eta \quad (2.189)$$

Now if $\max\{1, \beta + \eta\} = 1$. Then this means $\beta + \eta \leq 1$ which in turn implies $-1 < \beta < 1$ since $\beta > -1$ and $0 < \eta < 1$. Hence we have $|\beta| \leq 1$ and hence in this case we have

$$\frac{|\beta|}{n_0(n_0 - \beta)} \leq \frac{1}{\eta} \quad (2.190)$$

And if $\max\{1, \beta + \eta\} = \beta + \eta$ then then this mean $\beta + \eta \geq 1$ which in turn means $\beta > 0$ since $0 < \eta < 1$. Hence noting $\beta/(\beta + \eta) < 1$ we also have in this case that

$$\frac{|\beta|}{n_0(n_0 - \beta)} \leq \frac{1}{\eta} \quad (2.191)$$

Hence, using the above arguments we obtain

$$\sum_{n_0 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \frac{1}{\eta} + \sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \quad (2.192)$$

Let us now estimate the second term on the right hand side in (2.192). To do so we recall that n_1 is the smallest integer such that $n_1 \geq \max\{1, \beta + \eta\}$. Hence, $n_1 + 1$ is the smallest integer such that $n_1 + 1 \geq \{2, \beta + \eta + 1\}$.

First, suppose $\beta \leq 0$ then we have $|\beta| \leq 1$ and $\nu - \beta \geq \nu$ along with $n_1 + 1 = 2$ since in this case we'll get $\beta + \eta < 1$. Hence for $\beta \leq 0$ we have

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \sum_{2 \leq \nu < \infty} \frac{1}{\nu^2} = \frac{\pi^2}{6} - 1 \quad (2.193)$$

where the last inequality above follows because of the estimate $\sum 1/n^2 = \pi^2/6$.

Now suppose $\beta \geq 0$ and in addition $0 \leq \beta < 1$ so in this case we have

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \sum_{n_1+1 \leq \nu < \infty} \left(\frac{1}{\nu - \beta} - \frac{1}{\nu} \right) \quad (2.194)$$

Since $\nu - \beta \geq \nu - 1$ we have the estimate in (2.194) to be

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \sum_{n_1+1 \leq \nu < \infty} \left(\frac{1}{\nu - 1} - \frac{1}{\nu} \right) \leq \frac{1}{n_1} \leq \frac{1}{\max\{1, \beta + \eta\}} \quad (2.195)$$

Next, suppose $\beta \geq 0$ but now $\beta \geq 1$. In this case we have $n_1 + 1 \geq \beta + \eta + 1$ since $\beta + \eta > 1$ and

$$\begin{aligned} \sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} &\leq \sum_{n_1+1 \leq \nu < \infty} \left(\frac{1}{\nu - \beta} - \frac{1}{\nu} \right) \\ &\leq \sum_{\mu_0=1}^{\infty} \frac{1}{\mu_0} - \sum_{\nu=n_1+1}^{\infty} \frac{1}{\nu} \\ &= \sum_{\nu=1}^{n_1} \frac{1}{\nu} = \log(n_1) + \gamma + \frac{1}{2n_1 + 1/3} \\ &\leq \log(\beta + \eta + 1) + \gamma + \frac{3}{7} \end{aligned} \quad (2.196)$$

where the last inequality above follows because $1 < \beta + \eta \leq n_1 < \beta + \eta + 1$.

Hence gathering (2.193), (2.195) and (2.196)

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \begin{cases} \frac{\pi^2}{6} - 1 & \text{if } \beta \leq 0 \\ \frac{1}{\max\{1, \beta + \eta\}} & \text{if } 0 < \beta < 1 \\ \log(\beta + \eta + 1) + \gamma + \frac{3}{7} & \text{if } \beta \geq 1 \end{cases}$$

From the above estimates we can conclude that for $\beta > -1$ we have

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \log(\beta + 2) + \gamma + \frac{3}{7} \quad (2.197)$$

since

$$\frac{\pi^2}{6} - 1, \frac{1}{\max\{1, \beta + \eta\}} \leq 1; \quad \gamma + \frac{3}{7} > 1 \quad (2.198)$$

Hence substituting (2.197) in (2.192) and then the resulting expression in (2.187) we get

$$\sum_{n_0 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} \leq \frac{1}{\eta} + \log(\beta + 2) + \gamma + \frac{3}{7} \quad (2.199)$$

$$|G_{1,2}| \leq \frac{1}{\pi} \left(\frac{1}{\eta} + \log(\beta + 2) + \gamma + \frac{3}{7} \right) + \frac{1}{\pi} \sum_{n_0 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \quad (2.200)$$

It remains to bound the last sum on the right hand side in (2.200) and as in the previous computation, we do so by pulling out the main term and get

$$\sum_{n_0 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} = \frac{|\alpha|}{n_0(n_0 - \alpha)} + \sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \quad (2.201)$$

To bound the first term on the right hand side in (2.201) note that $|\alpha| \leq 1$ since $1 < \alpha < 1$ and that

$$n_0(n_0 - \alpha) = \max\{1, \beta + \eta\} \max\{1 - \alpha, \beta + \eta - \alpha\} \quad (2.202)$$

Now suppose that $\max\{1, \beta + \eta\} = 1$ and $\max\{1 - \alpha, \beta + \eta - \alpha\} = 1 - \alpha$. This means $\beta + \eta \leq 1$ which implies $\alpha + \eta \leq 1$ since $\alpha \leq \beta$. Hence we have, $1 - \alpha \geq \eta$ and in this case get

$$\frac{|\alpha|}{n_0(n_0 - \alpha)} \leq \frac{1}{\eta} \quad (2.203)$$

Next, suppose that $\max\{1, \beta + \eta\} = 1$ and $\max\{1 - \alpha, \beta + \eta - \alpha\} = \beta + \eta - \alpha$. Then since $\beta + \eta - \alpha \geq \eta$ because $\alpha \leq \beta$ we get

$$\frac{|\alpha|}{n_0(n_0 - \alpha)} \leq \frac{1}{\eta} \quad (2.204)$$

Now suppose that $\max\{1, \beta + \eta\} = \beta + \eta$ then we must have $\beta + \eta > 1$ which then forces us to have $\max\{1 - \alpha, \beta + \eta - \alpha\} = \beta + \eta - \alpha$. Then again using $\beta + \eta - \alpha \geq \eta$ we have

$$\frac{|\alpha|}{n_0(n_0 - \alpha)} \leq \frac{1}{\eta(\beta + \eta)} \leq \frac{1}{\eta} \quad (2.205)$$

where the last inequality above follows since $\beta + \eta > 1$. In any case, we have the estimate of the first term in (2.201) and thus obtaining

$$\sum_{n_0 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq \frac{1}{\eta} + \sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \quad (2.206)$$

Now to bound the last sum on the right hand side in (2.206) we first suppose $\alpha \leq 0$. Then $\nu - \alpha \geq \nu$ and thus

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq \sum_{n_1+1 \leq \nu < \infty} \frac{1}{\nu^2} \leq \frac{\pi^2}{6} - 1 \quad (2.207)$$

where the last inequality above is valid since n_1 is the smallest integer such that $n_1 \geq \{1, \beta + \eta\}$.

Now suppose that $\alpha > 0$ then since $0 < \alpha < 1$ we have

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq \sum_{n_1+1 \leq \nu < \infty} \left(\frac{1}{\nu - \alpha} - \frac{1}{\nu} \right) \quad (2.208)$$

and then using $\nu - \alpha > \nu - 1$ we have

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq \sum_{n_1+1 \leq \nu < \infty} \left(\frac{1}{\nu - 1} - \frac{1}{\nu} \right) = \frac{1}{n_1} \leq 1 \quad (2.209)$$

where the last inequality follows because $n_1 \geq \max\{1, \beta + \eta\} \geq 1$. Hence, gathering (2.207) and (2.209) we have

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq \begin{cases} 1 & \text{if } \alpha \leq 0 \\ \frac{1}{\max\{1, \beta + \eta\}} & \text{if } 0 < \alpha < 1 \end{cases}$$

and thus

$$\sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq 1 \quad (2.210)$$

Hence plugging in (2.210) in (2.206) we have

$$\sum_{n_0 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \leq \frac{1}{\eta} + 1 \quad (2.211)$$

Hence, substituting (2.211) in (2.200) we get

$$\begin{aligned} |G_{1,2}| &\leq \frac{1}{\pi} \left(\frac{1}{\eta} + \log(\beta + 2) + \gamma + \frac{3}{7} \right) + \frac{1}{\pi} \frac{1}{\eta} + \frac{1}{\pi} \\ &= \frac{1}{\pi} \log(\beta + 2) + \frac{2}{\pi} \frac{1}{\eta} + \frac{1}{\pi} \left(\gamma + \frac{3}{7} + 1 \right) \end{aligned} \quad (2.212)$$

We are now left with estimating the first sum on the right hand side of (2.185) and we do so by rewriting $f'(x)$ as $f'(x) - \nu + \nu$ in the integrand to split the sum into two sums as follows:

$$\sum_{1 \leq \nu < \beta + \eta} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x) - \nu x)} dx = G_{1,1} + \sum_{1 \leq \nu < \beta + \eta} \int_a^b e^{2\pi i(f(x) - \nu x)} dx \quad (2.213)$$

where

$$G_{1,1} = \sum_{1 \leq \nu < \beta + \eta} \frac{1}{\nu} \int_a^b (f'(x) - \nu) (e^{2\pi i(f(x) - \nu x)}) dx \quad (2.214)$$

Again, we recall the important remark that in case of $\beta + \eta < 1$ we note that the first sum on the left hand side in (2.213) is empty and thus would not yield any contribution. Thus we change the range of summation in $|G_{1,2}|$ given by (2.186) to be $1 \leq \nu \leq \infty$ which in turn gives us $|G_1| = |G_{1,2}|$ and this estimate is already computed in (2.212). Hence for the forthcoming calculations we may assume $\beta + \eta \geq 1$.

With this in mind, to estimate $G_{1,1}$ we evaluate the integral in its definition to get

$$\int_a^b (f'(x) - \nu) (e^{2\pi i(f(x) - \nu x)}) dx = \frac{1}{2\pi i} e^{2\pi i f(x) - \nu x} \Big|_a^b \quad (2.215)$$

Hence substituting (2.215) in (2.214) and then taking absolute value we get

$$|G_{1,1}| = \left| \sum_{1 \leq \nu < \beta + \eta} \frac{1}{\nu} \int_a^b (f'(x) - \nu) (e^{2\pi i(f(x) - \nu x)}) dx \right| \leq \frac{1}{\pi} \sum_{\nu=1}^{n_2} \frac{1}{\nu} \quad (2.216)$$

where n_2 is the largest integer such that $n_2 < \max\{2, \beta + \eta\}$. And now to bound the sum on the right hand side of the inequality in (2.216) we use Lemma 2.3 and get

$$|G_{1,1}| = \left| \sum_{1 \leq \nu < \beta + \eta} \frac{1}{\nu} \int_a^b (f'(x) - \nu) (e^{2\pi i(f(x) - \nu x)}) dx \right| \leq \frac{1}{\pi} \left(\log(n_2) + \gamma + \frac{1}{2n_2 + 1/3} \right) \quad (2.217)$$

And noting that $1 \leq n_2 < \max\{2, \beta + \eta\} < \beta + 3$ we can bound the log-term and the last term on the right hand side in (2.217) by $3/7$ and have

$$|G_{1,1}| = \left| \sum_{1 \leq \nu < \beta + \eta} \frac{1}{\nu} \int_a^b (f'(x) - \nu) (e^{2\pi i(f(x) - \nu x)}) dx \right| \leq \frac{1}{\pi} \left(\log(\beta + 3) + \gamma + \frac{3}{7} \right) \quad (2.218)$$

Now using the definition of C from (2.166) we have $|C| \leq 1$ since $|y - \lfloor y \rfloor - 1/2| \leq 1/2$.

Now using above sum splits we rewrite (2.172) as

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + \sum_{1 \leq \nu < \beta + \eta} \int_a^b e^{2\pi i (f(x) - \nu x)} dx + G_{1,1} + G_{1,2} - G_2 + C \quad (2.219)$$

From (2.218), (2.212), and (2.183) and then using triangle inequality we get

$$\begin{aligned} |G_{1,1} + G_{1,2} - G_2 + C| &\leq \frac{1}{\pi} \log(\beta + 3) + \frac{\gamma}{\pi} + \frac{3}{7\pi} + \frac{1}{\pi} \log(\beta + 2) + \frac{2}{\pi} \frac{1}{\eta} + \frac{\gamma}{\pi} + \frac{3}{7\pi} + \frac{1}{\pi} \\ &\quad + \frac{2}{\pi} \frac{1}{\eta} + \frac{1}{\pi} (\log 2 + 2\gamma) + \frac{1}{\pi} \log(\beta + 3) + 1 \\ &\leq \frac{3}{\pi} \log(\beta + 3) + \frac{4}{\pi} \frac{1}{\eta} + \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{13}{7} \right) \end{aligned} \quad (2.220)$$

Now combining the first two terms on the right hand side in (2.219) which allows us to extend the range of summation to $0 \leq \nu < \beta + \eta$ and recalling from the assumption $-1 < \alpha - \eta \leq 0$ along with $\beta - \alpha + 4 \geq \beta - \eta + 3 > \beta + 3$ we have

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha - \eta \leq \nu < \beta + \eta} \int_a^b e^{2\pi i (f(x) - \nu x)} dx + \mathfrak{J}_1 \log(\beta - \alpha + 4) + \mathfrak{J}_2 \quad (2.221)$$

where

$$|\mathfrak{J}_1| \leq \frac{3}{\pi}; \quad |\mathfrak{J}_2| \leq \frac{4}{\pi} \frac{1}{\eta} + \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{13}{7} \right)$$

□

At this stage we provide a clarification in [Tit86, Lemma 4.8]. We first state that lemma as given in [Tit86]:

Lemma 2.27. *Let $f(x)$ be a real differentiable function in the interval $[a, b]$, let $f'(x)$ be monotonic, and let $|f'(x)| \leq \theta < 1$. Then*

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + \mathcal{O}(1) \quad (2.222)$$

Remark 2.28. With the condition $|f'(x)| \leq \theta < 1$, the $\mathcal{O}(1)$ term in (2.222) should depend on θ and is not an absolute constant. To be precise (2.222) should read:

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + \mathcal{O}\left(\frac{1}{1 - \theta}\right)$$

Remark 2.29. If one were to wish $\mathcal{O}(1)$ to be absolute then the condition $|f'(x)| \leq \theta < 1$ should instead be changed to $|f'(x)| \leq \frac{1}{2}$.

Note that we can eliminate the term $\frac{1}{\eta}$ from $|\mathcal{J}_2|$ in (2.164) if we have additional information about a bound on $f''(x)$, that is, $\lambda_2 \leq |f''(x)| \leq \mathcal{A}_2 \lambda_2$ which we will come across this condition in Theorem 2.31. This result is given in the next corollary.

Corollary 2.30. *Let $f(x)$ be a real function with a continuous and steadily decreasing derivative $f'(x)$ in (a, b) with $f'(b) = \alpha, f'(a) = \beta$. Furthermore, suppose $\lambda_2 \leq |f''(x)| \leq \mathcal{A}_2 \lambda_2$. Then*

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha - \eta \leq \nu < \beta + \eta} \int_a^b e^{2\pi i (f(x) - \nu x)} dx + \mathcal{J}_1 \log(\beta - \alpha + 4) + \mathcal{J}_2 \quad (2.223)$$

where

$$|\mathcal{J}_1| \leq \frac{3}{\pi}; \quad |\mathcal{J}_2| \leq \frac{12}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{20}{7} \right) \quad (2.224)$$

Proof. From the proof of Lemma 2.26 we note that $\frac{1}{\eta}$ parameter arises because of estimates on $G_{1,2}$ and G_2 given by (2.212) and (2.183). Hence to remove this $\frac{1}{\eta}$ factor we will modify the estimates of these two quantities using the additional assumption that $\lambda_2 \leq |f''(x)| \leq \mathcal{A}_2 \lambda_2$.

We start with modifying the estimate for $G_{1,2}$. First we recall the definition of $G_{1,2}$ which is

$$G_{1,2} := \sum_{\beta + \eta \leq \nu < \infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i (f(x) - \nu x)} dx = \sum_{\beta + \eta \leq \nu < \infty} \frac{1}{2\pi i \nu} \int_a^b \frac{f'(x)}{f'(x) - \nu} d(e^{2\pi i (f(x) - \nu x)}) \quad (2.225)$$

And again we note the important observation that we made after equation (2.186) that the range of summation for $G_{1,2}$ is infact $n_0 \leq \nu < \infty$ where $n_0 = \max\{1, \beta + \eta\}$. Thus splitting $\nu = n_0$ terms and consider the rest of ν terms together we have

$$\begin{aligned} |G_{1,2}| &\leq \left| \frac{1}{n_0} \int_a^b f'(x) e^{2\pi i (f(x) - n_0 x)} dx \right| + \left| \sum_{n_1 + 1 \leq \nu < \infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i (f(x) - \nu x)} dx \right| \\ &= \left| \frac{1}{n_0} \int_a^b f'(x) e^{2\pi i (f(x) - n_0 x)} dx \right| + \left| \sum_{n_1 + 1 \leq \nu < \infty} \frac{1}{\nu} \int_a^b \frac{f'(x)}{f'(x) - \nu} d(e^{2\pi i (f(x) - \nu x)}) \right| \end{aligned} \quad (2.226)$$

where n_1 is the smallest integer such that $n_1 \geq \max\{1, \beta + \eta\}$.

We estimate the first term on the right hand side in (2.226) using the mean value theorem for complex integrals Lemma 2.21 and get

$$\left| \frac{1}{n_0} \int_a^b f'(x) e^{2\pi i (f(x) - n_0 x)} dx \right| \leq \left| \frac{\beta}{n_0} \int_{\xi_3}^b e^{2\pi i (f(x) - n_0 x)} dx \right| + \left| \frac{\alpha}{n_0} \int_a^{\xi_4} e^{2\pi i (f(x) - n_0 x)} dx \right| \quad (2.227)$$

where $\xi_3, \xi_4 \in [a, b]$. And to bound the two exponential integrals on the right hand side in (2.227) we use Lemma 2.23 since $(f(x) - \nu x)'' = f''(x)$ and thus have

$$\left| \int_a^{\xi_5} e^{2\pi i(f(x) - \nu_0 x)} dx \right| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}}; \quad \xi_5 \in \{\xi_3, \xi_4\} \quad (2.228)$$

Substituting (2.228) in (2.227) we get

$$\left| \frac{1}{n_0} \int_a^b f'(x) e^{2\pi i(f(x) - n_0 x)} dx \right| \leq \frac{|\beta|}{n_0} \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{|\alpha|}{n_0} \frac{4}{\sqrt{\pi}} \frac{1}{\lambda_2} \leq \frac{8}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \quad (2.229)$$

where the last inequality in (2.229) follows since $|\alpha|/n_0 \leq 1$ because $|\alpha| \leq 1$ and $n_0 \geq 1$. And similarly $|\beta|/n_0 \leq 1$ for $-1 < \beta \leq 1$. In case $\beta > 1$ then $n_0 = \beta + \eta > \beta$ and we still have $|\beta|/n_0 \leq 1$.

Plugging (2.229) in (2.226) we obtain

$$|G_{1,2}| \leq \frac{8}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \left| \sum_{n_1+1 \leq \nu < \infty} \frac{1}{\nu} \int_a^b \frac{f'(x)}{f'(x) - \nu} d(e^{2\pi i(f(x) - \nu x)}) \right| \quad (2.230)$$

Now to bound the second sum on the right hand side in (2.230) we again use the second mean value theorem for integrals Lemma 2.21 and get

$$\begin{aligned} \left| \sum_{n_1+1 \leq \nu < \infty} \frac{1}{\nu} \int_a^b \frac{f'(x)}{f'(x) - \nu} d(e^{2\pi i(f(x) - \nu x)}) \right| &\leq \frac{1}{\pi} \sum_{n_1+1 \leq \nu < \infty} \frac{|\beta|}{\nu(\nu - \beta)} + \frac{1}{\pi} \sum_{n_1+1 \leq \nu < \infty} \frac{|\alpha|}{\nu(\nu - \alpha)} \\ &\leq \frac{1}{\pi} \left(\log(\beta + 2) + \gamma + \frac{3}{7} + 1 \right) \end{aligned} \quad (2.231)$$

where the last inequality above follows because of (2.197) and (2.210) Hence using (2.231) in (2.230) we have

$$|G_{1,2}| \leq \frac{8}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \log(\beta + 2) + \frac{1}{\pi} \left(\gamma + \frac{3}{7} + 1 \right) \quad (2.232)$$

Thus we can compare the modification in estimate for $G_{1,2}$ by comparing (2.212) and (2.232).

Next, let us modify the estimate for G_2 . We first recall its definition from (2.171)

$$G_2 := \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x) + \nu x)} dx \quad (2.233)$$

and also an estimate for $|G_2|$ obtained in (2.176)

$$|G_2| \leq \frac{1}{\pi} \left(\frac{|\beta|}{\beta + 1} + \frac{|\alpha|}{\alpha + 1} \right) + \frac{1}{\pi} \sum_{\nu=2}^{\infty} \frac{|\beta|}{\nu(\beta + \nu)} + \frac{1}{\pi} \sum_{\nu=2}^{\infty} \frac{|\alpha|}{\nu(\alpha + \nu)} \quad (2.234)$$

Note that we cannot use (2.234) if either $\beta, \alpha < 0$ otherwise the $1/\eta$ term reappears. Hence we will use the above estimate only for the case: $\beta, \alpha \geq 0$. Thus for $\beta, \alpha \geq 0$ we note

$$\frac{|\beta|}{\beta + 1} + \frac{|\alpha|}{\alpha + 1} \leq 2 \quad (2.235)$$

Hence substituting (2.235) in (2.234) we get

$$|G_2| \leq \frac{2}{\pi} + \frac{1}{\pi} \sum_{\nu=2}^{\infty} \frac{|\beta|}{\nu(\beta + \nu)} + \frac{1}{\pi} \sum_{\nu=2}^{\infty} \frac{|\alpha|}{\nu(\alpha + \nu)} \quad (2.236)$$

Using the estimates from (2.180) and (2.181) to bound the two sums on the right hand side in (2.236) we have

$$|G_2| \leq \frac{1}{\pi} \log(\beta + 3) + \frac{2\gamma + \log 2 + 2}{\pi}, \quad \beta, \alpha \geq 0 \quad (2.237)$$

At this stage the following two cases remain: $\alpha, \beta < 0$ and $\alpha < 0, \beta > 0$. Recall that in these cases we cannot directly use (2.234). We also make the important remark that the $1/\eta$ term in the estimate of G_2 corresponds to $\nu = 1$ term.

We start with the case $\alpha, \beta < 0$ and we split the sum in the definition of G_2 in (2.233) from $v = 1$ and $2 \leq \nu < \infty$ and thus

$$G_2 = \int_a^b f'(x) e^{2\pi i(f(x)+x)} dx + \sum_{\nu=2}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x)+\nu x)} dx \quad (2.238)$$

To estimate the first term on the right hand side in (2.233) we note that $f'(x)$ is of constant sign in this case. Hence we use the second mean value theorem part (b) and get

$$\left| \int_a^b f'(x) e^{2\pi i(f(x)+x)} dx \right| \leq \left| f'(a) \int_{\xi_6}^b e^{2\pi i(f(x)+x)} dx \right|, \quad \xi_6 \in [a, b] \quad (2.239)$$

We bound the exponential integral on the right hand side in (2.239) using Lemma 2.23 since $(f(x) + x)'' = f''(x)$ to get

$$\left| \int_a^b f'(x) e^{2\pi i(f(x)+x)} dx \right| \leq |\beta| \frac{4}{\sqrt{\pi} \sqrt{\lambda_2}} \leq \frac{4}{\sqrt{\pi} \sqrt{\lambda_2}} \quad (2.240)$$

where the last inequality above follows since $|\beta| \leq 1$ because $-1 < \beta < 0$.

Furthermore, to bound the last term on the right hand side in (2.238). We first rewrite it as follows

$$\left| \sum_{\nu=2}^{\infty} \frac{1}{2\pi\nu} \int_a^b \frac{f'(x)}{f'(x) + \nu} d(e^{2\pi i(f(x)+\nu x)}) \right| \leq \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\beta|}{\nu(|\beta| + \nu)} + \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\alpha|}{\nu(|\alpha| + \nu)} \quad (2.241)$$

where the last inequality above follows after applying Lemma 2.21 and then carrying out similar computation done in (2.175).

Using the estimates in (2.180) and (2.181) in (2.241) we get

$$\left| \sum_{\nu=2}^{\infty} \frac{1}{2\pi\nu} \int_a^b \frac{f'(x)}{f'(x) + \nu} d(e^{2\pi i(f(x) + \nu x)}) \right| \leq \frac{1}{\pi} \log(\beta + 3) + \frac{2\gamma + \log 2}{\pi} \quad (2.242)$$

Gathering (2.240) and (2.242) and plugging them into (2.238) we obtain for $\alpha, \beta < 0$

$$|G_2| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \log(\beta + 3) + \frac{2\gamma + \log 2}{\pi} \quad (2.243)$$

It remains to modify the estimate for G_2 in the case $\alpha < 0, \beta \geq 0$. We apply the second mean value theorem for function with non-constant sign to (2.233) and get

$$G_2 = \int_a^b f'(x) e^{2\pi i(f(x) + x)} dx + \sum_{\nu=2}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x) + \nu x)} dx \quad (2.244)$$

We bound the first term on the right hand in (2.244) as follows. Let κ be such that $f'(\kappa) = 0$ and thus we have $a \leq \kappa \leq b$ and we partition the interval $[a, b]$ as $[a, \kappa)$ and $[\kappa, \beta]$ and thus we have

$$\left| \int_a^b f'(x) e^{2\pi i(f(x) + x)} dx \right| \leq \left| \int_a^{\kappa} f'(x) e^{2\pi i(f(x) + x)} dx \right| + \left| \int_{\kappa}^b f'(x) e^{2\pi i(f(x) + x)} dx \right| \quad (2.245)$$

The second integral on the right hand side in (2.245) is estimated using the complex mean value theorem Lemma 2.21 along with the conditions: $|\alpha| \leq 1$ and $\lambda_2 \leq |f''(x)| \leq \mathcal{A}_2 \lambda_2$ and we obtain

$$\left| \int_{\kappa}^b f'(x) e^{2\pi i(f(x) + x)} dx \right| \leq \left| f'(b) \int_{\kappa}^{\xi_7} e^{2\pi i(f(x) + x)} dx \right| + \left| f'(\kappa) \int_{\xi_8}^b e^{2\pi i(f(x) + x)} dx \right| \quad (2.246)$$

where $\xi_7, \xi_8 \in [a, b]$. We bound the exponential integral in the first term on the right hand side in (2.246) using Lemma 2.23 since $(f(x) + x)'' = f''(x)$ and use $f'(\kappa) = 0$ to bound the second term to get

$$\left| \int_{\kappa}^b f'(x) e^{2\pi i(f(x) + x)} dx \right| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \quad (2.247)$$

To bound the first integral on the right hand side in (2.245) we first rewrite it as

$$\left| \int_a^{\kappa} f'(x) e^{2\pi i(f(x) + x)} dx \right| = \left| \frac{1}{2\pi} \int_a^{\kappa} \frac{f'(x)}{f'(x) + 1} d(e^{2\pi i(f(x) + x)}) \right| \leq \frac{1}{\pi} \frac{\beta}{\beta + 1} \quad (2.248)$$

And noting that $\beta/\beta + 1 < 1$ since $\beta \geq 0$ we have

$$\left| \int_a^{\kappa} f'(x) e^{2\pi i(f(x) + x)} dx \right| \leq \frac{1}{\pi} \quad (2.249)$$

Hence, gathering (2.247), (2.249) and substituting it into (2.245)

$$\left| \int_a^b f'(x) e^{2\pi i(f(x) + x)} dx \right| \leq \frac{1}{\pi} + \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \quad (2.250)$$

Using (2.247) in (2.244) we have

$$|G_2| \leq \frac{1}{\pi} + \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \left| \sum_{\nu=2}^{\infty} \frac{1}{\nu} \int_a^b f'(x) e^{2\pi i(f(x)+\nu x)} dx \right| \quad (2.251)$$

Finally to bound the last term on the right hand side in (2.251). We first rewrite it as follows

$$\left| \sum_{\nu=2}^{\infty} \frac{1}{2\pi\nu} \int_a^b \frac{f'(x)}{f'(x)+\nu} d(e^{2\pi i(f(x)+\nu x)}) \right| \leq \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\beta|}{\nu(|\beta|+\nu)} + \frac{1}{\pi} \sum_{\nu=1}^{\infty} \frac{|\alpha|}{\nu(|\alpha|+\nu)} \quad (2.252)$$

where the last inequality above follows after applying Lemma 2.21 and then carrying out similar computation done in (2.175).

Using the estimates in (2.180) and (2.181) in (2.252) we get

$$\left| \sum_{\nu=2}^{\infty} \frac{1}{2\pi\nu} \int_a^b \frac{f'(x)}{f'(x)+\nu} d(e^{2\pi i(f(x)+\nu x)}) \right| \leq \frac{1}{\pi} \log(\beta+3) + \frac{2\gamma + \log 2}{\pi} \quad (2.253)$$

Now plugging (2.253) in (2.251) we get for $\alpha < 0, \beta \geq 0$ that

$$|G_2| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \log(\beta+3) + \frac{2\gamma + \log 2 + 1}{\pi} \quad (2.254)$$

Collecting (2.237), (2.243), and (2.254) we have that for $-1 < \eta - 1 < \alpha \leq \eta$ and $\beta \geq \alpha$ that

$$|G_2| \leq \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \log(\beta+3) + \frac{2\gamma + \log 2 + 1}{\pi} \quad (2.255)$$

Finally, recalling from (2.219) we have

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \int_a^b e^{2\pi i f(x)} dx + \sum_{1 \leq \nu < \beta + \eta} \int_a^b e^{2\pi i(f(x) - \nu x)} dx + G_{1,1} + G_{1,2} - G_2 + C \quad (2.256)$$

From the modified $G_{1,2}$ and G_2 estimate from (2.232), (2.255) respectively, and our earlier estimate for $G_{1,1}$ from (2.218) along with $|C| \leq 1$ and triangle inequality we get

$$\begin{aligned} |G_{1,1} + G_{1,2} - G_2 + C| &\leq \frac{1}{\pi} \left(\log(\beta+3) + \gamma + \frac{3}{7} \right) + \frac{8}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \log(\beta+2) + \frac{1}{\pi} \left(\gamma + \frac{3}{7} + 1 \right) \\ &\quad + \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \log(\beta+3) + \frac{2\gamma + \log 2 + 1}{\pi} + 1 \\ &= \frac{3}{\pi} \log(\beta+3) + \frac{12}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{20}{7} \right) \end{aligned} \quad (2.257)$$

We finish the proof by using a similar argument after (2.220) to get

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha - \eta \leq \nu < \beta + \eta} \int_a^b e^{2\pi i(f(x) - \nu x)} dx + \mathcal{J}_1 \log(\beta - \alpha + 4) + \mathcal{J}'_2 \quad (2.258)$$

where

$$|\mathcal{J}_1| \leq \frac{3}{\pi}; \quad |\mathcal{J}'_2| \leq \frac{12}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} + \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{20}{7} \right)$$

□

Theorem 2.31. *Let $f(x)$ be a real function with derivatives up to the third order. Let $f'(x)$ be steadily decreasing in $[a, b]$ and $f'(b) = \alpha, f'(a) = \beta$. Let x_ν be defined by*

$$f'(x_\nu) - \nu = 0 \quad (\alpha < \nu \leq \beta) \quad (2.259)$$

Let

$$\lambda_2 \leq |f''(x)| \leq \mathcal{A}_2 \lambda_2, \quad |f'''(x)| \leq \mathcal{A}_3 \lambda_3 \quad (2.260)$$

Then

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = e^{-\pi i/4} \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i(f(x_\nu) - \nu x_\nu)}}{\sqrt{f''(x_\nu)}} + \mathcal{P}_1 \lambda_2^{-1/2} + \mathcal{P}_2 \log(\beta - \alpha + 4) + \mathcal{P}_3 (b - a) \lambda_2^{1/2} \lambda_3^{1/5} + \mathcal{P}_4 \quad (2.261)$$

where

$$|\mathcal{P}_1| \leq \frac{40}{\sqrt{\pi}}; \quad |\mathcal{P}_2| \leq \frac{3 + 2\mathcal{A}_2}{\pi}; \quad |\mathcal{P}_3| \leq \Psi_1 \mathcal{A}_2 \mathcal{A}_3^{1/5}; \quad |\mathcal{P}_4| \leq \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{20}{7} \right) \quad (2.262)$$

where Ψ_1 is defined in (2.119).

Proof. We first start by applying Corollary 2.30 to transform the exponential sum into sum of exponential integrals

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha - \eta \leq \nu < \beta + \eta} \int_a^b e^{2\pi i(f(x) - \nu x)} dx + \mathcal{J}_1 \log(\beta - \alpha + 2) + \mathcal{J}'_2 \quad (2.263)$$

We now estimate the integral inside the sum on the right hand side of (2.263). Next, we shift the range of summation for ν from $[\alpha - \eta, \beta + \eta]$ to $(\alpha + 1, \beta - 1)$ and as a result pick up at most two points at each endpoint. To estimate the contributions of each of these points we note that $|f''(x)| \geq \lambda_2$. Thus, using Lemma 2.23 we find that the contribution of each of these points is at most $\frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}}$. Using this we modify (2.263) as follows:

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha + 1 < \nu < \beta - 1} \int_a^b e^{2\pi i(f(x) - \nu x)} dx + \mathcal{J}_3 \frac{1}{\sqrt{\lambda_2}} + \mathcal{J}_1 \log(\beta - \alpha + 2) + \mathcal{J}'_2 \quad (2.264)$$

where $|\mathcal{J}_3| \leq \frac{16}{\sqrt{\pi}}$. Furthermore, since $\mathcal{J}'_2$ contains a term involving $1/\sqrt{\lambda_2}$ we combine $\mathcal{J}_3$ and $\mathcal{J}'_2$ to modify (2.264) as

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = \sum_{\alpha + 1 < \nu < \beta - 1} \int_a^b e^{2\pi i(f(x) - \nu x)} dx + \mathcal{J}_4 \frac{1}{\sqrt{\lambda_2}} + \mathcal{J}_1 \log(\beta - \alpha + 4) + \mathcal{P}_4 \quad (2.265)$$

where

$$|\mathcal{J}_4| \leq |\mathcal{J}_3| + \frac{12}{\sqrt{\pi}} = \frac{28}{\sqrt{\pi}}; \quad |\mathcal{J}_4| \leq \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{20}{7} \right) \quad (2.266)$$

Next, to estimate the sum of the integrals on the right hand side in (2.264) we apply Lemma 2.25 and use triangle inequality to get

$$\begin{aligned} \sum_{\alpha+1 < \nu < \beta-1} \int_a^b e^{2\pi i(f(x)-\nu x)} dx &= \sum_{\alpha+1 < \nu < \beta-1} \frac{e^{2\pi i(f(x_\nu)-\nu x_\nu-1/8)}}{\sqrt{|f''(x_\nu)|}} + \sum_{\alpha+1 < \nu < \beta-1} \mathcal{J}_1(\lambda_2, \lambda_3) \\ &\quad + \sum_{\alpha+1 < \nu < \beta-1} \mathcal{J}_2(a, b, \lambda_2) \end{aligned} \quad (2.267)$$

We have that the length of the second sum on the right hand side in (2.267) is atmost $\beta - \alpha - 1 \leq \beta - \alpha$ and hence using (2.119), that second sum in (2.267) can be bounded by

$$|\mathcal{J}_1(\lambda_2, \lambda_3)| \sum_{\alpha+1 < \nu < \beta-1} 1 \leq \Psi_1 \mathcal{A}_3^{1/5} (\beta - \alpha) \lambda_2^{-4/5} \lambda_3^{1/5} \quad (2.268)$$

And to bound the third sum on the right hand side in (2.267) we use the bound for $|\mathcal{J}_2(a, b, \lambda_2)|$ derived in (2.118) and using $(f(x) - \nu x)' = f'(x) - \nu$ we get

$$\begin{aligned} \sum_{\alpha+1 < \nu < \beta-1} |\mathcal{J}_2(a, b, \lambda_2)| &\leq \sum_{\alpha+1 < \nu < \beta-1} \left(\min \left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(a) - \nu|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \right) \right. \\ &\quad \left. + \min \left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(b) - \nu|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \right) \right) \end{aligned} \quad (2.269)$$

Now let us bound the first sum on the right hand side in (2.269) by using $f'(a) = \beta$ and the fact that $|f'(a) - \nu| = \beta - \nu$ we have

$$\sum_{\alpha+1 < \nu < \beta-1} \min \left(\frac{1}{\pi} \frac{\mathcal{A}_2}{|f'(a) - \nu|}, \frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}} \right) \leq \frac{\mathcal{A}_2}{\pi} \sum_{\alpha+1 < \nu < \beta-1} \frac{1}{\beta - \nu} \quad (2.270)$$

Recognizing the sum on the right hand side in (2.270) is a harmonic sum which can be written as

$$\frac{\mathcal{A}_2}{\pi} \sum_{\alpha+1 < \nu < \beta-1} \frac{1}{\beta - \nu} \leq \frac{\mathcal{A}_2}{\pi} \sum_{1 \leq \mu_4 < \lceil \beta - \alpha \rceil - 1} \frac{1}{\mu_4} \quad (2.271)$$

Similarly we can obtain a bound for the second sum on the right hand side in (2.269) by noting that $f'(b) = \alpha$ and $|f'(b) - \nu| = \nu - \alpha$ and recognizing the resulting sum is a harmonic sum as before to get

$$\frac{\mathcal{A}_2}{\pi} \sum_{\alpha+1 < \nu < \beta-1} \frac{1}{\nu - \alpha} \leq \frac{\mathcal{A}_2}{\pi} \sum_{1 \leq \mu_5 < \lceil \beta - \alpha \rceil - 1} \frac{1}{\mu_5} \quad (2.272)$$

Note that the harmonic sums in (2.271) and (2.272) are non-empty and in turn yield a contribution only if $\beta - \alpha \geq 2$ since $\lceil \beta - \alpha \rceil - 1 \geq \beta - \alpha - 1 \geq 1$. Now the harmonic sums can be bounded using Lemma 2.1. Take for example,

$$\sum_{1 \leq \mu_4 < \lceil \beta - \alpha \rceil - 1} \frac{1}{\mu_4} = \psi(\lceil \beta - \alpha \rceil - 1) + \gamma \leq \log(\beta - \alpha) + \gamma \quad (2.273)$$

where the last inequality above follows by applying Lemma 2.1 and $\lceil x \rceil \leq x + 1$. Hence, substituting the bound obtained by carrying out the computation done for (2.273) in (2.271) and (2.272) we can bound (2.269) as follows

$$\sum_{\alpha+1 < \nu < \beta-1} |\mathcal{J}_2(a, b, \lambda_2)| \leq \frac{2}{\pi} \mathcal{A}_2 \log(\beta - \alpha) + \frac{2}{\pi} \mathcal{A}_2 \gamma \quad (2.274)$$

And now substituting (2.268) and (2.274) in (2.267)

$$\begin{aligned} \sum_{\alpha+1 < \nu < \beta-1} \int_a^b e^{2\pi i(f(x) - \nu x)} dx &= \sum_{\alpha+1 < \nu < \beta-1} \frac{e^{2\pi i(f(x_\nu) - \nu x_\nu - 1/8)}}{\sqrt{|f''(x_\nu)|}} + \theta_1 \Psi_1 \mathcal{A}_3^{1/5} (\beta - \alpha) \lambda_2^{-4/5} \lambda_3^{1/5} \\ &\quad + \theta_2 \frac{2}{\pi} \mathcal{A}_2 \log(\beta - \alpha) + \theta_2 \frac{2}{\pi} \mathcal{A}_2 \gamma \end{aligned} \quad (2.275)$$

where $|\theta_1|, |\theta_2| \leq 1$.

Finally, to bound $\beta - \alpha$ in terms of $b - a$ we use the second mean value theorem to get

$$\frac{f'(a) - f'(b)}{b - a} = |f''(\xi)| \leq \mathcal{A}_2 \lambda_2 \quad (2.276)$$

and hence plugging (2.276) in (2.275)

$$\begin{aligned} \sum_{\alpha+1 < \nu < \beta-1} \int_a^b e^{2\pi i(f(x) - \nu x)} dx &= \sum_{\alpha+1 < \nu < \beta-1} \frac{e^{2\pi i(f(x_\nu) - \nu x_\nu - 1/8)}}{\sqrt{|f''(x_\nu)|}} + \theta_1 \Psi_1 \mathcal{A}_2 \mathcal{A}_3^{1/5} (b - a) \lambda_2^{1/5} \lambda_3^{1/5} \\ &\quad + \theta_2 \frac{2}{\pi} \mathcal{A}_2 \log(\beta - \alpha) + \theta_2 \frac{2}{\pi} \mathcal{A}_2 \gamma \end{aligned} \quad (2.277)$$

Using $\log(\beta - \alpha) \leq \log(\beta - \alpha + 4)$ and then substituting (2.277) in (2.265) we have

$$\begin{aligned} \sum_{a < n \leq b} e^{2\pi i f(n)} &= \sum_{\alpha+1 < \nu < \beta-1} \frac{e^{2\pi i(f(x_\nu) - \nu x_\nu - 1/8)}}{\sqrt{|f''(x_\nu)|}} + \theta_1 \Psi_1 \mathcal{A}_2 \mathcal{A}_3^{1/5} (b - a) \lambda_2^{1/5} \lambda_3^{1/5} \\ &\quad + \theta_2 \frac{2}{\pi} \mathcal{A}_2 \log(\beta - \alpha) + \theta_2 \frac{2}{\pi} \mathcal{A}_2 \gamma + \mathcal{J}_4 \frac{1}{\sqrt{\lambda_2}} + \mathcal{J}_1 \log(\beta - \alpha + 4) + \mathcal{P}_4 \end{aligned} \quad (2.278)$$

Using similar analysis done in moving the range of summation from $[\alpha - \eta, \beta + \eta]$ to $(\alpha + 1, \beta - 1)$, we now move the range of summation from $(\alpha + 1, \beta - 1)$ to $(\alpha, \beta]$. In

doing so, we pick up atmost one point at the left-end point and atmost two points at the right-end point. And the contribution of each of these points as before is at most $\frac{4}{\sqrt{\pi}} \frac{1}{\sqrt{\lambda_2}}$ which we then add these contribution to the fifth term on the right hand side in (2.278) and eventually obtain

$$\sum_{a < n \leq b} e^{2\pi i f(n)} = e^{-\pi i/4} \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i(f(x_\nu) - \nu x_\nu)}}{\sqrt{f''(x_\nu)}} + \rho_1 \lambda_2^{-1/2} + \rho_2 \log(\beta - \alpha + 4) + \rho_3 (b - a) \lambda_2^{1/2} \lambda_3^{1/5} + \rho_4 \quad (2.279)$$

where

$$\begin{aligned} |\rho_1| &\leq |\jmath_4| + \frac{12}{\sqrt{\pi}} = \frac{40}{\sqrt{\pi}}; & |\rho_2| &\leq |\jmath_1| + \frac{2\mathcal{A}_2}{\pi} \leq \frac{3}{\pi} + \frac{2\mathcal{A}_2}{\pi} \\ |\rho_3| &\leq \Psi_1 \mathcal{A}_2 \mathcal{A}_3^{1/5}; & |\rho_4| &\leq \frac{1}{\pi} \left(4\gamma + \log 2 + \pi + \frac{20}{7} \right). \end{aligned} \quad (2.280)$$

□

CHAPTER 3

SUB-WEYL ESTIMATE FOR $|\zeta(1/2 + it)|$

3.1 Main Result

First, we state the Riemann–Siegel sum used to bound $|\zeta(1/2 + it)|$ which reduces the problem to bounding an exponential sum of about $\sqrt{t}$ length. This result can be found in [Hia16, Lemma 2.1].

Lemma 3.1. *If $t \geq 200$ and $N = \lfloor \sqrt{t/(2\pi)} \rfloor$ then*

$$|\zeta(1/2 + it)| \leq 2 \left| \sum_{n=1}^N n^{-1/2 - it} \right| + \mathcal{R}(t), \quad (3.1)$$

where $\mathcal{R}(t) = 1.48t^{-\frac{1}{4}} + 0.127t^{-\frac{3}{4}}$.

Using the above result as a starting point we obtain the following theorem:

Theorem 3.2 (sub-Weyl Bound). *If $t \geq 3$, then*

$$|\zeta(1/2 + it)| \leq 307.098t^{27/164}.$$

Before we begin the proof of Theorem 3.2 we list the main steps taken in this proof. Recall that the $\{A, B\}$ –words are applied to exponent pairs right-to-left, as in function composition, so $AB(0, 1) = A(B(0, 1))$. On the other hand, if one carries out the corresponding proof explicitly, then the $\{A, B\}$ –words are applied left-to-right. So the order of steps in the proof of our explicit version of the process ABA^3B is:

- (i) Transform the sum obtained from (3.1) using A (explicit Weyl differencing) and this is carried out in equation (3.44)
- (ii) Transform the sums resulting from (i) using B (explicit truncated Poisson summation) which is carried out in equation (3.55), and
- (iii) Transform the sum resulting from (ii) using A^3B (explicit fifth derivative test) and this is done in equation (3.90).

3.2 Proof of Theorem 3.2

The main sum in (3.1) is divided into four smaller pieces namely,

$$\begin{aligned}
2 \left| \sum_{n=1}^{M_2} n^{-1/2-it} \right| &\leq 2 \left| \sum_{n=1}^m n^{-1/2-it} \right| + 2 \left| \sum_{n=m+1}^{\lfloor jt^{1/4} \rfloor - \delta_0} n^{-1/2-it} \right| \\
&\quad + 2 \left| \sum_{n=\lceil jt^{1/4} \rceil}^{\lfloor t^{7/17} \rfloor} n^{-1/2-it} \right| + 2 \left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{t/2\pi} \rfloor} n^{-1/2-it} \right|
\end{aligned} \tag{3.2}$$

where $m, j \in \mathbb{Z}_{>0}$ to be chosen later and

$$\delta_0 := \begin{cases} 1 & jt^{1/4} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

The first sum above in the range $1 \leq n \leq m$ is estimated trivially using the triangle inequality and by direct numerical computation. The second sum in the range $m+1 \leq n \leq \lfloor jt^{1/4} \rfloor - \delta_0$ is estimated using the triangle inequality and then bounding it with an integral. Lastly, the third and fourth sum in the ranges, $\lceil jt^{1/4} \rceil \leq n \leq \lfloor t^{7/17} \rfloor$ and $\lceil t^{7/17} \rceil \leq n \leq \lfloor \sqrt{t/2\pi} \rfloor$ respectively are divided into dyadic pieces.

In the sections that follow, namely, in Section 3.2.1 we carry out the computations to bound the third sum using the explicit fourth derivative test, that is Lemma 2.14 (Explicit A^2B Process). And in Section 3.2.2 we bound the fourth sum where we use Weyl differencing given by Lemma 2.6, explicit truncated poisson summation formula stated in Lemma 2.31 and explicit fifth derivative test given by Lemma 2.15 (Explicit ABA^3B Process).

We start by estimating the second sum on the right hand side in (3.2) using integral bound and this gives us

$$\begin{aligned}
2 \left| \sum_{n=m+1}^{\lfloor jt^{1/4} \rfloor - \delta_0} n^{-1/2-it} \right| &\leq 2 \sum_{n=m+1}^{\lfloor jt^{1/4} \rfloor - \delta_0} n^{-1/2} \leq 2 \int_m^{\lfloor jt^{1/4} \rfloor - \delta_0} \frac{1}{\sqrt{x}} dx \\
&\leq 4 \sqrt{\lfloor jt^{1/4} \rfloor - \delta_0 - 4m} \leq 4 \sqrt{j} t^{1/8} - 4m
\end{aligned} \tag{3.3}$$

After using the technical lemmas described above for the third and fourth sum we obtain

from triangle inequality, (3.3), (3.29), and (3.182) that

$$|\zeta(1/2 + it)| \leq 2 \left| \sum_{n=1}^{M_2} n^{-1/2-it} \right| + \mathcal{R}(t) \quad (3.4)$$

$$\begin{aligned} &\leq 2 \left| \sum_{n=1}^m n^{-1/2-it} \right| + 2 \left| \sum_{n=m+1}^{\lfloor jt^{1/4} \rfloor - \delta_0} n^{-1/2-it} \right| + 2 \left| \sum_{n=\lceil jt^{1/4} \rceil}^{\lfloor t^{7/17} \rfloor} n^{-1/2-it} \right| \\ &\quad + 2 \left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| + \mathcal{R}(t) \end{aligned} \quad (3.5)$$

$$\begin{aligned} &\leq 2 \sum_{n=1}^m n^{-1/2} + 4\sqrt{j}t^{1/8} - 4m + A_1 t^{19/119} + A_2 t^{-1/8} + A_3 t^{295/1904} + A_4 t^{-3/16} \\ &\quad + 2B_3 t^{\frac{27}{164}} + 2H_3 t^{\frac{175}{1312}} + 2\mathcal{N}_1 t^{27/328} + 2\mathcal{N}_2 t^{95/697} + 2\mathcal{N}_3 \log^{3/2}(t) + 2\mathcal{N}_4 \log t \\ &\quad + 2F_3 t^{443/2788} + 2F_4 t^{411/2788} + \mathcal{R}(t_0) \end{aligned} \quad (3.6)$$

Eventually we choose $m = 10000, j = 5, t \geq t_0 = 4.41 \times 10^{105}$ and obtain the numerical value of the constants in (3.6) from (3.209) and (3.213).

Finally to find the sub-Weyl constant, we label the numerical estimate obtained for $|\zeta(1/2 + it)|$ in (3.6) as $(*)$ and solve: $(*) - xt^{27/164} \leq 0$ for x . Using Mathematica, first we obtain that at $t_0 = 4.41 \times 10^{105}$ and $x = 307.098$ we have $(*) - xt^{27/164} = 0$. And then we also verify that $(*) - 307.098t^{27/164} < 0$ for all $t > t_0 = 4.41 \times 10^{105}$.

For $3 \leq t \leq 4.41 \times 10^{105}$ we have the van der Corput bound:

$$|\zeta(1/2 + it)| \leq 0.77t^{1/6} \log t \quad (3.7)$$

which is sharper than our sub-Weyl constant for that range.

In the remaining sections let us carry out our computation for explicitly bounding the third and fourth sum on the right hand side in (3.2)

3.2.1 Third Sum Computation Using Explicit A^2B Process

Lemma 3.3. *For $t \geq 200$, we have*

$$2 \left| \sum_{n=\lceil jt^{1/4} \rceil}^{\lfloor t^{7/17} \rfloor} n^{-1/2-it} \right| \leq A_1 t^{19/119} + A_2 t^{-1/8} + A_3 t^{295/1904} + A_4 t^{-3/16}$$

where

$$A_1 = \mathcal{C}_1 \frac{(1+\epsilon)^{3/14}}{j^{3/14}((1+\epsilon)^{3/14} - 1)}, \quad A_2 = \mathcal{C}_2 \frac{(1+\epsilon)^{11/14}}{(1+\epsilon)^{11/14} - 1}, \quad (3.8)$$

$$A_3 = \mathcal{C}_3 \frac{(1+\epsilon)^{4/7}}{j^{4/7}((1+\epsilon)^{4/7} - 1)}, \quad A_4 = \mathcal{C}_4 \frac{(1+\epsilon)^{3/7}}{(1+\epsilon)^{3/7} - 1}. \quad (3.9)$$

Proof. The dyadic subdivision for the third sum on the right hand side in (3.2) is carried out in the following manner and then using triangle inequality subject to $\epsilon > \frac{3}{jt^{1/4}}$:

$$2 \left| \sum_{n=\lceil jt^{1/4} \rceil}^{\lfloor t^{7/17} \rfloor} n^{-1/2-it} \right| \leq 2 \sum_{r=0}^{R(\epsilon)-1} \left| \sum_{n=\lceil (1+\epsilon)^r jt^{1/4} \rceil}^{N_1} n^{-1/2-it} \right| \quad (3.10)$$

where $N_1 = \min\{\lfloor t^{7/17} \rfloor, \lfloor (1+\epsilon)^{r+1} jt^{1/4} \rfloor - \delta_r\}$ and

$$\delta_r := \begin{cases} 1 & (1+\epsilon)^{r+1} jt^{1/4} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

Here $R(\epsilon)$ is the largest integer such that $\lceil (1+\epsilon)^{R(\epsilon)-1} jt^{1/4} \rceil \leq \lfloor t^{7/17} \rfloor$. So,

$$R(\epsilon) \leq \left\lfloor \frac{\frac{11}{68} \log t - \log j}{\log(1+\epsilon)} \right\rfloor + 1 \quad (3.11)$$

along with $R(\epsilon) \geq 1$ because as later on we will see that based on our choice of j we get $t \geq j^{68/11}$. Essentially, $R(\epsilon)$ gives the number of dyadic pieces of our main sum obtained using the parameter ϵ . This bound on $R(\epsilon)$ is obtained using the following inequality and solving for $R(\epsilon)$:

$$(1+\epsilon)^{R(\epsilon)-1} jt^{1/4} \leq \lceil (1+\epsilon)^{R(\epsilon)-1} jt^{1/4} \rceil \leq \lfloor t^{7/17} \rfloor \leq t^{7/17}.$$

To bound the right hand side in (3.10) we first use Lemma 2.4 and thus we get

$$S_3 := 2 \sum_{r=0}^{R(\epsilon)-1} \left| \sum_{n=\lceil (1+\epsilon)^r jt^{1/4} \rceil}^{N_1} n^{-1/2-it} \right| \leq 2 \sum_{r=0}^{R(\epsilon)-1} \frac{1}{\sqrt{\lceil (1+\epsilon)^r jt^{1/4} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon)^r jt^{1/4} \rceil}^{\lceil (1+\epsilon)^{r+1} jt^{1/4} \rceil + \Delta - 1} n^{-it} \right| \quad (3.12)$$

where $L := L_r = \lfloor (1+\epsilon)^{r+1} jt^{1/4} \rfloor - \delta_r - \lceil (1+\epsilon)^r jt^{1/4} \rceil + 1$. Here $L > 1$ since $\epsilon > \frac{3}{jt^{1/4}}$.

We will apply explicit fourth derivative test given by Lemma 2.14 to the inner sum on the right hand side in (3.12). To do so, first note,

$$f(x) := \frac{-t}{2\pi} \log x, \quad \lceil (1+\epsilon)^r jt^{1/4} \rceil \leq x \leq \lceil (1+\epsilon)^{r+1} jt^{1/4} \rceil + (\Delta - 1),$$

then

$$f^{(4)}(x) := \frac{3t}{\pi x^4}, \quad \lceil (1+\epsilon)^r jt^{1/4} \rceil \leq x \leq \lceil (1+\epsilon)^{r+1} jt^{1/4} \rceil + (\Delta - 1).$$

From this we have using $1 \leq \Delta \leq L$ and the definition of L ,

$$\frac{3t}{\pi(\lceil (1+\epsilon)^r jt^{1/4} \rceil + \Delta - 1)^4} \leq |f^{(4)}(x)| \leq \frac{3t}{\pi \lceil (1+\epsilon)^r jt^{1/4} \rceil^4}$$

Therefore,

$$\frac{3t}{\pi(\lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor)^4} \leq |f^{(4)}(x)| \leq \frac{3t}{\pi\lceil(1+\epsilon)^r jt^{1/4}\rceil^4} \quad (3.13)$$

Thus, now defining

$$W := W_r = \frac{\pi(\lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor)^4}{3t}, \quad \lambda := \lambda_r = \frac{(\lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor)^4}{(\lceil(1+\epsilon)^r jt^{1/4}\rceil)^4}$$

we write (3.13) as

$$\frac{1}{W} \leq |f^{(4)}(x)| \leq \lambda \frac{1}{W}$$

Noting $\Delta \leq L$ and $\lambda \geq 1$ we apply explicit fourth derivative test Lemma 2.14 to the sum on the right hand side of (3.12) and get

$$S_3 \leq 2 \sum_{r=0}^{R(\epsilon)-1} \frac{1}{\sqrt{\lceil(1+\epsilon)^r jt^{1/4}\rceil}} \left(\alpha_4 L^2 W^{-1/7} + \gamma_4 L^{3/2} W^{1/7} + \beta_4 L W^{2/7} + \eta_4 \alpha_4 L \right. \\ \left. + \eta_4 \gamma_4 \sqrt{L} W^{2/7} + \eta_4 \beta_4 W^{3/7} \right)^{1/2} \quad (3.14)$$

Factoring $L^2 W^{-1/7}$ we get

$$S_3 \leq 2 \sum_{r=0}^{R(\epsilon)-1} \frac{1}{\sqrt{\lceil(1+\epsilon)^r jt^{1/4}\rceil}} \sqrt{L^2 W^{-1/7}} \left(\alpha_4 + \gamma_4 \frac{W^{2/7}}{L^{1/2}} + \beta_4 \frac{W^{3/7}}{L} + \eta_4 \alpha_4 \frac{W^{1/7}}{L} \right. \\ \left. + \eta_4 \gamma_4 \frac{W^{3/7}}{L^{3/2}} + \eta_4 \beta_4 \frac{W^{4/7}}{L^2} \right)^{1/2} \quad (3.15)$$

Next let us bound $L^2 W^{-1/7}$ using the definition of L and W , and the fact that $\delta_r \geq 0$ and $\lfloor x \rfloor \leq x$, $\lceil x \rceil \geq x$:

$$L = \lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor - \delta_r - \lceil(1+\epsilon)^r jt^{1/4}\rceil + 1 \\ \leq \epsilon(1+\epsilon)^r jt^{1/4} \psi_{\epsilon,r,t} \quad (3.16)$$

where

$$\psi_{\epsilon,r,t} := 1 + \frac{1}{\epsilon(1+\epsilon)^r jt^{1/4}} \quad (3.17)$$

With this at hand we now bound $L^2 W^{-1/7}$ from above using $\lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor \geq \lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor - \delta_r \geq \lceil(1+\epsilon)^r jt^{1/4}\rceil \geq (1+\epsilon)^r jt^{1/4}$

$$L^2 W^{-1/7} \leq L \epsilon (1+\epsilon)^r jt^{1/4} \psi_{\epsilon,r,t} \left(\frac{\pi(\lfloor(1+\epsilon)^{r+1}jt^{1/4}\rfloor)^4}{3t} \right)^{-1/7} \\ \leq L \epsilon \psi_{\epsilon,r,t} \left(\frac{3}{\pi} \right)^{1/7} j^{3/7} (1+\epsilon)^{3r/7} t^{1/4} \quad (3.18)$$

Substituting (3.18) in (3.15) and simplifying along with the inequality $\sqrt{x_1 + x_2 + \dots} \leq \sqrt{x_1} + \sqrt{x_2} + \dots$ for $x_1, x_2, \dots \geq 0$ we have

$$S_3 \leq \frac{2\epsilon 3^{1/14} j^{3/14} t^{1/8}}{\pi^{1/14}} \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{3r/14} \psi_{\epsilon,r,t} \sqrt{\alpha_4} + \frac{2\epsilon 3^{1/14} j^{3/14} t^{1/8}}{\pi^{1/14}} \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{3r/14} \psi_{\epsilon,r,t} \mathcal{B}_r \quad (3.19)$$

where

$$\mathcal{B}_r = \sqrt{\gamma_4 \frac{W^{2/7}}{L^{1/2}} + \beta_4 \frac{W^{3/7}}{L} + \eta_4 \alpha_4 \frac{W^{1/7}}{L} + \eta_4 \gamma_4 \frac{W^{3/7}}{L^{3/2}} + \eta_4 \beta_4 \frac{W^{4/7}}{L^2}} \quad (3.20)$$

Now let us focus on bounding $\mathcal{B}_r$ from above. To do so we will first find good upper and lower bounds for W and L respectively.

$$W = \frac{\pi \lfloor (1+\epsilon)^{r+1} j t^{1/4} \rfloor^4}{3t} \leq \frac{\pi}{3} (1+\epsilon)^4 j^4 (1+\epsilon)^{4r} \quad (3.21)$$

and using $\lfloor x \rfloor \geq x - 1$, $\lceil x \rceil \leq x + 1$, $(1+\epsilon)^r \geq 1$ and $\delta_r \leq 1$ we obtain

$$\begin{aligned} L &= \lfloor (1+\epsilon)^{r+1} j t^{1/4} \rfloor - \delta_r - \lceil (1+\epsilon)^r j t^{1/4} \rceil + 1 \\ &\geq (1+\epsilon)^r j t^{1/4} \left(\epsilon - \frac{2}{(1+\epsilon)^r j t^{1/4}} \right) \\ &\geq \phi_{\epsilon,j,t} (1+\epsilon)^r j t^{1/4} \end{aligned} \quad (3.22)$$

where $\phi_{\epsilon,j,t} = \left(\epsilon - \frac{2}{j t^{1/4}} \right)$

Now using (3.21) and (3.22) we can obtain a bound for the following quantity for $a, b > 0$ to be

$$\frac{W^{a/7}}{L^b} \leq \left(\frac{\pi}{3} \right)^{a/7} \frac{(1+\epsilon)^{4a/7} j^{4a/7-b}}{\phi_{\epsilon,j,t}^b} (1+\epsilon)^{(4a/7-b)r} \frac{1}{t^{b/4}} \quad (3.23)$$

Using (3.23) we can bound the following quantities

$$\frac{W^{2/7}}{L^{1/2}}, \frac{W^{3/7}}{L}, \frac{W^{1/7}}{L}, \frac{W^{3/7}}{L^{3/2}}, \frac{W^{4/7}}{L^2} \quad (3.24)$$

by substituting $a = 1, 2, 3, 4$ and $b = 1/2, 1, 3/2, 2$ and we finally obtain the bound for $\mathcal{B}_r$ of the form

$$\begin{aligned} \mathcal{B}_r &\leq \left(\gamma_4 \left(\frac{\pi}{3} \right)^{2/7} (1+\epsilon)^{8/7} j^{9/14} \frac{(1+\epsilon)^{9r/14}}{\sqrt{\phi_{\epsilon,j,t} t^{1/8}}} + \beta_4 \left(\frac{\pi}{3} \right)^{3/7} (1+\epsilon)^{12/7} j^{5/7} \frac{(1+\epsilon)^{5r/7}}{\phi_{\epsilon,j,t} t^{1/4}} \right. \\ &\quad + \eta_4 \alpha_4 \left(\frac{\pi}{3} \right)^{1/7} (1+\epsilon)^{4/7} j^{-3/7} \frac{1}{\phi_{\epsilon,j,t} (1+\epsilon)^{3r/7} t^{1/4}} + \eta_4 \gamma_4 \left(\frac{\pi}{3} \right)^{3/7} (1+\epsilon)^{12/7} j^{3/14} \frac{(1+\epsilon)^{3r/14}}{\phi_{\epsilon,j,t}^{3/2} t^{3/8}} \\ &\quad \left. + \eta_4 \beta_4 \left(\frac{\pi}{3} \right)^{4/7} (1+\epsilon)^{16/7} j^{2/7} \frac{(1+\epsilon)^{2r/7}}{\phi_{\epsilon,j,t}^2 t^{1/2}} \right)^{1/2} \end{aligned}$$

At this stage we factor $\frac{(1+\epsilon)^{5r/7}}{t^{1/8}}$ from the right hand side above and use $(1+\epsilon) \geq 1$ and get a simplified bound for $\mathcal{B}_r$ of the form

$$\mathcal{B}_r \leq c(\alpha_4, \beta_4, \gamma_4, \epsilon, j, t)(1+\epsilon)^{5r/14}t^{-1/16} \quad (3.25)$$

where

$$\begin{aligned} c(\alpha_4, \beta_4, \gamma_4, \epsilon, j, t) := & \left(\gamma_4 \left(\frac{\pi}{3} \right)^{2/7} (1+\epsilon)^{8/7} j^{9/14} \frac{1}{\sqrt{\phi_{\epsilon,j,t}}} + \beta_4 \left(\frac{\pi}{3} \right)^{3/7} (1+\epsilon)^{12/7} j^{5/7} \frac{1}{\phi_{\epsilon,j,t}^{1/8}} \right. \\ & + \eta_4 \alpha_4 \left(\frac{\pi}{3} \right)^{1/7} (1+\epsilon)^{4/7} j^{-3/7} \frac{1}{\phi_{\epsilon,j,t}^{1/8}} + \eta_4 \gamma_4 \left(\frac{\pi}{3} \right)^{3/7} (1+\epsilon)^{12/7} j^{3/14} \frac{1}{\phi_{\epsilon,j,t}^{3/2} t^{1/4}} \\ & \left. + \eta_4 \beta_4 \left(\frac{\pi}{3} \right)^{4/7} (1+\epsilon)^{16/7} j^{2/7} \frac{1}{\phi_{\epsilon,j,t}^2 t^{3/8}} \right)^{1/2} \end{aligned}$$

Finally plugging in the simplified estimate for $\mathcal{B}_r$ from (3.25) in (3.19) we have

$$\begin{aligned} S_3 \leq & \left(\frac{3j^3}{\pi} \right)^{1/14} 2\epsilon \sqrt{\alpha_4} t^{1/8} \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{3r/14} \psi_{\epsilon,r,t} \\ & + \left(\frac{3j^3}{\pi} \right)^{1/14} 2\epsilon t^{1/16} c(\alpha_4, \beta_4, \gamma_4, \epsilon, j, t) \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{8r/14} \psi_{\epsilon,r,t} \end{aligned} \quad (3.26)$$

Now using the definition of $\psi_{\epsilon,r,t}$ from (3.17) and substituting it in (3.26) we get

$$\begin{aligned} S_3 \leq & \mathcal{C}_1 t^{1/8} \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{3r/14} + \mathcal{C}_2 t^{-1/8} \sum_{r=0}^{R(\epsilon)-1} \frac{1}{(1+\epsilon)^{11r/14}} \\ & + \mathcal{C}_3 t^{1/16} \sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{8r/14} + \mathcal{C}_4 t^{-3/16} \sum_{r=0}^{R(\epsilon)-1} \frac{1}{(1+\epsilon)^{3r/7}} \end{aligned} \quad (3.27)$$

where

$$\begin{aligned} \mathcal{C}_1 &= 2\epsilon \left(\frac{3j^3}{\pi} \right)^{1/14} \sqrt{\alpha_4}, \quad \mathcal{C}_2 = 2 \left(\frac{3}{\pi j^{11}} \right)^{1/14} \sqrt{\alpha_4}, \quad \mathcal{C}_3 = 2\epsilon \left(\frac{3j^3}{\pi} \right)^{1/14} c(\alpha_4, \beta_4, \gamma_4, \epsilon, j, t) \\ \mathcal{C}_4 &= 2 \left(\frac{3}{\pi j^{11}} \right)^{1/14} c(\alpha_4, \beta_4, \gamma_4, \epsilon, j, t) \end{aligned}$$

Now to bound the sums in (3.27) we derive the following inequalities that are valid for $c, d > 0$ and using (3.11)

$$\sum_{r=0}^{R(\epsilon)-1} (1+\epsilon)^{cr} \leq \frac{(1+\epsilon)^c}{j^c((1+\epsilon)^c - 1)} t^{11c/68}, \quad \sum_{r=0}^{R(\epsilon)-1} \frac{1}{(1+\epsilon)^{dr}} \leq \frac{(1+\epsilon)^d}{(1+\epsilon)^d - 1} \quad (3.28)$$

To bound the second and fourth sum in (3.27) we use the second inequality in (3.28) and get

$$2 \left| \sum_{n=\lceil jt^{1/4} \rceil}^{\lfloor t^{7/17} \rfloor} n^{-1/2-it} \right| \leq 2 \sum_{r=0}^{R(\epsilon)-1} \left| \sum_{n=\lceil (1+\epsilon)^r jt^{1/4} \rceil}^{N_1} n^{-1/2-it} \right| \leq A_1 t^{19/119} + A_2 t^{-1/8} + A_3 t^{295/1904} + A_4 t^{-3/16} \quad (3.29)$$

where

$$A_1 = \mathcal{C}_1 \frac{(1+\epsilon)^{3/14}}{j^{3/14}((1+\epsilon)^{3/14} - 1)}, \quad A_2 = \mathcal{C}_2 \frac{(1+\epsilon)^{11/14}}{(1+\epsilon)^{11/14} - 1} \quad (3.30)$$

$$A_3 = \mathcal{C}_3 \frac{(1+\epsilon)^{4/7}}{j^{4/7}((1+\epsilon)^{4/7} - 1)}, \quad A_4 = \mathcal{C}_4 \frac{(1+\epsilon)^{3/7}}{(1+\epsilon)^{3/7} - 1} \quad (3.31)$$

□

3.2.2 Fourth Sum Computation Using Explicit ABA^3B Process

In this section let us find a bound for the last sum on the right hand side in (3.2).

Lemma 3.4. *For $t \geq 200$ we have*

$$\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| \leq B_3 t^{\frac{27}{164}} + H_3 t^{\frac{175}{1312}} + \mathcal{N}_1 t^{27/328} + \mathcal{N}_2 t^{95/697} + \mathcal{N}_3 \log^{3/2}(t) + \mathcal{N}_4 \log t + F_3 t^{443/2788} + F_4 t^{411/2788} \quad (3.32)$$

where the constants $B_3, H_3, \mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_3, \mathcal{N}_4, F_3, F_4$ are listed in Section 3.2.4.

Proof. To bound the sum on the left hand side in (3.32) we first split it into dyadic pieces subject to $\epsilon_1 > \frac{3}{t^{7/17}}$:

$$\sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} = \sum_{j=0}^{J(\epsilon_1)-1} \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{N_2} n^{-1/2-it} \quad (3.33)$$

where $N_2 = \min\{\lfloor (1+\epsilon_1)^{j+1} t^{7/17} \rfloor - \delta_r, \lfloor \sqrt{t/2\pi} \rfloor\}$ and $J(\epsilon_1)$ is the largest integer such that $\lceil (1+\epsilon_1)^{J(\epsilon_1)-1} t^{7/17} \rceil \leq \lfloor \sqrt{t/2\pi} \rfloor$.

Next, applying partial summation using Lemma 2.4 along with triangle inequality we get

$$\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| \leq \sum_{j=0}^{J(\epsilon_1)-1} \frac{1}{\sqrt{\lceil (1+\epsilon_1)^j t^{7/17} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^{j+1} t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \quad (3.34)$$

where

$$L = \lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor - \delta_j - \lceil (1 + \epsilon_1)^j t^{7/17} \rceil + 1 \quad (3.35)$$

and since $\epsilon_1 > \frac{3}{t^{7/17}}$ we have $\lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor - \delta_j > \lceil (1 + \epsilon_1)^j t^{7/17} \rceil$. Also, here

$$J(\epsilon_1) \leq \left\lfloor \frac{\frac{3}{34} \log t - \log \sqrt{2\pi}}{\log(1 + \epsilon_1)} \right\rfloor + 1 \quad (3.36)$$

along with $J(\epsilon_1) \geq 1$ since we will use this estimate for $t \geq (2\pi)^{34/6}$. We note that the bound for $J(\epsilon_1)$ which gives the total number of dyadic pieces in the sum is obtained using

$$(1 + \epsilon_1)^{J(\epsilon_1)-1} t^{7/17} \leq \lceil (1 + \epsilon_1)^{J(\epsilon_1)-1} t^{7/17} \rceil \leq \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor \leq \sqrt{\frac{t}{2\pi}}. \quad (3.37)$$

Before proceeding, we split the range over j in (3.34) into two parts: $0 \leq j \leq J_1(\epsilon_1) - 1$ and $J_1(\epsilon_1) \leq j \leq J(\epsilon_1)$ where $J_1(\epsilon_1)$ is the largest integer such that $\lceil (1 + \epsilon_1)^{J_1(\epsilon_1)} t^{7/17} \rceil \leq \left\lfloor (t/u_1)^{4/9} \right\rfloor$ and

$$u_1 = \left(\frac{2688\pi^4}{25\sqrt{5}} \right)^{1/4} \quad (3.38)$$

Thus, we obtain

$$\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| \leq \left(\sum_{j=0}^{J_1(\epsilon_1)-1} + \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} \right) \frac{1}{\sqrt{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}^{\lceil (1 + \epsilon_1)^{j+1} t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \quad (3.39)$$

where

$$J_1(\epsilon_1) \leq \left\lfloor \frac{\frac{5}{153} \log t - \frac{4}{9} \log u_1}{\log(1 + \epsilon_1)} \right\rfloor \quad (3.40)$$

We consider a split such as (3.39) to ensure that applying the explicit fifth derivative test, Lemma 2.15 does not give us a worse bound than trivial bound. This remark is elaborated further in after equation (3.83) below.

Setting

$$M_1 = \sum_{j=0}^{J_1(\epsilon_1)-1} \frac{1}{\sqrt{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}^{\lceil (1 + \epsilon_1)^{j+1} t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \quad (3.41)$$

$$M_2 = \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} \frac{1}{\sqrt{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}^{\lceil (1 + \epsilon_1)^{j+1} t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \quad (3.42)$$

we rewrite (3.39) as

$$\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| \leq M_1 + M_2 \quad (3.43)$$

First, we bound the innermost sum in a way such that the resulting estimate is applicable for all relevant j . We do this by first applying Weyl differencing given by Lemma 2.6 and then taking square-roots along with the inequalities: $\sqrt{x_1 + x_2 + \cdots} \leq \sqrt{x_1} + \sqrt{x_2} + \cdots$ for $x_1, x_2, \cdots > 0$ and $\Delta \leq L$ to get

$$\begin{aligned} & \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \\ & \leq w_1 \frac{L}{q^{1/2}} + w_2 \max_{1 \leq \Delta \leq L} \left(\left(\frac{L}{q} \sum_{r=1}^{q-1} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r} e^{-it(\log(n+r) - \log n)} \right| \right)^{1/2} \right) \end{aligned} \quad (3.44)$$

where

$$1 \leq q \leq L$$

Note the double sum on the right hand side in (3.44) is empty when $\Delta - 1 - r < 0$. Hence we assume $\Delta - 1 - r \geq 0$ and obtain a bound for the sum on the left hand side of (3.44) which also holds in case that $\Delta - 1 - r < 0$. The constants w_1 and w_2 are the constants that appear because of applying the differencing Lemma 2.6. With this in mind, consider that innermost sum on the right hand side in (3.44), that is

$$\left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r} e^{-it(\log(n+r) - \log n)} \right| \quad (3.45)$$

and let us apply the truncated poisson summation result given by Theorem 2.31 to bound it. To do so, we first set

$$f(x) = \frac{t}{2\pi} (\log(x+r) - \log(x)), \quad \text{for } x \in [\lceil (1+\epsilon_1)^j t^{7/17} \rceil, \lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r] \quad (3.46)$$

which gives us

$$|f''(x)| = \left| \frac{tr}{2\pi} \frac{2x+r}{x^2(x+r)^2} \right| \quad (3.47)$$

One thing to note here is that the amount of variation of $f''(x)$ plays a significant role in controlling the main term in Theorem 2.31. Hence, let us bound $|f''(x)|$ from above and below using (3.47) and the range of x in (3.46). To do so we first bound the quantity:

$$\frac{2x+r}{x^2(x+r)^2} \quad (3.48)$$

from above and below. We achieve this by first noting that (3.48) is a decreasing function in $x, r > 0$. So, for $x \in [\lceil (1+\epsilon_1)^j t^{7/17} \rceil, \lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r]$ and $r \in [1, q -$

$1] \subset [1, \Delta - 1] \subset [1, L - 1]$, we conclude that the maximum value of (3.48) occurs when $r = 1$, $x = \lceil (1 + \epsilon_1)^j t^{7/17} \rceil$. Therefore, we have

$$\frac{2x + r}{x^2(x + r)^2} \leq \frac{2\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + 1}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2 (\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + 1)^2} \leq \frac{2}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^3} \quad (3.49)$$

Next, we find a lower bound for (3.48) using the inequalities:

$$\lceil (1 + \epsilon_1)^j t^{7/17} \rceil \leq x \leq \lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r, \quad 1 \leq r \leq q - 1, \quad 1 \leq \Delta \leq L \quad (3.50)$$

along with $\lfloor x \rfloor \leq x \leq \lceil x \rceil$

$$\begin{aligned} \frac{2x + r}{x^2(x + r)^2} &\geq \frac{2\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + 1}{(\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r)^2 (\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1)^2} \\ &\geq \frac{2}{1 + \epsilon_1} \frac{1}{\lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor^3} \end{aligned} \quad (3.51)$$

Hence gathering (3.49) and (3.51) and substituting them in (3.47) we get

$$\frac{tr}{(1 + \epsilon_1)\pi} \frac{1}{\lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor^3} \leq |f''(x)| \leq \frac{tr}{\pi} \frac{1}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^3} \quad (3.52)$$

Setting

$$a_2 = (1 + \epsilon_1) \frac{\lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor^3}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^3} \leq (1 + \epsilon_1)^4, \quad A_2 = \frac{tr}{(1 + \epsilon_1)\pi} \frac{1}{\lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor^3} \quad (3.53)$$

we rewrite (3.52) as

$$A_2 \leq |f''(x)| \leq a_2 A_2 \quad (3.54)$$

Next, applying explicit truncated poisson summation given in Lemma 2.31 to (3.45) with $\mathcal{A}_2 = a_2$ and $\lambda_2 = A_2$ we get

$$\left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r} e^{-it(\log(n+r) - \log n)} \right| \leq \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| + |\mathcal{P}(\epsilon_1, r, t)| \quad (3.55)$$

where

$$\phi(\nu) = f(x_\nu) - \nu x_\nu, \alpha = f'(\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r), \beta = f'(\lceil (1 + \epsilon_1)^j t^{7/17} \rceil) \quad (3.56)$$

and

$$\mathcal{P}(\epsilon_1, r, t) = \mathfrak{p}_1 A_2^{-1/2} + \mathfrak{p}_2 \log(\beta - \alpha + 2) + \mathfrak{p}_3 (\Delta - r) A_2^{1/5} A_3^{1/5} + \mathfrak{p}_4 \quad (3.57)$$

Note that

$$\nu = \frac{tr}{2\pi x_\nu(x_\nu + r)}, \quad \nu \in (\alpha, \beta], \quad x_\nu \in [\lceil (1 + \epsilon_1)^j t^{7/17} \rceil, \lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r] \quad (3.58)$$

Hence, substituting (3.55) in (3.44) and then using $\sqrt{x_1 + x_2 + \dots} \leq \sqrt{x_1} + \sqrt{x_2} + \dots$ for $x_1, x_2, \dots \geq 0$ we have that

$$\begin{aligned}
& \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \\
& \leq w_1 \frac{L}{q^{1/2}} + w_2 \max_{1 \leq \Delta \leq L} \left(\left(\frac{L}{q} \sum_{r=1}^{q-1} \left(\left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| + |\mathcal{P}(\epsilon_1, r, t)| \right) \right)^{1/2} \right) \\
& \leq w_1 \frac{L}{q^{1/2}} + w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=1}^{q-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \right)^{1/2} + w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=1}^{q-1} |\mathcal{P}(\epsilon_1, r, t)| \right)^{1/2}
\end{aligned} \tag{3.59}$$

At this stage, we split the sum in the second term in (3.59) over r into two ranges: $1 \leq r \leq \lceil u_1^4(1 + \epsilon_1)^{9j/4} t^{-5/68} \rceil - 1$ and $\lceil u_1^4(1 + \epsilon_1)^{9j/4} t^{-5/68} \rceil \leq r \leq q - 1$. Again, this is done to ensure that applying the fifth derivative test given in Lemma 2.15 does not yield a bound that is worse than the trivial bound and this will be elaborated later in the section. With this remark in mind and using $\sqrt{x_1 + x_2 + \dots} \leq \sqrt{x_1} + \sqrt{x_2} + \dots$ for $x_1, x_2, \dots > 0$, we bound (3.59) as follows:

$$\max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \leq S_{4,1} + S_{4,2} + \mathcal{R}_1 \tag{3.60}$$

where

$$S_{4,1} = w_1 \frac{L}{q^{1/2}} + w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=r_1}^{q-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \right)^{1/2} \tag{3.61}$$

$$S_{4,2} = w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=1}^{r_1-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \right)^{1/2} \tag{3.62}$$

$$\mathcal{R}_1 = w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=1}^{q-1} |\mathcal{P}(\epsilon_1, r, t)| \right)^{1/2} \tag{3.63}$$

where $r_1 = \lceil u_1^4(1 + \epsilon_1)^{9j/4} t^{-5/68} \rceil$.

Now we focus on bounding $S_{4,1}$ first and we start by estimating the inner sum in the second term in (3.61). We do so by applying partial summation, Lemma 2.4 since $|f''(x_\nu)|^{-1/2}$ is monotonic decreasing in ν and obtain

$$\left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq \frac{1}{\sqrt{f''(x_{\alpha_0})}} \max_{\alpha < \gamma \leq \beta} \left| \sum_{\alpha < \nu \leq \gamma} e^{2\pi i \phi(\nu)} \right| \tag{3.64}$$

where α_0 is the smallest integer such that $\alpha_0 > \alpha$. At this stage, we estimate the exponential sum on the right hand side in (3.64) using explicit fifth derivative, Lemma 2.15. To do so, we first note that $f'(x_\nu) = \nu$ and from (3.47) we see that the numbers x_ν are given by

$$\nu = \frac{tr}{2\pi x_\nu(x_\nu + r)}, \quad \text{for } x_\nu \in [\lceil(1 + \epsilon_1)^j t^{7/17}\rceil, \lceil(1 + \epsilon_1)^j t^{7/17}\rceil + \Delta - 1 - r] \quad (3.65)$$

and then solving for x_ν and rearranging we find that

$$x_\nu = \frac{1}{2} \left(r^2 + \frac{2tr}{\pi\nu} \right)^{1/2} - \frac{1}{2}r \quad (3.66)$$

Using (3.66) we can compute $\phi'(\nu)$ by noting the definition of $\phi(\nu)$ in (3.56) and that $f'(x_\nu) - \nu = 0$ we get

$$\phi'(\nu) = (f'(x_\nu) - \nu) \frac{dx_\nu}{d\nu} - x_\nu = -x_\nu = \frac{1}{2}r - \frac{1}{2} \left(r^2 + \frac{2tr}{\pi\nu} \right)^{1/2} \quad (3.67)$$

which we then use to find

$$\phi^{(5)}(\nu) = -\frac{3t\sqrt{r}}{2\sqrt{\pi\nu\nu^4}(\pi r\nu + 2t)^{7/2}} (8\pi^3 r^3 \nu^3 + 36\pi^2 r^2 t \nu^2 + 60\pi r \nu t^2 + 35t^3) \quad (3.68)$$

and after taking absolute values we get

$$|\phi^{(5)}(\nu)| = \frac{3t\sqrt{r}}{2\sqrt{\pi\nu\nu^4}(\pi r\nu + 2t)^{7/2}} (8\pi^3 r^3 \nu^3 + 36\pi^2 r^2 t \nu^2 + 60\pi r \nu t^2 + 35t^3) \quad (3.69)$$

Now, with (3.69) at hand, we will find upper and lower bounds for $|\phi^{(5)}(\nu)|$.

We start with the upper bound and to obtain this we first use the inequality $\pi r\nu + 2t > 2t$, write $t^{\frac{7}{2}} = t^{3+\frac{1}{2}}$ and plugging this in (3.69) gives us after some simplification

$$|\phi^{(5)}(\nu)| \leq \frac{3}{64\sqrt{2\pi}} \left(\frac{8\pi^3 r^3 \nu^3}{t^3} + \frac{36\pi^2 r^2 \nu^2}{t^2} + \frac{60\pi r \nu}{t} + 35 \right) \frac{(tr)^{1/2}}{\nu^{9/2}} \quad (3.70)$$

To further simplify (3.70) we note that

$$\begin{aligned} r\nu &\leq r\beta = rf'(\lceil(1 + \epsilon_1)^j t^{7/17}\rceil) = r \frac{tr}{2\pi \lceil(1 + \epsilon_1)^j t^{7/17}\rceil (\lceil(1 + \epsilon_1)^j t^{7/17}\rceil + r)} \\ &\leq \frac{t}{2\pi} \frac{r^2}{\lceil(1 + \epsilon_1)^j t^{7/17}\rceil^2} \leq \frac{t}{2\pi} \end{aligned} \quad (3.71)$$

The last inequality above follows because $r \leq q \leq \lceil(1 + \epsilon_1)^j t^{7/17}\rceil$. This is valid because from the choice of q that we make later in (3.103) we get that $\lceil(1 + \epsilon_1)^{36j/41} t^{65/697}\rceil \leq \lceil(1 + \epsilon_1)^j t^{7/17}\rceil$.

Similarly we find the lower bound for $|\phi^{(5)}(\nu)|$ using

$$\begin{aligned}
r\nu &\geq r\alpha = rf'(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1 - r) \\
&= r \frac{tr}{2\pi(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1 - r)(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1)} \\
&\geq \frac{t}{2\pi} \frac{r^2}{(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1)^2} \\
&\geq \frac{t}{2\pi} \frac{r^2}{(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + L - 1)^2} \\
&\geq \frac{t}{2\pi} \frac{r^2}{(\lfloor(1+\epsilon_1)^{j+1} t^{7/17}\rfloor - \delta_r)^2} \\
&\geq \frac{t}{2\pi} \frac{r^2}{((1+\epsilon_1)^{j+1} t^{7/17})^2} \\
&= \frac{t}{2\pi(1+\epsilon_1)^2} \frac{r^2}{((1+\epsilon_1)^j t^{7/17})^2} \\
&\geq \frac{t}{2\pi(1+\epsilon_1)^2} \frac{r^2}{\lceil(1+\epsilon_1)^j t^{7/17}\rceil^2} \\
&\geq \frac{t}{2\pi(1+\epsilon_1)^2} \frac{1}{\lfloor\sqrt{\frac{t}{2\pi}}\rfloor^2} \\
&\geq \frac{t}{2\pi(1+\epsilon_1)^2} \frac{2\pi}{t} \\
&\geq \frac{1}{(1+\epsilon_1)^2}
\end{aligned} \tag{3.72}$$

Hence combining we get (3.71) and (3.72) we have the following inequality

$$0 < \frac{\pi}{t(1+\epsilon_1)^2} \leq \frac{\pi r\nu}{t} \leq \frac{1}{2} \tag{3.73}$$

From (3.73) we get that

$$\pi r\nu + 2t \leq \frac{5}{2}t \tag{3.74}$$

and plugging (3.74) into (3.69) we get a lower bound for $|\phi^{(5)}(\nu)|$ of the form

$$|\phi^{(5)}(\nu)| \geq \frac{12\sqrt{2}}{125\sqrt{5}\pi} \left(\frac{8\pi^3 r^3 \nu^3}{t^3} + \frac{36\pi^2 r^2 \nu^2}{t^2} + \frac{60\pi r\nu}{t} + 35 \right) \frac{(tr)^{1/2}}{\nu^{9/2}} \tag{3.75}$$

Moreover, using (3.73) we simplify (3.70) and (3.75) and get

$$\frac{84\sqrt{2}}{25\sqrt{5}\pi} \frac{(tr)^{1/2}}{\nu^{9/2}} \leq |\phi^{(5)}(\nu)| \leq \frac{225}{64\sqrt{2}\pi} \frac{(tr)^{1/2}}{\nu^{9/2}} \tag{3.76}$$

At this stage, we further modify (3.76) by bounding ν from above and below. To do so, we use the fact that

$$x_\nu \in [\lceil(1+\epsilon_1)^j t^{7/17}\rceil, \lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1 - r] \tag{3.77}$$

along with the definition of ν from (3.58) to get

$$\begin{aligned}\nu &\leq \frac{tr}{2\pi} \frac{1}{(\lceil(1+\epsilon_1)^j t^{7/17}\rceil)(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + r)} \\ \nu &\geq \frac{tr}{2\pi} \frac{1}{(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1 - r)(\lceil(1+\epsilon_1)^j t^{7/17}\rceil + \Delta - 1)}\end{aligned}\quad (3.78)$$

Next, using the following inequalities

$$\Delta - 1 - r \leq \Delta - 1 \leq L - 1, \quad \lceil(1+\epsilon_1)^j t^{7/17}\rceil \leq \lceil(1+\epsilon_1)^j t^{7/17}\rceil + r \quad (3.79)$$

in (3.78) we further simplify it to obtain

$$\frac{tr}{2\pi(\lfloor(1+\epsilon_1)^{j+1} t^{7/17}\rfloor)^2} \leq \nu \leq \frac{tr}{2\pi(\lceil(1+\epsilon_1)^j t^{7/17}\rceil)^2} \quad (3.80)$$

from which we conclude

$$\left(\frac{2\pi}{tr}\right)^{9/2} \lceil(1+\epsilon_1)^j t^{7/17}\rceil^9 \leq \frac{1}{\nu^{9/2}} \leq \left(\frac{2\pi}{tr}\right)^{9/2} \lfloor(1+\epsilon_1)^{j+1} t^{7/17}\rfloor^9 \quad (3.81)$$

And substituting (3.81) in (3.76)

$$\frac{2688\pi^4}{25\sqrt{5}} \frac{\lceil(1+\epsilon_1)^j t^{7/17}\rceil^9}{(tr)^4} \leq |\phi^{(5)}(\nu)| \leq \frac{225\pi^4}{4} \frac{\lfloor(1+\epsilon_1)^{j+1} t^{7/17}\rfloor^9}{(tr)^4} \quad (3.82)$$

Setting

$$\lambda_5 = \frac{1875\sqrt{5}}{3584}(1+\epsilon_1)^9, \quad W_5 = \frac{25\sqrt{5}}{2688\pi^4} \frac{(tr)^4}{\lceil(1+\epsilon_1)^j t^{7/17}\rceil^9} \quad (3.83)$$

At this stage we make several important observations about the size of W_5 which is defined in (3.83). Notice that for $j \in \{0, 1, \dots, J(\epsilon_1) - 1\}$ we have we have the order of $\lceil(1+\epsilon_1)^j t^{7/17}\rceil^9$ ranging from $\mathcal{O}(t^{\frac{63}{17}}), \dots, \mathcal{O}(t^{\frac{9}{2}})$.

Consider the special case when r is of the order $\mathcal{O}(1)$ and $\lceil(1+\epsilon_1)^j t^{7/17}\rceil^9$ of the order $\mathcal{O}(t^{\frac{9}{2}})$, then we get W_5 of the order $\mathcal{O}(t^{-\frac{1}{2}})$. If we apply explicit fifth derivative test that is Lemma 2.15 in this case then we get a bound on the exponential sum in (3.64) which is worse than the trivial bound since $W_5 < 1$. Hence, to ensure that Lemma 2.15 does not give us a bound worse than the trivial bound we will only apply Lemma 2.15 for “small” ranges of j . These “small” ranges of j are: $0 \leq j \leq J_1(\epsilon_1) - 1$ as seen in (3.39). And for “large” ranges of j given by $J_1(\epsilon_1) \leq j \leq J(\epsilon_1) - 1$ we only apply for large “ r ” which are precisely the values when $r \geq r_1 = \lceil u_1^4(1+\epsilon_1)^{9j/4} t^{-5/68} \rceil$ and thus avoiding the problematic case described above. To summarize, for $0 \leq j \leq J_1(\epsilon_1) - 1$ with $1 \leq r \leq q - 1$ and $J_1(\epsilon_1) \leq j \leq J(\epsilon_1) - 1$ with $r_1 \leq r \leq q - 1$ we have $W_5 > 1$. One can see that $W_5 > 1$ in $0 \leq j \leq J_1(\epsilon_1) - 1$ with $1 \leq r \leq q - 1$ because $r \geq r_1$ and

$$W_5 = \frac{25\sqrt{5}}{2688\pi^4} \frac{(tr)^4}{\lceil(1+\epsilon_1)^j t^{7/17}\rceil^9} \geq \frac{25\sqrt{5}}{2688\pi^4} \frac{t^4}{\lceil(1+\epsilon_1)^j t^{7/17}\rceil^9} > 1 \quad (3.84)$$

where the middle inequality above follows because of $r \geq 1$ and $\lceil (1 + \epsilon_1)^j t^{7/17} \rceil < \frac{t^{4/9}}{u_1^{4/9}}$ and the last inequality follows because of the definition of u_1 in (3.38). And $W_5 > 1$ for $J_1(\epsilon_1) \leq j \leq J(\epsilon_1) - 1$ with $r_1 \leq r \leq q - 1$ because $r \geq r_1$ and $\lceil (1 + \epsilon_1)^j t^{7/17} \rceil \leq \lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor$.

Therefore, we get a bound better than the trivial bound when we apply Lemma 2.15 in the above ranges. Moreover, when we apply the explicit fifth derivative test from Lemma 2.15 we will bound the constants α_5, τ_5 defined in (2.36) and thus get

$$\begin{aligned}\alpha_5 &\leq \tilde{\alpha}_5 = \frac{1}{\eta_5} + \frac{392}{435} \sqrt{\tilde{\alpha}_4} (\eta_5 + 1)^{1/14} \\ \tau_5 &= \frac{392}{351} \sqrt{\tilde{\gamma}_4} \eta_5^{-1/14}\end{aligned}\tag{3.85}$$

For the remaining range which is: $J_1(\epsilon_1) \leq j \leq J(\epsilon_1) - 1$ with $1 \leq r \leq r_1 - 1$ we will use triangle inequality.

Now we apply Lemma 2.15 whenever possible and we start with $J_1(\epsilon_1) \leq j \leq J(\epsilon_1) - 1$ and $r \geq \lceil u_1^4 (1 + \epsilon_1)^{9j/4} t^{-5/68} \rceil$. To do so we first bound the length of the sum in (3.64). We do this by applying the mean value theorem to f' and get

$$\gamma - \alpha \leq \beta - \alpha = |f'(\lceil (1 + \epsilon_1)^j t^{7/17} \rceil) - f'(\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r)| = |f''(\xi)| |(r + 1 - \Delta)|$$

where $\xi \in (\lceil (1 + \epsilon_1)^j t^{7/17} \rceil, \lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r)$

We know that $\Delta - 1 - r \geq 0$ which means $r + 1 \leq \Delta$. Thus, we have $|r + 1 - \Delta| = -(r + 1 - \Delta) = \Delta - r - 1$. Since $r \geq 1$ we see that $\Delta - r - 1 \leq \Delta - 2 \leq \Delta - 1 \leq L - 1$.

To complete the bound, let us first find an upper bound of $L - 1$ using the inequalities

$$\lfloor x \rfloor \leq x \leq \lceil x \rceil, \quad \delta_j \geq 0\tag{3.86}$$

and get

$$\begin{aligned}L - 1 &= \lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor - \delta_j - \lceil (1 + \epsilon_1)^j t^{7/17} \rceil \\ &\leq (1 + \epsilon_1)^{j+1} t^{7/17} - (1 + \epsilon_1)^j t^{7/17} \\ &= \epsilon_1 (1 + \epsilon_1)^j t^{7/17}\end{aligned}\tag{3.87}$$

Thus using an upper bound on $|f''(\xi)|$ from (3.52) and then using (3.87) we obtain

$$\beta - \alpha \leq |f''(\xi)| |r + 1 - \Delta| \leq |f''(\xi)| (L - 1) \leq \frac{\epsilon_1}{\pi} \frac{tr}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2}\tag{3.88}$$

In this case, using $\lceil (1 + \epsilon_1)^j t^{7/17} \rceil \leq \sqrt{t/2\pi}$ and $r \geq 1$ along with (3.88) we first compute a bound on the length of the exponential sum on the right hand side in (3.64)

$$\gamma - \alpha + 1 \leq \beta - \alpha + 1 \leq \frac{\epsilon_1}{\pi} \frac{tr}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2} + 1 \leq b_1 \frac{tr}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2}\tag{3.89}$$

where $b_1 = \frac{1+2\epsilon_1}{2\pi}$. Note that this value of b_1 can be lowered by using a slightly better bound for r which is $r \geq r_1$ instead of $r \geq 1$ but we make the choice of using the worse lower bound to ensure that the constant b_1 can be used when we apply this lemma in the range $0 \leq j \leq J_1(\epsilon_1) - 1$ with $1 \leq r \leq q - 1$. This is acceptable because of the small size of b_1 and the fact that it will keep computation write-up cleaner.

Thus now using explicit fifth derivative test Lemma 2.15 and using $\sqrt{x_1 + x_2 + \dots} \leq \sqrt{x_1} + \sqrt{x_2} + \dots$ we have an estimate on the exponential sum in (3.64)

$$\max_{\alpha < \gamma \leq \beta} \left| \sum_{\alpha < \nu \leq \gamma} e^{2\pi i \phi(\nu)} \right| \leq \max_{\alpha < \nu \leq \gamma} \left(\sqrt{\tilde{\alpha}_5}(\gamma - \alpha + 1)W_5^{-1/30} + \sqrt{\tau_5}(\gamma - \alpha + 1)^{7/8}W_5^{1/30} + \mathcal{E}(\epsilon_1, r, t) \right) \quad (3.90)$$

where $\mathcal{E}(\epsilon_1, r, t)$ are the lower order terms in Lemma 2.15 given by

$$\begin{aligned} \mathcal{E}(\epsilon_1, r, t) = & (\gamma_5 L_1^{3/2} W^{2/15} + \omega_5 L_1^{5/4} W^{2/15} + \beta_5 L_1 W^{3/15} + \eta_5 \alpha_5 L_1 + \eta_5 \tau_5 L_1^{3/4} W^{2/15} \\ & + \eta_5 \gamma_5 L_1^{1/2} W^{3/15} + \eta_5 \omega_5 L_1^{1/4} W^{1/5} + \eta_5 \beta_5 W^{4/15})^{1/2} \end{aligned} \quad (3.91)$$

where $L_1 = \gamma - \alpha + 1 > 1$.

Let us focus on bounding the first two terms on the right hand side in (3.90). We do so by substituting (3.83) and (3.89) in (3.90) we have

$$\begin{aligned} & \max_{\alpha < \gamma \leq \beta} \left(\sqrt{\tilde{\alpha}_5}(\gamma - \alpha + 1)W_5^{-1/30} + \sqrt{\tau_5}(\gamma - \alpha + 1)^{7/8}W_5^{1/30} + \mathcal{E}(\epsilon_1, r, t) \right) \\ & \leq \sqrt{\tilde{\alpha}_5} b_1 b_2^{-1/30} \frac{(tr)^{13/15}}{\lceil (1 + \epsilon_1) j t^{7/17} \rceil^{17/10}} + \sqrt{\tau_5} b_1^{7/8} b_2^{1/30} \frac{(tr)^{121/120}}{\lceil (1 + \epsilon_1) j t^{7/17} \rceil^{41/20}} + \mathcal{E}_{max}(\epsilon_1, r, t) \end{aligned} \quad (3.92)$$

where $b_2 = \frac{25\sqrt{5}}{2688\pi^4}$ and $\mathcal{E}_{max}(\epsilon_1, r, t) = \max_{\alpha < \gamma \leq \beta} \mathcal{E}(\epsilon_1, r, t)$.

Now we substitute (3.92) in (3.90) and then in (3.64) we get the following bound

$$\begin{aligned} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| & \leq \frac{1}{\sqrt{|f''(x_{\alpha_0})|}} \left(\sqrt{\tilde{\alpha}_5} b_1 b_2^{-1/30} \frac{(tr)^{13/15}}{\lceil (1 + \epsilon_1) j t^{7/17} \rceil^{17/10}} \right. \\ & \quad \left. + \sqrt{\tau_5} b_1^{7/8} b_2^{1/30} \frac{(tr)^{121/120}}{\lceil (1 + \epsilon_1) j t^{7/17} \rceil^{41/20}} + \mathcal{E}_{max}(t) \right) \end{aligned} \quad (3.93)$$

To simplify the bound in (3.93) we first simplify the bound for $|f''(\alpha)|$ using (3.52) and $\lfloor x \rfloor \leq x$

$$\frac{1}{\sqrt{|f''(x)|}} \leq \sqrt{\pi} (1 + \epsilon_1)^2 \frac{a^{3/2}}{(tr)^{1/2}} \quad (3.94)$$

where $a = (1 + \epsilon_1) j t^{7/17}$.

Thus, substituting (3.94) in (3.93) and using $x \leq \lceil x \rceil$ and simplifying we get

$$\left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq \frac{1}{\sqrt{f''(x_{\alpha_0})}} \max_{\alpha < \gamma \leq \beta} \left| \sum_{\alpha < \nu \leq \gamma} e^{2\pi i \phi(\nu)} \right| \quad (3.95)$$

$$\leq B_1 a^{-1/5} (tr)^{11/30} + D_1 a^{-11/20} (tr)^{61/120} + \tilde{\mathcal{E}}_{max}. \quad (3.96)$$

where

$$B_1 = \sqrt{\pi} (1 + \epsilon_1)^2 b_1 b_2^{-1/30} \sqrt{\alpha_5}, \quad D_1 = \sqrt{\pi} \sqrt{\tau_5} (1 + \epsilon_1)^2 b_1^{7/8} b_2^{1/30} \quad (3.97)$$

$$\tilde{\mathcal{E}}_{max} = \sqrt{\pi} (1 + \epsilon_1)^2 \frac{a^{3/2}}{(tr)^{1/2}} \mathcal{E}_{max}(\epsilon_1, r, t) \quad (3.98)$$

Hence plugging in (3.96) in (3.61) along with $r \leq q$ and the square-root inequality we have that

$$\begin{aligned} S_{4,1} &= w_1 \frac{L}{q^{1/2}} + w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=r_1}^{q-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \right)^{1/2} \\ &\leq w_1 \frac{L}{q^{1/2}} + w_2 \sqrt{L} \left(\sqrt{B_1} a^{-1/10} (tq)^{11/60} + \sqrt{D_1} a^{-11/40} (tq)^{61/240} \right) + \mathcal{R}_2 \end{aligned} \quad (3.99)$$

where

$$\mathcal{R}_2 = w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=r_1}^{q-1} \tilde{\mathcal{E}}_{max} \right)^{1/2} \quad (3.100)$$

We will bound this term, $\mathcal{R}_2$ in the next section. For the remainder of this section we will focus on performing our calculations using the first three terms on the right hand side in (3.99).

At this stage we bound L using upper bound on $L - 1$ given in (3.87) and $(1 + \epsilon_1)^j \geq 1$

$$L \leq b_3 (1 + \epsilon_1)^j t^{7/17} = b_3 a \quad (3.101)$$

where $b_3 = \epsilon_1 + \frac{1}{t^{7/17}}$ and a is defined in (3.94) as $a = (1 + \epsilon_1)^j t^{7/17}$.

Thus plugging in (3.101) in (3.99) and simplifying we get

$$S_{4,1} \leq w_1 b_3 \frac{a}{q^{1/2}} + w_2 \sqrt{b_3} \sqrt{B_1} a^{2/5} (tq)^{11/60} + w_2 \sqrt{b_3} \sqrt{D_1} a^{9/40} (tq)^{61/240} + \mathcal{R}_2 \quad (3.102)$$

Now we choose q such that the first two terms in (3.102) are of the same magnitude and this gives us

$$q = \lceil t^{-11/41} a^{36/41} \rceil \quad (3.103)$$

Before proceeding we make a couple of remarks about the size of q . First that q is of size $\mathcal{O}((1 + \epsilon_1)^{36j/41} t^{65/697})$ and using $\lceil x \rceil \leq x + 1$

$$q \leq b_4 t^{-11/41} a^{36/41}, \quad q - 1 \leq t^{-11/41} a^{36/41} \quad (3.104)$$

where $b_4 = 1 + \frac{t^{11/41}}{a^{36/41}} \leq 1 + \frac{1}{t^{65/697}}$. We also note that the with the choice of q in (3.103) we indeed have $q \geq 2$ since

$$t^{-11/41} a^{36/41} > 1 \iff a^{36/41} > t^{11/41} \iff a > t^{11/36} \iff a \geq t^{7/17} > t^{11/36} \quad (3.105)$$

We now bound w_1 and w_2 which are the constants that arise from applying Weyl differencing Lemma 2.6 and before we do so we note

$$L \geq \phi_{\epsilon_1, t}(1 + \epsilon_1)^j t^{7/17} = \phi_{\epsilon_1, t} a \quad (3.106)$$

where $\phi_{\epsilon_1, t} = \epsilon_1 - \frac{2}{t^{7/17}}$ and recall that since we started with $\epsilon_1 > \frac{3}{t^{7/17}}$ we still have $L \geq 1$. Hence, using (3.104) and (3.106) we bound the quantity

$$\frac{q-1}{L} \leq \frac{t^{-11/41} a^{36/41}}{\phi_{\epsilon_1, t} a} \leq \frac{1}{\phi_{\epsilon_1, t} t^{222/697}} \leq \frac{1}{\phi_{\epsilon_1, t} t_0^{222/697}} \quad (3.107)$$

where the last inequality follows because $t \geq t_0$.

Now using (3.107) we bound w_1 and w_2 and we get

$$w_1 = \sqrt{1 + \frac{q-1}{L}} \leq \sqrt{1 + \frac{1}{\phi_{\epsilon_1, t} t_0^{222/697}}}; \quad w_2 = \sqrt{2 + \frac{2(q-1)}{L}} \leq \sqrt{2 + \frac{2}{\phi_{\epsilon_1, t} t_0^{222/697}}} \quad (3.108)$$

Hence, substituting (3.103) in (3.102) along with $x \leq [x] \leq x+1$ and (3.104), we get

$$\begin{aligned} S_{4,1} &= w_1 \frac{L}{q^{1/2}} + w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=r_1}^{q-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \right)^{1/2} \\ &\leq B_2 a^{23/41} t^{11/82} + D_2 a^{147/328} t^{61/328} + \mathcal{R}_2 \end{aligned} \quad (3.109)$$

where $B_2 = w_1 b_3 + w_2 \sqrt{B_1 b_3} b_4^{11/60}$, $D_2 = w_2 \sqrt{D_1 b_3} b_4^{61/240}$.

It remains to estimate $S_{4,2}$ given in (3.62) using triangle inequality and to do so we bound the exponential sum on the right hand side in (3.64) with its length and using $\gamma \leq \beta$ we have

$$\left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq \frac{1}{\sqrt{f''(x_{\alpha_0})}} (\beta - \alpha + 1) \quad (3.110)$$

Now simplifying the bound for f'' in (3.52) we get

$$\frac{1}{\sqrt{|f''(x)|}} \leq \frac{\sqrt{\pi}(1 + \epsilon_1)^2(1 + \epsilon_1)^{3j/2} t^{2/17}}{r^{1/2}} \quad (3.111)$$

Now substituting (3.111) in (3.110) and then using (3.89) to bound $\beta - \alpha + 1$ we get that

$$\left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq h_1 \frac{t^{5/17}}{(1 + \epsilon_1)^{j/2}} r^{1/2} \quad (3.112)$$

where $h_1 = \sqrt{\pi}(1 + \epsilon_1)^2 b_1$.

Now plugging in (3.112) in (3.62) and then taking the sum over r we have

$$\sum_{r=1}^{r_1-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq h_1 \frac{t^{5/17}}{(1 + \epsilon_1)^{j/2}} \sum_{r=1}^{r_1-1} r^{1/2} \quad (3.113)$$

We note that the sum on the right hand side in (3.112) is estimated using the Cauchy-Schwarz inequality and the formula

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (3.114)$$

to get

$$\sum_{i=1}^n i^{1/2} \leq \frac{n^{3/2}}{\sqrt{2}} \sqrt{1 + \frac{1}{n}} \leq n^{3/2} \quad (3.115)$$

where the last inequality follows assuming $n \geq 1$.

Taking $n = r_1 - 1$ and using $r_1 - 1 \leq u_1^4(1 + \epsilon_1)^{9j/4} t^{-5/68}$ and then simplifying we estimate the right hand side of (3.113) as

$$\sum_{r=1}^{r_1-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq H_1 (1 + \epsilon_1)^{23j/8} t^{25/136} \quad (3.116)$$

where $H_1 = h_1 u_1^6$.

Next, we estimate the term

$$\frac{L}{q} \sum_{r=1}^{r_1-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \quad (3.117)$$

To do so, we substitute (3.116) and $L \leq b_3 a$ from (3.101) in (3.117) along with $q \geq t^{-11/41} a^{36/41} = (1 + \epsilon_1)^{36j/41} t^{65/697}$ we get

$$\frac{L}{q} \sum_{r=1}^{r_1-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \leq b_3 H_1 a (1 + \epsilon_1)^{655j/328} t^{505/5576} \quad (3.118)$$

Taking square root and plugging in (3.62) we obtain that

$$\begin{aligned} S_{4,2} &= w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=1}^{r_1-1} \left| \sum_{\alpha < \nu \leq \beta} \frac{e^{2\pi i \phi(\nu)}}{\sqrt{f''(x_\nu)}} \right| \right)^{1/2} \\ &\leq H_2 \sqrt{a} (1 + \epsilon_1)^{655j/656} t^{505/11152} \end{aligned} \quad (3.119)$$

where $H_2 = w_2 \sqrt{H_1}$.

Finally we substitute (3.109) and (3.119) in (3.60) we get

$$\begin{aligned} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1} n^{-it} \right| &\leq B_2 a^{23/41} t^{11/82} + D_2 a^{147/328} t^{61/328} \\ &\quad + H_2 \sqrt{a} (1 + \epsilon_1)^{655j/656} t^{505/11152} + \mathcal{R}_1 + \mathcal{R}_2 \end{aligned} \quad (3.120)$$

We remark that the last two terms in (3.120), $\mathcal{R}_1$ and $\mathcal{R}_2$, are the lower order terms present because of applying Theorem 2.31 and Lemma 2.15 respectively and will be bounded in the next section.

In the end we substitute (3.120) in (3.42) and using $a \leq \lceil (1 + \epsilon_1)^j t^{7/17} \rceil$ to obtain

$$\begin{aligned} M_2 &= \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} \frac{1}{\sqrt{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \\ &\leq B_2 t^{11/82} \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} a^{5/82} + D_2 t^{61/328} \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} a^{-17/328} \\ &\quad + H_2 t^{505/11152} \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} (1 + \epsilon_1)^{655j/656} + \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} + \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \end{aligned} \quad (3.121)$$

It remains to bound M_1 which is the range $0 \leq j \leq J_1(\epsilon_1) - 1$ with $1 \leq r \leq q - 1$. Note that the bound obtained in (3.120) also holds for the dyadic piece in this range of j but without the third term as it was obtained using triangle inequality which is not required in this range. Also, we can ensure that the constants remain unchanged since in the case of M_2 whenever we needed to bound r from below we used a slightly worse bound $r \geq 1$ instead of $r \geq r_1 - 1$. Similarly, for this range whenever we need to use $j \leq J_1(\epsilon_1) - 1$ we use a slightly worse bound $j \leq J(\epsilon_1) - 1$ instead. Hence we obtain a bound similar to (3.121) for M_1

$$\begin{aligned} M_1 &= \sum_{j=0}^{J_1(\epsilon_1)-1} \frac{1}{\sqrt{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil}} \max_{1 \leq \Delta \leq L} \left| \sum_{n=\lceil (1+\epsilon_1)^j t^{7/17} \rceil}^{\lceil (1+\epsilon_1)^j t^{7/17} \rceil + \Delta - 1} n^{-it} \right| \\ &\leq B_2 t^{11/82} \sum_{j=0}^{J_1(\epsilon_1)-1} a^{5/82} + D_2 t^{61/328} \sum_{j=0}^{J_1(\epsilon_1)-1} a^{-17/328} \\ &\quad + \sum_{j=0}^{J_1(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} + \sum_{j=0}^{J_1(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \end{aligned} \quad (3.122)$$

Hence, gathering (3.122) and (3.121) and combining the sums over j we obtain

$$\begin{aligned}
\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| &\leq M_1 + M_2 \\
&\leq B_2 t^{11/82} \sum_{j=0}^{J(\epsilon_1)-1} a^{5/82} + D_2 t^{61/328} \sum_{j=0}^{J(\epsilon_1)-1} a^{-17/328} \\
&\quad + H_2 t^{505/11152} \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} (1+\epsilon_1)^{655j/656} + \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} + \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \quad (3.123)
\end{aligned}$$

To estimate the sums in (3.123) we use the following bounds

$$\sum_{j=0}^{J(\epsilon_1)-1} (1+\epsilon_1)^{v_1 j} \leq \frac{(1+\epsilon_1)^{v_1}}{(2\pi)^{v_1/2}((1+\epsilon_1)^{v_1} - 1)} t^{3v_1/34}, \quad \sum_{j=0}^{J(\epsilon_1)-1} \frac{1}{(1+\epsilon_1)^{v_2 j}} \leq \frac{(1+\epsilon_1)^{v_2}}{(1+\epsilon_1)^{v_2} - 1} \quad (3.124)$$

Now using $a = (1+\epsilon_1)^j t^{7/17}$ in first three sums in (3.123) we get

$$\begin{aligned}
\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| &\leq M_1 + M_2 \\
&\leq B_2 t^{111/697} \sum_{j=0}^{J(\epsilon_1)-1} (1+\epsilon_1)^{5j/82} + D_2 t^{27/164} \sum_{j=0}^{J(\epsilon_1)-1} (1+\epsilon_1)^{-17j/328} \\
&\quad + H_2 t^{505/11152} \sum_{j=J_1(\epsilon_1)}^{J(\epsilon_1)-1} (1+\epsilon_1)^{655j/656} + \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} + \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \quad (3.125)
\end{aligned}$$

And the last two sums will be estimated in the next section. Using (3.124) to bound the first three sums on the right hand side in (3.125) we get

$$\left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| \leq B_3 t^{\frac{27}{164}} + H_3 t^{\frac{175}{1312}} + \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} + \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \quad (3.126)$$

where

$$B_3 = B_2 \frac{(1+\epsilon_1)^{5/82}}{(2\pi)^{5/164}((1+\epsilon_1)^{5/82} - 1)} + D_2 \frac{(1+\epsilon_1)^{17/328}}{(1+\epsilon_1)^{17/328} - 1} \quad (3.127)$$

$$H_3 = H_2 \frac{(1+\epsilon_1)^{655/656}}{(2\pi)^{655/1312}((1+\epsilon_1)^{655/656} - 1)} \quad (3.128)$$

To finish our proof we use the bound in equations (3.161) and (3.180) for the last two sums in (3.126) respectively. $\square$

3.2.3 Bounding Lower Order Terms $\mathcal{R}_1$ and $\mathcal{R}_2$

Let us start with estimating $\mathcal{R}_1$ first. Recalling the definition from (3.63) we have

$$\mathcal{R}_1 = w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=1}^{q-1} |\mathcal{P}(\epsilon_1, r, t)| \right)^{1/2} \quad (3.129)$$

where

$$\mathcal{P}(\epsilon_1, r, t) = \mathfrak{p}_1 A_2^{-1/2} + \mathfrak{p}_2 \log(\beta - \alpha + 4) + \mathfrak{p}_3 (b - a)(A_2 A_3)^{1/5} + \mathfrak{p}_4 \quad (3.130)$$

Hence plugging (3.130) in (3.129) and using triangle inequality and expanding after using $L = b_3 a$ we have

$$\begin{aligned} \mathcal{R}_1 \leq w_2 \sqrt{b_3} \sqrt{a} \max_{1 \leq \Delta \leq L} & \left(\frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_1| A_2^{-1/2} + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_2| \log(\beta - \alpha + 4) \right. \\ & \left. + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_3| (\Delta - r) A_2^{1/2} A_3^{1/5} + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_4| \right)^{1/2} \end{aligned} \quad (3.131)$$

We will bound each sum in (3.131) and in order to do so we start with the first sum and note that

$$A_2 = \frac{tr}{(1 + \epsilon_1)\pi} \frac{1}{[(1 + \epsilon_1)^{j+1} t^{7/17}]^3} \geq \frac{tr}{\pi a^3 (1 + \epsilon_1)^4} \quad (3.132)$$

Thus using (3.132) we obtain

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_1| A_2^{-1/2} \leq |\mathfrak{p}_1| \sqrt{\pi} (1 + \epsilon_1)^2 a^{3/2} t^{-1/2} \frac{1}{q} \sum_{r=1}^{q-1} \frac{1}{\sqrt{r}} \quad (3.133)$$

Next we use the following estimate to bound the sum on the right hand side in (3.133)

$$\sum_{r=1}^{q-1} \frac{1}{\sqrt{r}} \leq 2\sqrt{q} \quad (3.134)$$

Hence substituting (3.134) for the sum on the right hand side in (3.133) we obtain

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_1| A_2^{-1/2} \leq 2|\mathfrak{p}_1| \sqrt{\pi} (1 + \epsilon_1)^2 a^{3/2} t^{-1/2} \frac{1}{\sqrt{q}} \quad (3.135)$$

Now, using $q \geq t^{-11/41} a^{36/41}$ from (3.104), $a = (1 + \epsilon_1)^j t^{7/17}$ we simplify the bound in (3.135) to

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_1| A_2^{-1/2} \leq \mathcal{M}_1 (1 + \epsilon_1)^{87j/82} t^{99/1394} \quad (3.136)$$

where $\mathcal{M}_1 = 2|\mathfrak{p}_1| \sqrt{\pi} (1 + \epsilon_1)^2$.

Next, let us bound the second sum in (3.131). From (3.88) we have that

$$\beta - \alpha + 4 \leq b'_1 \frac{tr}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2} \leq b'_1 t^{188/697} \quad (3.137)$$

where $b'_1 = \frac{2\epsilon_1 + 1}{2\pi}$ and the last inequality is derived using $1 \leq r \leq q - 1$, (3.104), $a = (1 + \epsilon_1)^j t^{7/17}$, $(1 + \epsilon_1)^j \geq 1$ and simplifying. which gives us

$$\log(\beta - \alpha + 2) \leq \log(b'_1) + \frac{188}{697} \log(t) \leq \frac{188}{697} \log(t) \quad (3.138)$$

where the last inequality follows because $\log(b'_1) < 0$ since $b'_1 < 1$ and $\epsilon_1 < 1$. Hence, plugging in (3.138) in the second sum in (3.131) we get that

$$\frac{1}{q} \sum_{r=1}^{q-1} |\rho_2| \log(\beta - \alpha + 4) \leq |\rho_2| \frac{188}{697} \log(t) \quad (3.139)$$

Now we will bound the third sum in (3.131) using $r \geq 1$ whence $\Delta - r \leq \Delta - 1$ and obtain

$$\frac{1}{q} \sum_{r=1}^{q-1} |\rho_3| (\Delta - r) A_2^{1/5} A_3^{1/5} \leq |\rho_3| (L - 1) \frac{1}{q} \sum_{r=1}^{q-1} A_2^{1/2} A_3^{1/5} \quad (3.140)$$

We will now bound $(L - 1)(A_2 A_3)^{1/5}$. To do so, we first recall that:

$$f(x) = \frac{t}{2\pi} (\log(x + r) - \log(x)), \quad \text{for } x \in [\lceil (1 + \epsilon_1)^j t^{7/17} \rceil, \lceil (1 + \epsilon_1)^j t^{7/17} \rceil + \Delta - 1 - r]$$

and hence

$$f'''(x) = \frac{tr}{\pi} \frac{3x^2 + 3xr + r^2}{x^3(x + r)^3}. \quad (3.141)$$

Now consider the function

$$g_3(x, r) = \frac{3x^2 + 3xr + r^2}{x^3(x + r)^3}. \quad (3.142)$$

We observe that $g_3(x, r)$ is a decreasing function for $x, r > 0$ and thus its maximum over the range of x and r under consideration occurs when $x = \lceil (1 + \epsilon_1)^j t^{7/17} \rceil$ and $r = 1$. Hence using this observation we have that

$$|f'''(x)| \leq \frac{tr}{\pi} \frac{3\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2 + 3\lceil (1 + \epsilon_1)^j t^{7/17} \rceil + 1}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^6}. \quad (3.143)$$

Upon factoring out $\lceil (1 + \epsilon_1)^j t^{7/17} \rceil$ from above and recalling that $\lceil (1 + \epsilon_1)^j t^{7/17} \rceil \geq a$ and $(1 + \epsilon_1)^j \geq 1$, we get

$$|f'''(x)| \leq \frac{c_1}{\pi} \frac{tr}{a^4}, \quad (3.144)$$

where for $t \geq t_0$:

$$c_1 \leq \left(3 + \frac{3}{t_0^{7/17}} + \frac{1}{t_0^{14/17}} \right). \quad (3.145)$$

Let us set

$$\mathcal{A}_3 = a_3 = \frac{c_1}{\pi}, \quad A_3 = \frac{tr}{a^4}. \quad (3.146)$$

Also, recall from (3.53) that

$$A_2 = \frac{tr}{(1 + \epsilon_1)\pi} \frac{1}{[(1 + \epsilon_1)^{j+1} t^{7/17}]^3}.$$

Using (3.146) and $\lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor \geq \lfloor (1 + \epsilon_1)^{j+1} t^{7/17} \rfloor - \delta_r \geq \lceil (1 + \epsilon_1)^j t^{7/17} \rceil \geq a$ we can compute

$$\begin{aligned} A_2 A_3 &= \frac{tr}{(1 + \epsilon_1)\pi} \frac{1}{[(1 + \epsilon_1)^{j+1} t^{7/17}]^3} \frac{tr}{a^4} \\ &\leq \frac{1}{\pi(1 + \epsilon_1)} \frac{(tr)^2}{a^7}. \end{aligned} \quad (3.147)$$

Thus using (3.147) we obtain

$$(A_2 A_3)^{1/5} \leq \frac{1}{\pi^{1/5}(1 + \epsilon_1)^{1/5}} \frac{(tr)^{2/5}}{a^{7/5}}. \quad (3.148)$$

Now we have $L - 1 \leq \epsilon_1 a$ using (3.87) which combined with (3.148) gives

$$(L - 1)(A_2 A_3)^{1/5} \leq \frac{\epsilon_1}{\pi^{1/5}(1 + \epsilon_1)^{1/5}} \frac{(tr)^{2/5}}{a^{2/5}} \quad (3.149)$$

Hence substituting (3.149) in (3.140) we have

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathcal{P}_3| (\Delta - r) A_2^{1/5} A_3^{1/5} \leq |\mathcal{P}_3| \frac{\epsilon_1}{\pi^{1/5}(1 + \epsilon_1)^{1/5}} \frac{t^{2/5}}{a^{2/5}} \frac{1}{q} \sum_{r=1}^{q-1} r^{2/5}. \quad (3.150)$$

We bound the sum in (3.150) by replacing it with an integral and get the estimate

$$\sum_{r=1}^{q-1} r^{2/5} \leq \int_1^q x^{2/5} dx \leq \frac{5}{7} q^{7/5}. \quad (3.151)$$

Substituting (3.151) in (3.150) and then using upper bound on q from (3.104) to simplify we get

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathcal{P}_3| (\Delta - r) A_2^{1/5} A_3^{1/5} \leq |\mathcal{P}_3| \frac{5}{7} \frac{\epsilon_1}{\pi^{1/5}(1 + \epsilon_1)^{1/5}} \frac{t^{2/5}}{a^{2/5}} (b_4 t^{-11/41} a^{36/41})^{2/5} \quad (3.152)$$

$$= \frac{5\epsilon_1 |\mathcal{P}_3| b_4^{2/5}}{7\pi^{1/5}(1 + \epsilon_1)^{1/5}} \frac{t^{12/41}}{a^{2/41}}. \quad (3.153)$$

Using once again $a = (1 + \epsilon_1)^j t^{7/17}$ in (3.153) we have that the third sum in (3.131) is bounded by

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathcal{P}_3| (\Delta - r) A_2^{1/5} A_3^{1/5} \leq \mathcal{M}_2 t^{190/697} \frac{1}{(1 + \epsilon_1)^{2j/41}}. \quad (3.154)$$

where

$$\mathcal{M}_2 = \frac{5\epsilon_1 |\mathfrak{p}_3| b_4^{2/5}}{7\pi^{1/5} (1 + \epsilon_1)^{1/5}}. \quad (3.155)$$

Finally, the fourth sum in (3.131) is estimated simply as

$$\frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_4| \leq |\mathfrak{p}_4| \quad (3.156)$$

since $\mathfrak{p}_4$ does not depend on r .

Hence gathering the bounds for the four sums from (3.136), (3.139), (3.154), and (3.156)

$$\begin{aligned} & \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_1| A_2^{-1/2} + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_2| \log(\beta - \alpha + 2) + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_3| (\Delta - r) A_2^{1/2} A_3^{1/5} + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_4| \\ & \leq \mathcal{M}_1 (1 + \epsilon_1)^{87j/82} t^{99/1394} + |\mathfrak{p}_2| \frac{188}{697} \ln(t) + \mathcal{M}_2 t^{190/697} \frac{1}{(1 + \epsilon_1)^{2j/41}} + |\mathfrak{p}_4|. \end{aligned} \quad (3.157)$$

Substituting (3.157) in (3.131) and using square root inequality we get

$$\begin{aligned} \mathcal{R}_1 & \leq w_2 \sqrt{b_3} \sqrt{a} \max_{1 \leq \Delta \leq L} \left(\frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_1| A_2^{-1/2} + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_2| \log(\beta - \alpha + 2) \right. \\ & \quad \left. + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_3| (\Delta - r) A_2^{1/2} A_3^{1/5} + \frac{1}{q} \sum_{r=1}^{q-1} |\mathfrak{p}_4| \right)^{1/2} \\ & \leq w_2 \sqrt{b_3} \sqrt{a} \left(\sqrt{\mathcal{M}_1} (1 + \epsilon_1)^{87j/164} t^{99/2788} + \sqrt{|\mathfrak{p}_2| \frac{188}{697}} \sqrt{\log(t)} + \sqrt{\mathcal{M}_2} \frac{t^{95/697}}{(1 + \epsilon_1)^{j/41}} + \sqrt{|\mathfrak{p}_4|} \right). \end{aligned} \quad (3.158)$$

Using (3.158) we take the sum over j and get the estimate

$$\begin{aligned} \sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} & \leq \sum_{j=0}^{J(\epsilon_1)-1} w_2 \sqrt{b_3} \left(\sqrt{\mathcal{M}_1} (1 + \epsilon_1)^{87j/164} t^{99/2788} + \sqrt{|\mathfrak{p}_2| \frac{188}{697}} \sqrt{\log(t)} \right. \\ & \quad \left. + \sqrt{\mathcal{M}_2} \frac{t^{95/697}}{(1 + \epsilon_1)^{j/41}} + \sqrt{|\mathfrak{p}_4|} \right) \\ & = w_2 \sqrt{b_3} \sqrt{\mathcal{M}_1} t^{99/2788} \sum_{j=0}^{J(\epsilon_1)-1} (1 + \epsilon_1)^{87j/164} + w_2 \sqrt{b_3} \sqrt{|\mathfrak{p}_2| \frac{188}{697}} \sqrt{\log(t)} \sum_{j=0}^{J(\epsilon_1)-1} 1 \\ & \quad + w_2 \sqrt{b_3} \sqrt{\mathcal{M}_2} t^{95/697} \sum_{j=0}^{J(\epsilon_1)-1} \frac{1}{(1 + \epsilon_1)^{j/41}} + w_2 \sqrt{b_3} \sqrt{|\mathfrak{p}_4|} \sum_{j=0}^{J(\epsilon_1)-1} 1. \end{aligned} \quad (3.159)$$

Now we estimate the sums on the right hand side in (3.159) using (3.124) and also the estimate of number of dyadic pieces, $J(\epsilon_1)$ using the upper bound on $J(\epsilon_1)$ from (3.36) and $\lfloor x \rfloor \leq x$

$$J(\epsilon_1) \leq \left\lfloor \frac{\frac{3}{34} \log t - \log \sqrt{2\pi}}{\log(1 + \epsilon_1)} \right\rfloor + 1 \leq \frac{3}{34 \log(1 + \epsilon_1)} \log t + 1 \leq J_0 \log t. \quad (3.160)$$

where $J_0 = \frac{3}{34 \log(1 + \epsilon_1)} + \frac{1}{\log t_0}$. This gives us

$$\sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_1}{\sqrt{a}} \leq \mathcal{N}_1 t^{27/328} + \mathcal{N}_2 t^{95/697} + \mathcal{N}_3 \log^{3/2}(t) + \mathcal{N}_4 \log t \quad (3.161)$$

where

$$\mathcal{N}_1 = w_2 \sqrt{b_3 \mathcal{M}_1} \frac{(1 + \epsilon_1)^{87/164}}{(2\pi)^{87/328} ((1 + \epsilon_1)^{87/164} - 1)}, \quad \mathcal{N}_2 = w_2 \sqrt{b_3 \mathcal{M}_2} \frac{(1 + \epsilon_1)^{1/41}}{(1 + \epsilon_1)^{1/41} - 1}, \quad (3.162)$$

$$\mathcal{N}_3 = w_2 J_0 \sqrt{b_3 |\mathcal{P}_2|} \frac{188}{697}, \quad \mathcal{N}_4 = w_2 J_0 \sqrt{b_3} \sqrt{|\mathcal{P}_4|}. \quad (3.163)$$

It remains to estimate $\mathcal{R}_2$ in (3.100)

$$\mathcal{R}_2 = w_2 \max_{1 \leq \Delta \leq L} \left(\frac{L}{q} \sum_{r=r_1}^{q-1} \tilde{\mathcal{E}}_{max} \right)^{1/2}, \quad (3.164)$$

where

$$\tilde{\mathcal{E}}_{max} = \sqrt{\pi} (1 + \epsilon_1)^2 \frac{a^{3/2}}{(tr)^{1/2}} \mathcal{E}_{max}(t) \quad (3.165)$$

and

$$\begin{aligned} \mathcal{E}_{max}(\epsilon, r, t) = \max_{\alpha < \gamma \leq \beta} & (\gamma_5 L_1^{3/2} W^{2/15} + \omega_5 L_1^{5/4} W^{2/15} + \beta_5 L_1 W^{3/15} + \eta_5 \alpha_5 L_1 + \eta_5 \tau_5 L_1^{3/4} W^{2/15} \\ & + \eta_5 \gamma_5 L_1^{1/2} W^{3/15} + \eta_5 \omega_5 L_1^{1/4} W^{1/5} + \eta_5 \beta_5 W^{4/15})^{1/2}, \end{aligned} \quad (3.166)$$

where $L_1 = \gamma - \alpha + 1 > 1$.

The way we obtain an estimate for (3.166) is using square-root inequality and then factoring out the term of the form:

$$L_1^{a_1/2} W_5^{a_2/2}, \quad \text{where } a_1, a_2 > 0, \quad (3.167)$$

such that the remainder terms gives a non-increasing function in t . Here

$$L_1 = \gamma - \alpha + 1 \leq \beta - \alpha + 1 \leq b_1 \frac{tr}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^2}, \quad W_5 \leq \frac{25\sqrt{5}}{2688\pi^4} \frac{(tr)^4}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^9}. \quad (3.168)$$

Therefore,

$$\begin{aligned} L^{a_1/2} W_5^{a_2/2} & \leq b_1^{a_1/2} \frac{(tr)^{a_1/2}}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^{a_1}} b_2^{a_2/2} \frac{(tr)^{2a_2}}{\lceil (1 + \epsilon_1)^j t^{7/17} \rceil^{9a_2/2}} \\ & \leq b_1^{a_1/2} b_2^{a_2/2} \frac{(tr)^{a_1/2 + 2a_2}}{a^{a_1 + 9a_2/2}} \\ & \leq (b_1 b_2)^{(a_1 + a_2)/2} \frac{t^{(3a_1 + 5a_2)/34} r^{a_1/2 + 2a_2}}{(1 + \epsilon_1)^{(a_1 + 9a_2/2)j}}. \end{aligned} \quad (3.169)$$

Now multiplying both sides in (3.169) by $\frac{1}{\sqrt{|f''(x)|}}$ and bounding the second derivative factor using (3.94) along with $a = (1 + \epsilon_1)^j t^{7/17}$ we get that

$$\begin{aligned} \frac{1}{\sqrt{|f''(x)|}} L^{a_1/2} W_5^{a_2/2} &\leq \sqrt{\pi} (1 + \epsilon_1)^2 \frac{a^{3/2}}{(tr)^{1/2}} (b_1 b_2)^{(a_1+a_2)/2} \frac{t^{(3a_1+5a_2)/34} r^{a_1/2+2a_2}}{(1 + \epsilon_1)^{(a_1+9a_2/2)j}} \\ &\leq \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{(a_1+a_2)/2} \frac{t^{(3a_1+5a_2+4)/34} r^{(a_1+4a_2-1)/2}}{(1 + \epsilon_1)^{(2a_1+9a_2-3)/2}}. \end{aligned} \quad (3.170)$$

Moreover, we use (3.170) and $r \leq q - 1 \leq t^{-11/41} a^{36/41}$ along with (3.104) and $a = (1 + \epsilon_1)^j t^{7/17}$ we bound the quantity

$$\begin{aligned} \frac{1}{q} \sum_{r=r_1}^{q-1} \frac{1}{\sqrt{|f''(x)|}} L^{a_1/2} W_5^{a_2/2} &\leq \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{(a_1+a_2)/2} \frac{t^{(3a_1+5a_2+4)/34} (q-1)^{(a_1+4a_2-1)/2}}{(1 + \epsilon_1)^{(2a_1+9a_2-3)/2}} \\ &\leq \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{(a_1+a_2)/2} t^{\frac{188a_1+465a_2+99}{1394}} \frac{1}{(1 + \epsilon_1)^{\frac{46a_1+225a_2-87}{82}}}. \end{aligned} \quad (3.171)$$

Next, we use the square root inequality to bound (3.166) along with appropriate choices for a_1, a_2 in (3.171) as follows:

$$\begin{aligned} \mathcal{E}_{\max}(\epsilon, r, t) &= \max_{\alpha < \gamma \leq \beta} (\gamma_5 L_1^{3/2} W^{2/15} + \omega_5 L_1^{5/4} W^{2/15} + \beta_5 L_1 W^{3/15} + \eta_5 \alpha_5 L_1 + \eta_5 \tau_5 L_1^{3/4} W^{2/15} \\ &\quad + \eta_5 \gamma_5 L_1^{1/2} W^{3/15} + \eta_5 \omega_5 L_1^{1/4} W^{1/5} + \eta_5 \beta_5 W^{4/15})^{1/2} \\ &\leq \sqrt{L_1^{3/2} W_5^{2/15}} \left(\gamma_5 + \omega_5 \frac{1}{L_1^{1/4}} \right)^{1/2} + \sqrt{L_1 W_5^{4/15}} \left(\beta_5 \frac{1}{W_5^{1/15}} + \eta_5 \alpha_5 \frac{1}{W_5^{4/15}} \right. \\ &\quad \left. + \eta_5 \tau_5 \frac{1}{L_1^{1/4} W_5^{2/15}} + \eta_5 \gamma_5 \frac{1}{L_1^{1/2} W_5^{1/15}} + \eta_5 \omega_5 \frac{1}{L_1^{3/4} W_5^{1/15}} + \eta_5 \beta_5 \frac{1}{L_1} \right)^{1/2}. \end{aligned}$$

Now using $L_1 > 1$ and $W_5 > 1$ we have

$$\mathcal{E}_{\max}(\epsilon, r, t) \leq f_1 \sqrt{L_1^{3/2} W_5^{2/15}} + f_2 \sqrt{L_1 W_5^{4/15}}, \quad (3.172)$$

where

$$f_1 = \sqrt{\tilde{\gamma}_5 + \omega_5}; \quad f_2 = \sqrt{\tilde{\beta}_5 + \eta_5 \tilde{\alpha}_5 + \eta_5 \tau_5 + \eta_5 \tilde{\gamma}_5 + \eta_5 \omega_5 + \eta_5 \beta_5}, \quad (3.173)$$

and because of the definition of constants given in (2.36) we note

$$\alpha_5 \leq \tilde{\alpha}_5; \quad \gamma_5 \leq \tilde{\gamma}_5 = \frac{98}{78} \sqrt{\beta_4} \eta_5^{-1/7} + \sqrt{\eta_4 \tilde{\alpha}_4}. \quad (3.174)$$

Therefore, plugging (3.172) in (3.165) and using (3.171) we get that

$$\begin{aligned} &\frac{1}{q} \sum_{r=r_1}^{q-1} \frac{1}{\sqrt{|f''(x)|}} \left(f_1 \sqrt{L_1^{3/2} W_5^{2/15}} + f_2 \sqrt{L_1 W_5^{4/15}} \right) \\ &= f_1 \frac{1}{q} \sum_{r=r_1}^{q-1} \frac{1}{\sqrt{|f''(x)|}} \sqrt{L_1^{3/2} W_5^{2/15}} + f_2 \frac{1}{q} \sum_{r=r_1}^{q-1} \frac{1}{\sqrt{|f''(x)|}} \sqrt{L_1 W_5^{4/15}}, \end{aligned} \quad (3.175)$$

and now using (3.171) with $a_1 = 3/2, a_2 = 2/15$ for the first sum in (3.175) and $a_1 = 1, a_2 = 4/15$ for the second sum we obtain the estimate

$$\begin{aligned} & \frac{1}{q} \sum_{r=r_1}^{q-1} \frac{1}{\sqrt{|f''(x)|}} \left(f_1 \sqrt{L_1^{3/2} W_5^{2/15}} + f_2 \sqrt{L_1 W_5^{4/15}} \right) \\ & \leq F_1 t^{443/1394} \frac{1}{(1 + \epsilon_1)^{6/41}} + F_2 t^{411/1394} \frac{1}{(1 + \epsilon_1)^{19/82}}, \end{aligned} \quad (3.176)$$

where

$$F_1 = f_1 \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{49/60}, \quad F_2 = f_2 \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{19/30}. \quad (3.177)$$

Hence substituting (3.176) in (3.164) and using $L \leq b_3 a$ from (3.101) along with the square root inequality we get

$$\begin{aligned} \mathcal{R}_2 & \leq w_2 \sqrt{L} \left(F_1 t^{443/1394} \frac{1}{(1 + \epsilon_1)^{6/41}} + F_2 t^{411/1394} \frac{1}{(1 + \epsilon_1)^{19/82}} \right)^{1/2} \\ & \leq w_2 \sqrt{b_3} \sqrt{a} \left(\sqrt{F_1} t^{443/2788} \frac{1}{(1 + \epsilon_1)^{3/41}} + \sqrt{F_2} t^{411/2788} \frac{1}{(1 + \epsilon_1)^{19/164}} \right). \end{aligned} \quad (3.178)$$

Using (3.178) we obtain a bound for the following quantity:

$$\sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \leq w_2 \sqrt{b_3} F_1 t^{443/2788} \sum_{j=0}^{J(\epsilon_1)-1} \frac{1}{(1 + \epsilon_1)^{3/41}} + w_2 \sqrt{b_3} F_2 t^{411/2788} \sum_{j=0}^{J(\epsilon_1)-1} \frac{1}{(1 + \epsilon_1)^{19/164}}. \quad (3.179)$$

Now we estimate the two sums on the right hand side in (3.179) using (3.124) and obtain

$$\sum_{j=0}^{J(\epsilon_1)-1} \frac{\mathcal{R}_2}{\sqrt{a}} \leq F_3 t^{443/2788} + F_4 t^{411/2788} \quad (3.180)$$

where

$$F_3 = w_2 \sqrt{b_3} F_1 \frac{(1 + \epsilon_1)^{3/41}}{(1 + \epsilon_1)^{3/41} - 1}, \quad F_4 = w_2 \sqrt{b_3} F_2 \frac{(1 + \epsilon_1)^{19/164}}{(1 + \epsilon_1)^{19/164} - 1}. \quad (3.181)$$

Finally, substituting the two lower order terms estimate from (3.161) and (3.180) in (3.126) gives

$$\begin{aligned} \left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| & \leq B_3 t^{\frac{27}{164}} + H_3 t^{\frac{175}{1312}} + \mathcal{N}_1 t^{27/328} + \mathcal{N}_2 t^{95/697} + \mathcal{N}_3 \log^{3/2}(t) + \mathcal{N}_4 \log t \\ & \quad + F_3 t^{443/2788} + F_4 t^{411/2788} \end{aligned} \quad (3.182)$$

as claimed.

3.2.4 List of Constants

For convenience of the reader we list all the definition of constants used to obtain the sub-Weyl constant in this section below.

First, we state the constants for B_3 in $t^{27/164}$:

$$B_3 = B_2 \frac{(1 + \epsilon_1)^{5/82}}{(2\pi)^{5/164}((1 + \epsilon_1)^{5/82} - 1)} + D_2 \frac{(1 + \epsilon_1)^{17/328}}{(1 + \epsilon_1)^{17/328} - 1}, \quad (3.183)$$

$$B_2 = w_1 b_3 + w_2 \sqrt{B_1 b_3 b_4^{11/60}}, \quad D_2 = w_2 \sqrt{D_1 b_3 b_4^{61/240}}, \quad (3.184)$$

$$B_1 = \sqrt{\pi}(1 + \epsilon_1)^2 b_1 b_2^{-1/30} \sqrt{\alpha_5}, \quad D_1 = \sqrt{\pi} \sqrt{\tau_5} (1 + \epsilon_1)^2 b_1^{7/8} b_2^{1/30}. \quad (3.185)$$

Next, we state constants for H_3 in $t^{175/1312}$:

$$H_3 = H_2 \frac{(1 + \epsilon_1)^{655/656}}{(2\pi)^{655/1312}((1 + \epsilon_1)^{655/656} - 1)}, \quad (3.186)$$

$$H_2 = w_2 \sqrt{H_1}, \quad (3.187)$$

$$H_1 = h_1 u_1^6, \quad (3.188)$$

$$h_1 = \sqrt{\pi}(1 + \epsilon_1)^2 b_1. \quad (3.189)$$

We state constants for $\mathcal{N}_1$ in $t^{27/328}$:

$$\mathcal{N}_1 = w_2 \sqrt{b_3 \mathcal{M}_1} \frac{(1 + \epsilon_1)^{87/164}}{(2\pi)^{87/328}((1 + \epsilon_1)^{87/164} - 1)}, \quad (3.190)$$

$$\mathcal{M}_1 = 2|\mathcal{P}_1| \sqrt{\pi}(1 + \epsilon_1)^2. \quad (3.191)$$

We state constants for $\mathcal{N}_2$ in $t^{95/697}$:

$$\mathcal{N}_2 = w_2 \sqrt{b_3 \mathcal{M}_2} \frac{(1 + \epsilon_1)^{1/41}}{(1 + \epsilon_1)^{1/41} - 1}, \quad (3.192)$$

$$\mathcal{M}_2 = \frac{5\epsilon_1 |\mathcal{P}_3| b_4^{2/5}}{7\pi^{1/5} (1 + \epsilon_1)^{1/5}}, \quad (3.193)$$

$$c_1 = 3 + \frac{3}{t_0^{7/17}} + \frac{1}{t_0^{14/17}}. \quad (3.194)$$

We state constants for $\mathcal{N}_3$ in $\log^{3/2} t$:

$$\mathcal{N}_3 = w_2 J_0 \sqrt{b_3 |\mathcal{P}_2| \frac{188}{697}}, \quad (3.195)$$

$$J_0 = \frac{3}{34 \log(1 + \epsilon_1)} + \frac{1}{\log t_0}. \quad (3.196)$$

$$(3.197)$$

We state constants for $\mathcal{N}_4$ in $\log t$:

$$\mathcal{N}_4 = w_2 J_0 \sqrt{b_3} \sqrt{|p_4|}, \quad (3.198)$$

$$J_0 = \frac{3}{34 \log(1 + \epsilon_1)} + \frac{1}{\log t_0}, \quad (3.199)$$

$$b'_1 = \frac{2\epsilon_1 + 1}{2\pi}. \quad (3.200)$$

We state constants for F_3 in $t^{443/2788}$:

$$F_3 = w_2 \sqrt{b_3} F_1 \frac{(1 + \epsilon_1)^{3/41}}{(1 + \epsilon_1)^{3/41} - 1}, \quad (3.201)$$

$$F_1 = f_1 \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{49/60}, \quad (3.202)$$

$$f_1 = \sqrt{\tilde{\gamma}_5 + \omega_5}. \quad (3.203)$$

We state constants for F_4 in $t^{411/2788}$:

$$F_4 = w_2 \sqrt{b_3} F_2 \frac{(1 + \epsilon_1)^{19/164}}{(1 + \epsilon_1)^{19/164} - 1}, \quad (3.204)$$

$$F_2 = f_2 \sqrt{\pi} (1 + \epsilon_1)^2 (b_1 b_2)^{19/30}, \quad (3.205)$$

$$f_2 = \sqrt{\beta_5 + \eta_5 \tilde{\alpha}_5 + \eta_5 \tau_5 + \eta_5 \tilde{\gamma}_5 + \eta_5 \omega_5 + \eta_5 \beta_5}. \quad (3.206)$$

3.2.5 Numerical Values of Parameters

Finally, we choose the following numerical values for our parameters to obtain numerical values for $B_3, H_3, \mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_3, \mathcal{N}_4, F_3, F_4$:

$$t \geq t_0 = 4.41 \times 10^{105}, \quad \epsilon_1 = 0.342, \quad \eta_3 = \left(\frac{15\sqrt{\pi}}{32} \right)^{2/3}, \quad \eta_4 = \left(\frac{91}{72\sqrt{23}} \right)^{6/7}, \quad \eta_5 = 2.222 \quad (3.207)$$

and we obtain

$$2 \left| \sum_{n=\lceil t^{7/17} \rceil}^{\lfloor \sqrt{\frac{t}{2\pi}} \rfloor} n^{-1/2-it} \right| \leq 2B_3 t^{\frac{27}{164}} + 2H_3 t^{\frac{175}{1312}} + 2\mathcal{N}_1 t^{27/328} + 2\mathcal{N}_2 t^{95/697} + 2\mathcal{N}_3 \log^{3/2}(t) \\ + 2\mathcal{N}_4 \log t + 2F_3 t^{443/2788} + 2F_4 t^{411/2788}, \quad (3.208)$$

where

$$2B_3 = 298.01005 \dots, \quad 2H_3 = 2325.05735 \dots, \quad 2\mathcal{N}_1 = 84.39712 \dots, \quad 2\mathcal{N}_2 = 292.31305 \dots, \\ 2\mathcal{N}_3 = 0.45391 \dots, \quad 2\mathcal{N}_4 = 0.85131 \dots, \quad 2F_3 = 4.11705 \dots, \quad 2F_4 = 9.78740 \dots \quad (3.209)$$

Similarly, to compute A_1, A_2, A_3, A_4 and the first two terms on the right hand side of the inequality in (3.6) we choose

$$t \geq t_0 = 4.41 \times 10^{105}, \quad \epsilon = 0.1, \quad \eta_3 = 1, \quad \eta_4 = 1, \quad j = 5, \quad m = 10000. \quad (3.210)$$

Hence we have

$$2 \left| \sum_{n=1}^m n^{-1/2-it} \right| + 2 \left| \sum_{n=m+1}^{\lfloor jt^{1/4} \rfloor - \delta_0} n^{-1/2-it} \right| + 2 \left| \sum_{n=\lceil jt^{1/4} \rceil}^{\lfloor t^{7/17} \rfloor} n^{-1/2-it} \right| + \mathcal{R}(t) \quad (3.211)$$

$$\leq 2 \sum_{n=1}^m n^{-1/2} + 4\sqrt{j}t^{1/8} - 4m + A_1 t^{19/119} + A_2 t^{-1/8} + A_3 t^{295/1904} + A_4 t^{-3/16} + \mathcal{R}(t_0), \quad (3.212)$$

where

$$\begin{aligned} A_1 &= 16.05388 \dots; \quad A_2 = 12.70136 \dots; \quad ; A_3 = 17.66801 \dots; \\ A_4 &= 117.39097 \dots; \quad 4\sqrt{j} = 4\sqrt{5}. \end{aligned} \quad (3.213)$$

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